

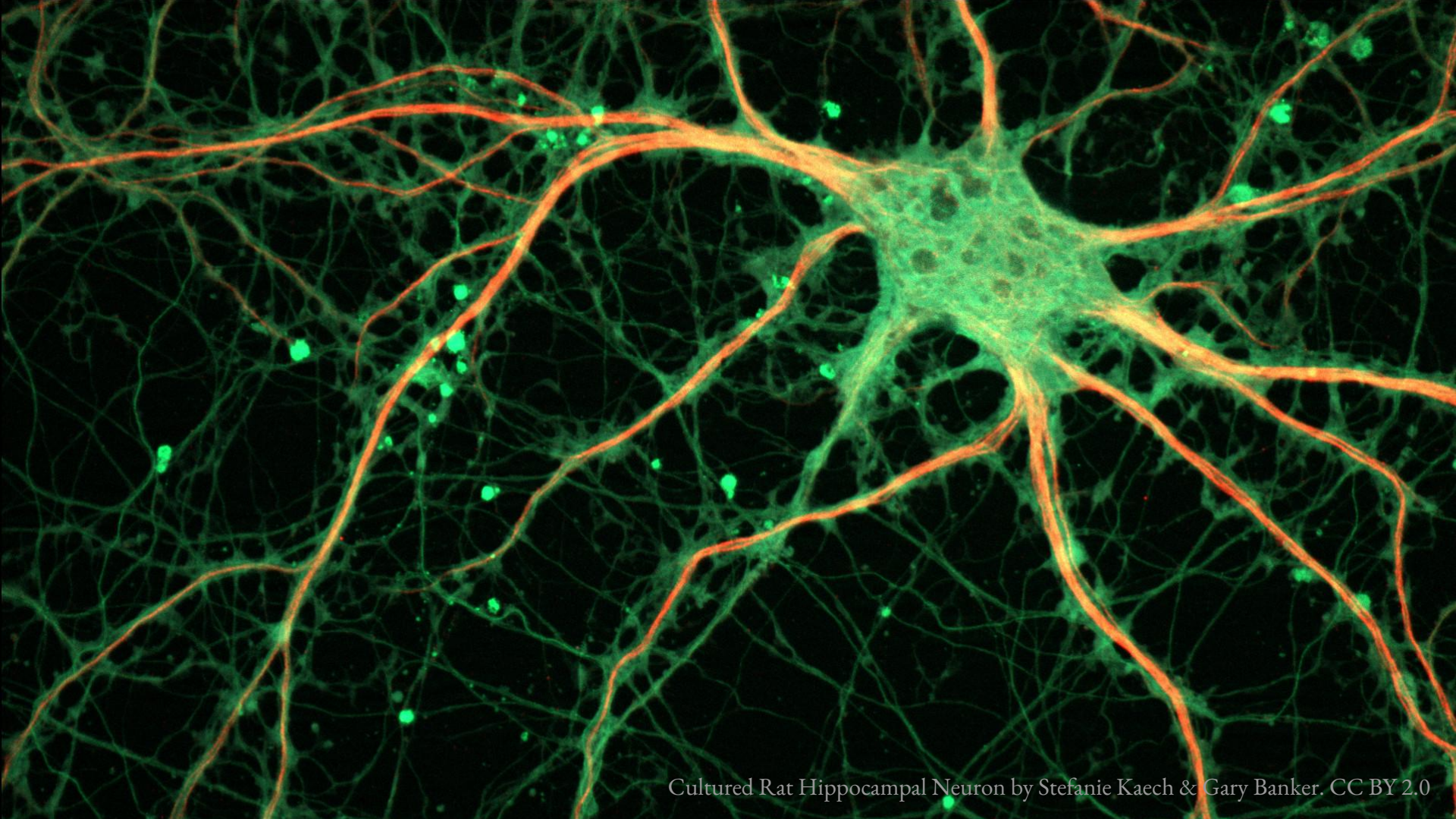
Scale and Intelligence

CSCI-GA.2590: Natural Language Processing, March 5th, 2025

Nicholas Lourie



<https://nyu-cs2590.github.io/spring2025/>



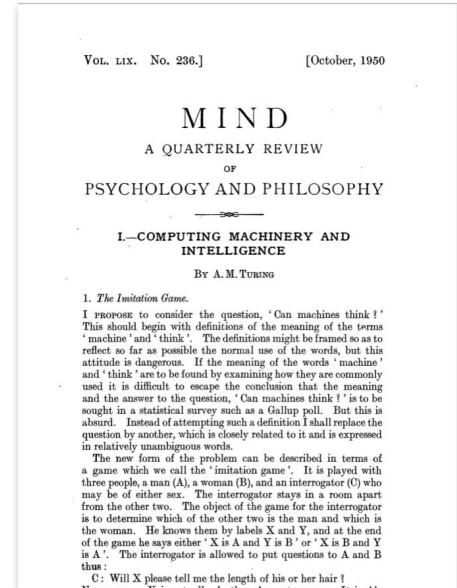
Cultured Rat Hippocampal Neuron by Stefanie Kaech & Gary Banker. CC BY 2.0





Intelligence
needs **scale.**

“Is there a corresponding phenomenon for minds, and is there one for machines?”



Computing Machinery and Intelligence, Alan M. Turing



OpenAI



deepseek

Scale might
lead to **intelligence**.

In the Beginning...

How did scaling laws come about?

Imagine you're
an *NLP* researcher

Imagine you're
an *NLP* researcher
in **2018**.



CONTROL DATA | 6600

CONTROL DATA | 6600

THE FATHER OF THE SUPERCOMPUTER

Superior C-100 features a central CPU, and the two large circular displays show the computer's internal circuitry. In the background, the CDC 6600 is visible, showing its massive size and the complex wiring that connects it to the rest of the system.

A LOT OF COMPUTER

The CDC 6600 was installed in the basement of the Department of Science Building at Purdue University in 1962. In its memory, it stored over 100,000 words of data. It was the first computer to use a central processor to manage the flow of data between the various units of the system.

WHY SO FAST?

Designers of the CDC 6600 series realized that the central processor was the bottleneck in the system. To get around this, they added 16 peripheral processors to handle all the data processing. This allowed the CDC 6600 to process data at a rate of 3 million words per second.

In 2018, GPT1 arrives!

Improving Language Understanding by Generative Pre-Training

Alec Radford	Karthik Narasimhan	Tim Salimans	Ilya Sutskever
OpenAI	OpenAI	OpenAI	OpenAI
alec@openai.com	karthikn@openai.com	tim@openai.com	ilyasu@openai.com

Abstract

Natural language understanding comprises a wide range of diverse tasks such as textual entailment, question answering, semantic similarity assessment, and document classification. Although large unlabeled text corpora are abundant, labeled data for learning these specific tasks is scarce, making it challenging for discriminatively trained models to perform adequately. We demonstrate that large gains on these tasks can be realized by *generative pre-training* of a language model on a diverse corpus of unlabeled text, followed by *discriminative fine-tuning* on each specific task. In contrast to previous approaches, we make use of task-aware input transformations during fine-tuning to achieve effective transfer while requiring minimal changes to the model architecture. We demonstrate the effectiveness of our approach on a wide range of benchmarks for natural language understanding. Our general task-agnostic model outperforms discriminatively trained models that use architectures specifically crafted for each task, significantly improving upon the state of the art in 9 out of the 12 tasks studied. For instance, we achieve absolute improvements of 8.9% on commonsense reasoning (Stories Cloze Test), 5.7% on question answering (RACE), and 1.5% on textual entailment (MatchNLI).

1 Introduction

The ability to learn effectively from raw text is crucial to alleviating the dependence on supervised learning in natural language processing (NLP). Most deep learning methods require substantial amounts of manually labeled data, which restricts their applicability in many domains that suffer from a dearth of annotated resources [31]. In these situations, models that can leverage linguistic information from unlabeled data provide a valuable alternative to gathering more annotation, which can be time-consuming and expensive. Further, even in cases where considerable supervision is available, learning good representations in an unsupervised fashion can provide a significant performance boost. The most compelling evidence for this so far has been the extensive use of pre-trained word embeddings [10, 39, 42] to improve performance on a range of NLP tasks [8, 13, 26, 45]. Leveraging more than word-level information from unlabeled text, however, is challenging for two main reasons. First, it is unclear what type of optimization objectives are most effective at learning text representations that are useful for transfer. Recent research has looked at various objectives such as language modeling [64], machine translation [48], and discourse coherence [72], with each method outperforming the others on different tasks¹. Second, there is no consensus on the most effective way to transfer these learned representations to the target task. Existing techniques involve a combination of making task-specific changes to the model architecture [63, 44], using inductive learning schemes [21] and adding auxiliary learning objectives [59]. These uncertainties have made it difficult to develop effective semi-supervised learning approaches for language processing.

¹<https://gluebenchmark.com/leaderboard>

Preprint. Work in progress.



In the Beginning...

Improving Language Understanding by Generative Pre-Training (Radford et al., 2018)

Method	MNLI-m	MNLI-mm	SNLI	SciTail	QNLI	RTE
ESIM + ELMo [44] (5x)	-	-	<u>89.3</u>	-	-	-
CAFE [58] (5x)	80.2	79.0	<u>89.3</u>	-	-	-
Stochastic Answer Network [35] (3x)	<u>80.6</u>	<u>80.1</u>	-	-	-	-
CAFE [58]	78.7	77.9	88.5	<u>83.3</u>		
GenSen [64]	71.4	71.3	-	-	<u>82.3</u>	59.2
Multi-task BiLSTM + Attn [64]	72.2	72.1	-	-	<u>82.1</u>	61.7
Finetuned Transformer LM (ours)	82.1	81.4	89.9	88.3	88.1	56.0

Old World

—

—

In the Beginning...

Summarization

Sentiment Analysis

Mathematics

Programming

Translation

Relation Extraction

Dialogue

Semantic Parsing

Syntactic Parsing

Common Sense

Entailment

Multimodality

Slot Filling

Logical Reasoning

Question Answering

Search & Retrieval

Creative Writing

Paraphrasing

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- **Supervised** learning *works*.



Old World

- **Supervised** learning *works*.
- Labeled **data** is *expensive*.



\$???

Dolma's price with CommonsenseQA's cost per word?

\$60B

Dolma's price with CommonsenseQA's cost per word.

\$600B

Modern corpus's price with CommonsenseQA's cost per word.

Old World

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Old World

- **Supervised** learning *works*.
- Labeled **data** is *expensive*.
- Evaluate on **fixed** *benchmarks*.



In the Beginning...

Methodology:

Hold the data fixed, iterate on the model.

In the Beginning...

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In the Beginning...

Semi-supervised Sequence Tagging with Bidirectional Language Models (Peters et al., 2017)

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Semi-supervised Sequence Tagging with Bidirectional Language Models (Peters et al., 2017)

Model	External resources	F_1 Without	F_1 With	Δ
Yang et al. (2017)	transfer from CoNLL 2000/PTB-POS	91.2	91.26	+0.06
Chiu and Nichols (2016)	with gazetteers	90.91	91.62	+0.71
Collobert et al. (2011)	with gazetteers	88.67	89.59	+0.92
Luo et al. (2015)	joint with entity linking	89.9	91.2	+1.3
Ours	no LM vs TagLM <i>unlabeled data only</i>	90.87	91.93	+1.06

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In the Beginning...

Keep the data fixed
for **practical** reasons.

In the Beginning...

Keep the data fixed
for **scientific** reasons.

Old World

- **Supervised** learning *works*.
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Unsupervised
learning begins
to work.



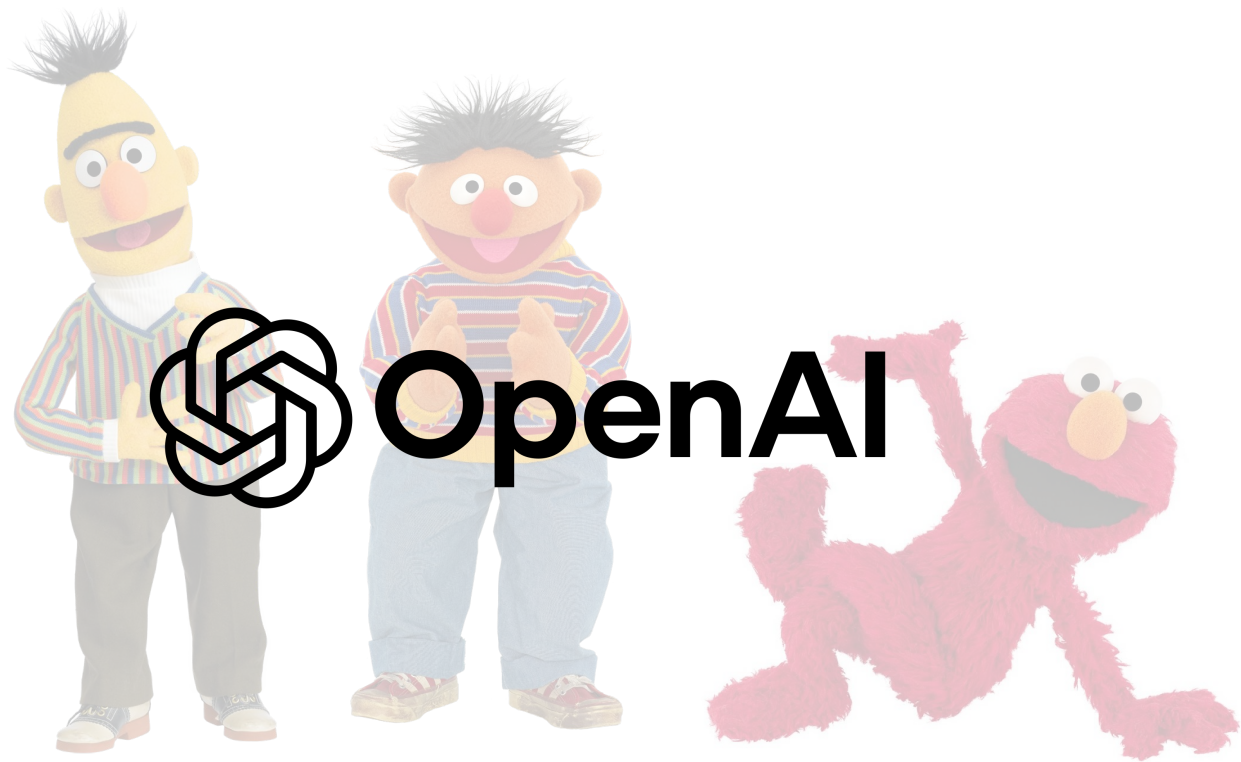
Deep Contextualized Word Representations (Peters et al., 2018)



BERT: Pre-training of Deep Bidirectional Transformers (Devlin et al., 2019)



ERNIE: Enhanced Language Representation (Zhang et al., 2019)



GPT 2, 3, 4, ...

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New World

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In the Beginning...

In NLP, you just...

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use structured prediction.

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train different neural architectures from scratch.

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finetune a pretrained language model.

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just ask the language model a question.

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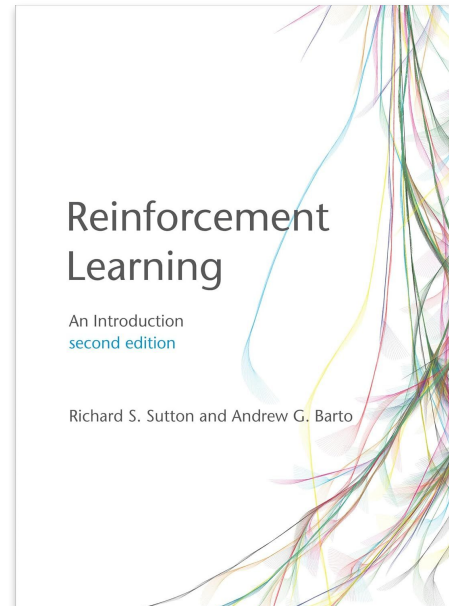
~~finetune a pretrained language model.~~

just ask the language model a question.

...

Scale is
all you *need*.

“... general methods that leverage computation are ultimately the most effective, and by a large margin.”



The Bitter Lesson, Richard S. Sutton

In the Beginning...

We want “... methods that continue to scale...
even as the available computation becomes very great.”

The Bitter Lesson, Richard S. Sutton

In the Beginning...

“... the actual contents of minds are tremendously,
irredeemably complex;”

The Bitter Lesson, Richard S. Sutton

Old World

- **Supervised** learning *works*.
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-

New World

- **Unsupervised** learning *works*.
 - Unlabeled **data** is *cheap*.
-

Try to build the **best model**

Try to build ~~the best model~~
simple methods that **scale efficiently**.

How do you
build *at* **scale**?

In the Beginning...

Before you scale it:

In the Beginning...

Before you scale it:

- What's the cost?

In the Beginning...

Before you scale it:

- What's the cost?
- What's the benefit?

In the Beginning...

Before you scale it:

- What's the cost?
- What's the benefit?
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In the Beginning...

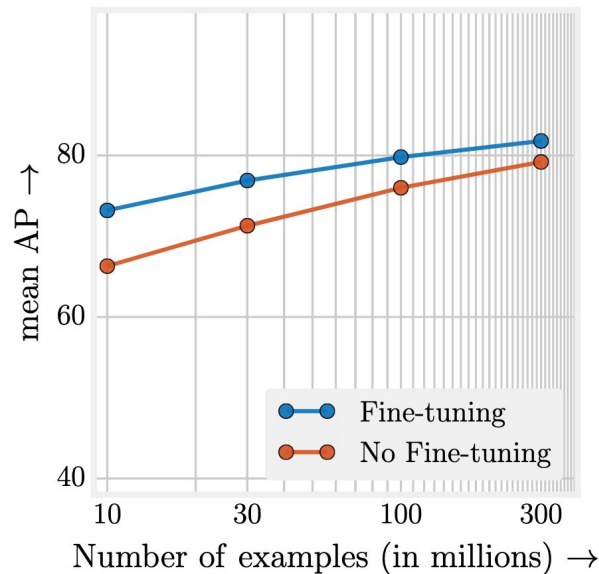
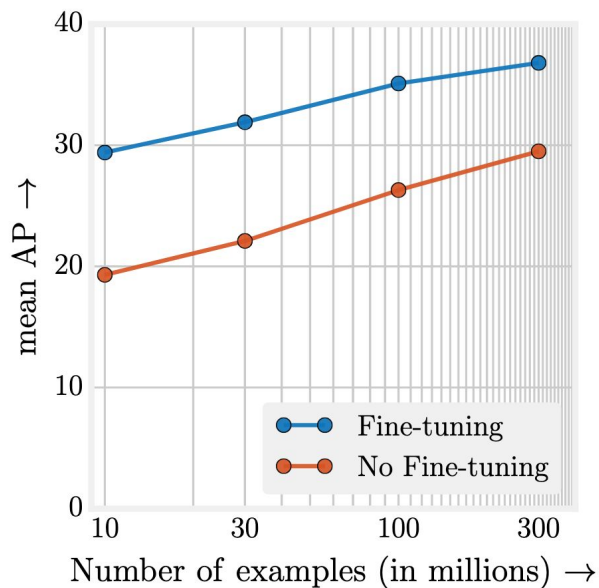
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Deep learning
scales **predictably.**

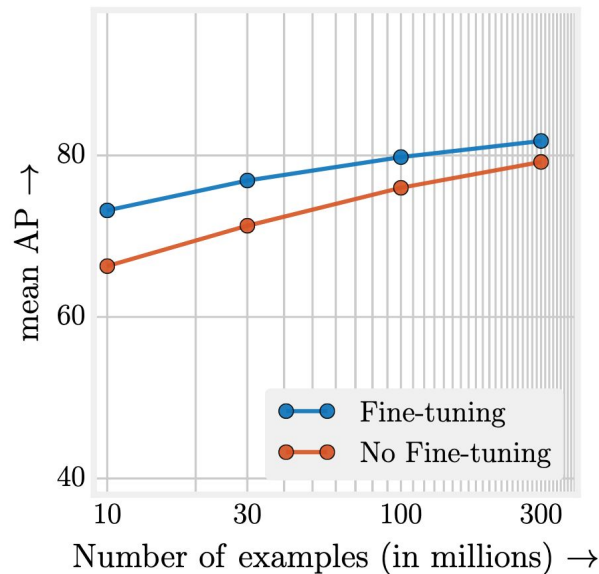
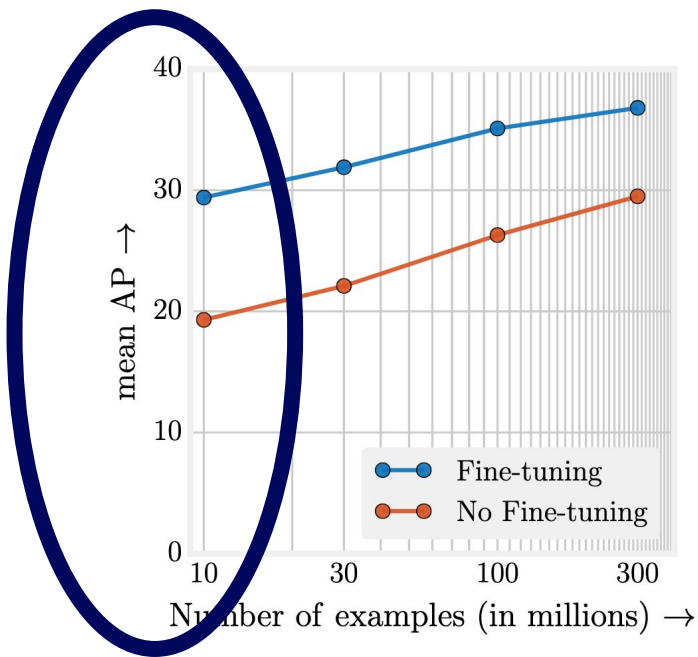
In the Beginning...

Revisiting Unreasonable Effectiveness of Data in Deep Learning Era (Sun et al., 2017)



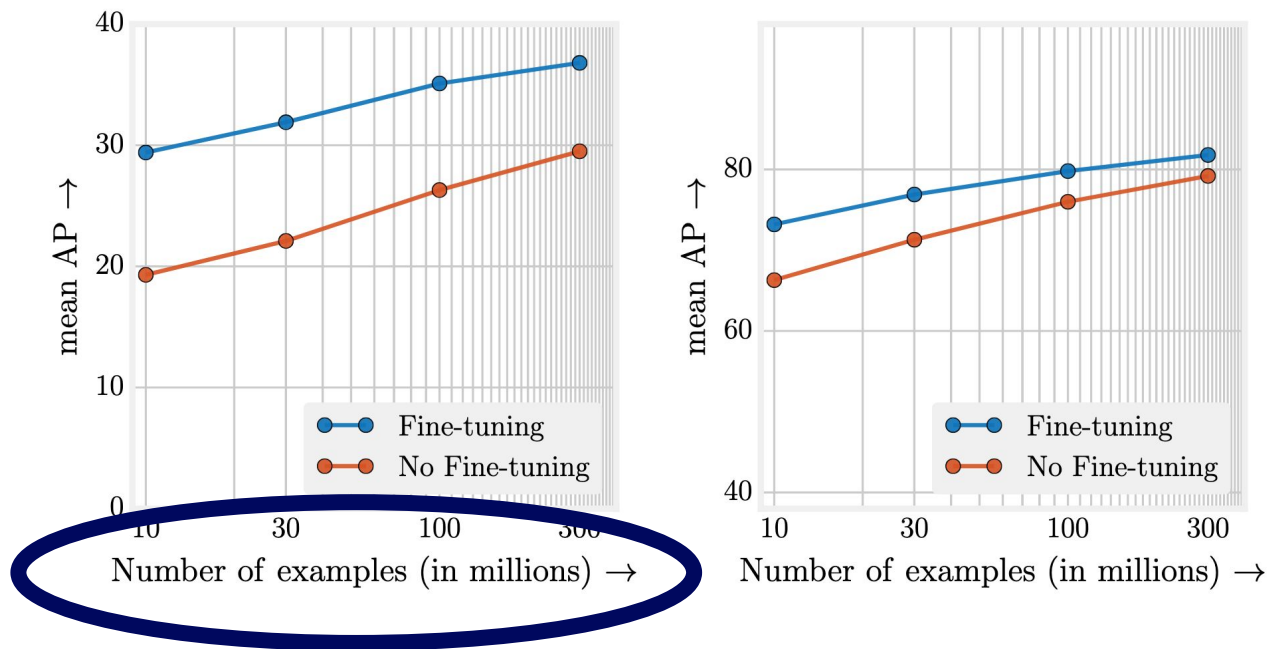
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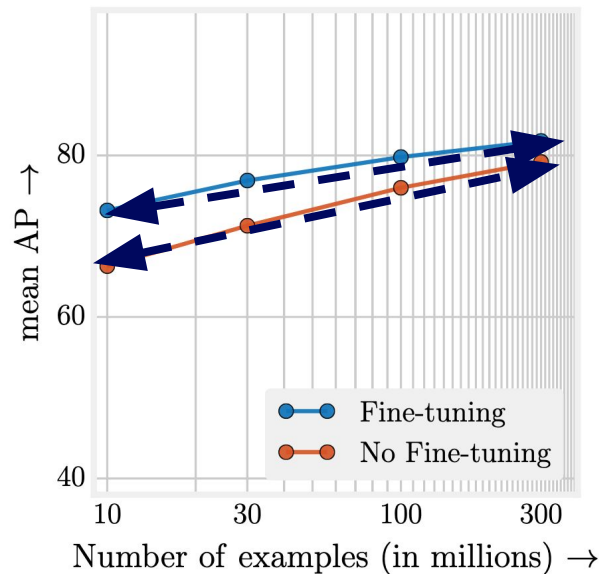
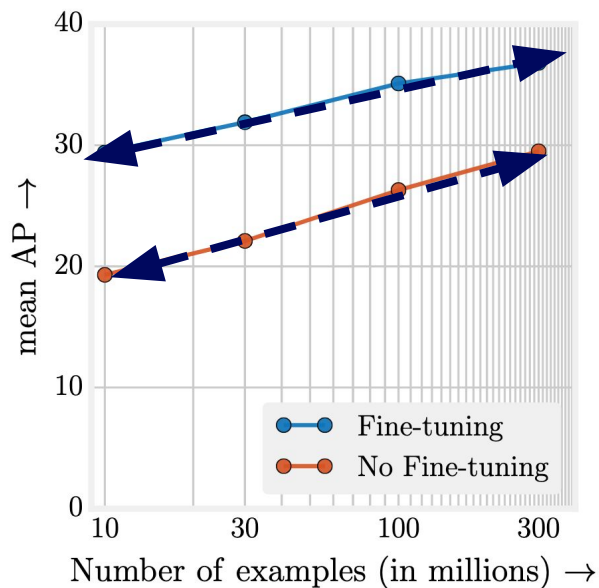
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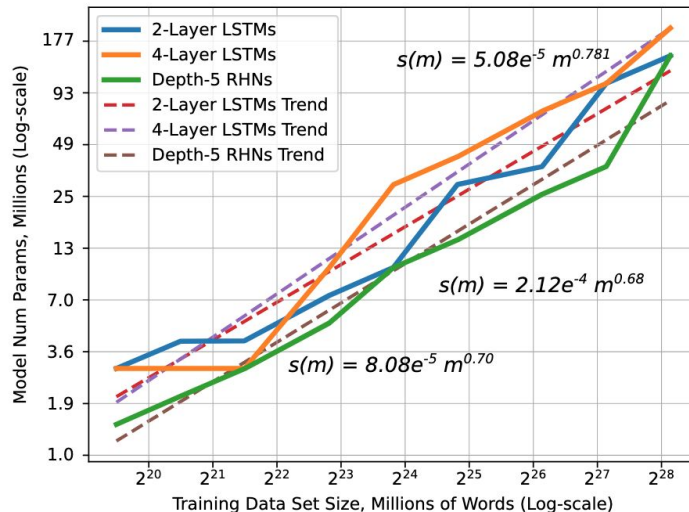
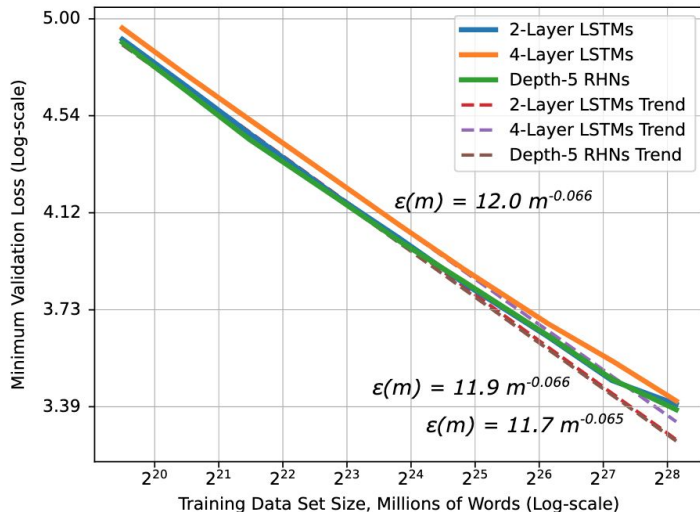
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Without a theory, a scaling law's

Without a theory, a scaling law's
functional form is only an **approximation**.

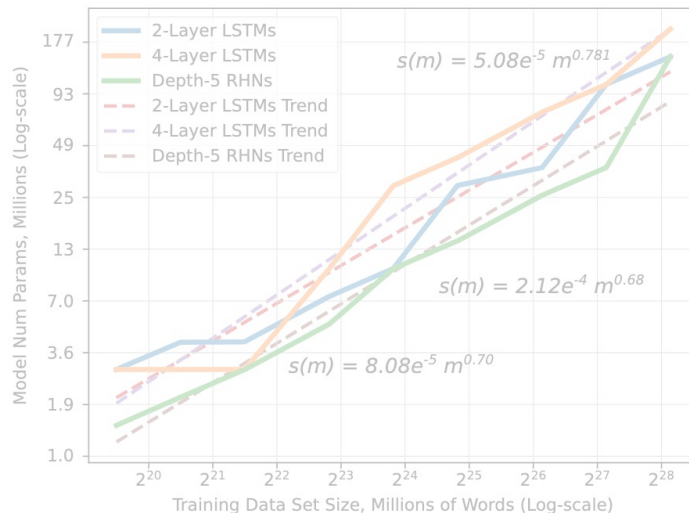
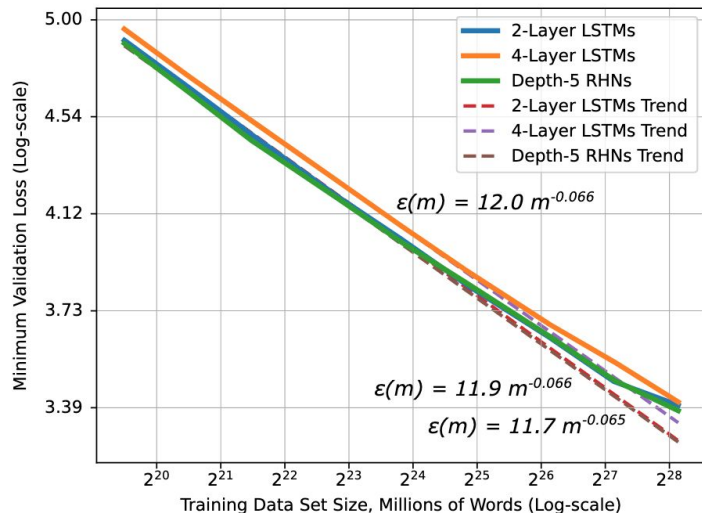
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Deep Learning Scaling is Predictable, Empirically (Hestness et al., 2017)



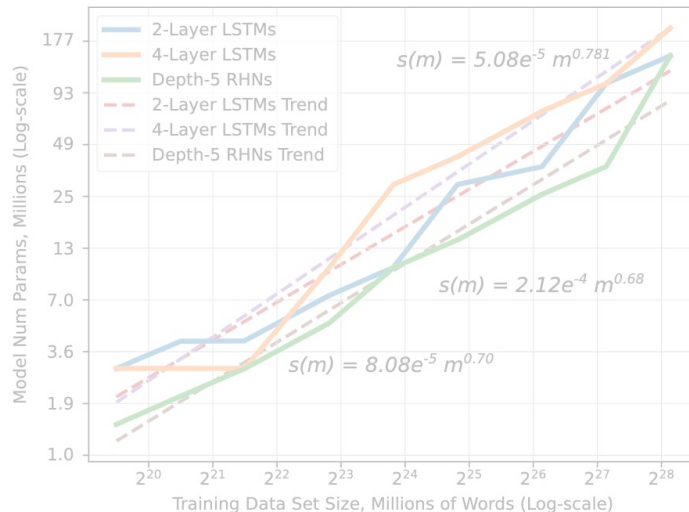
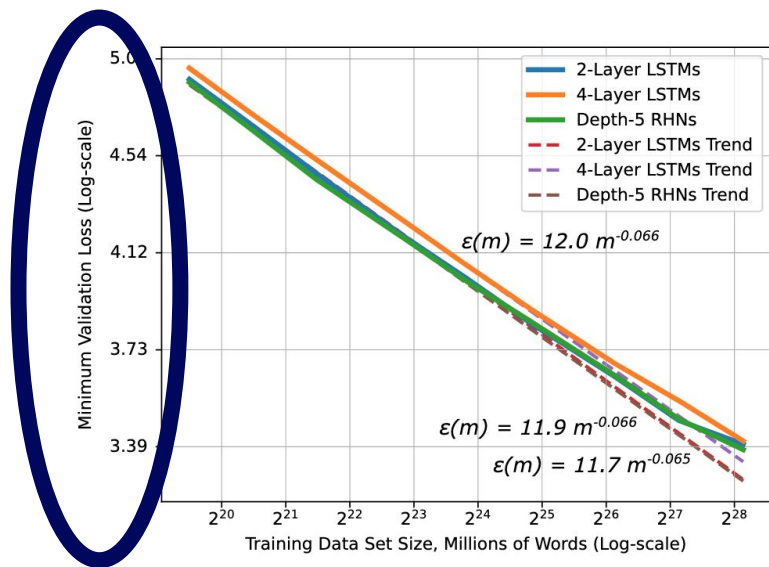
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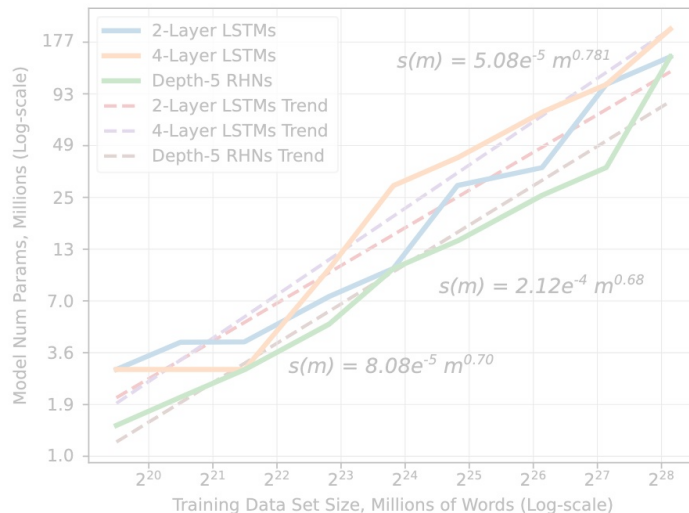
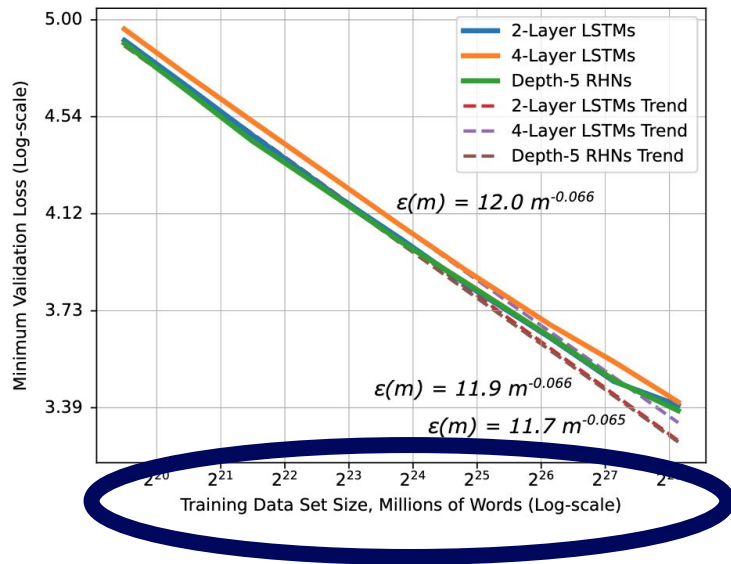
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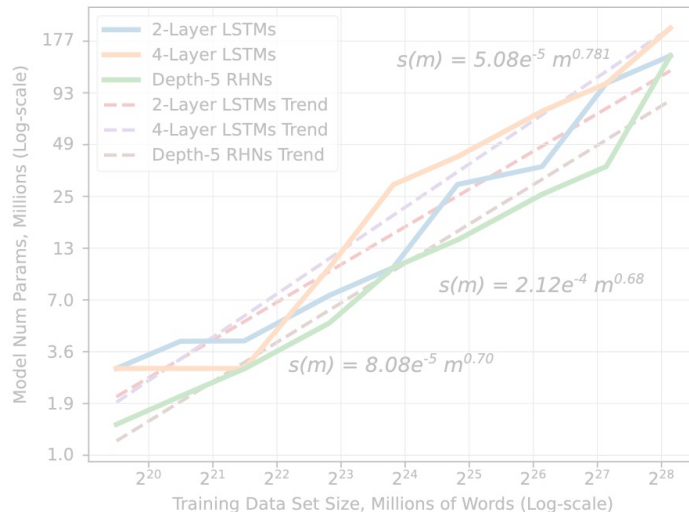
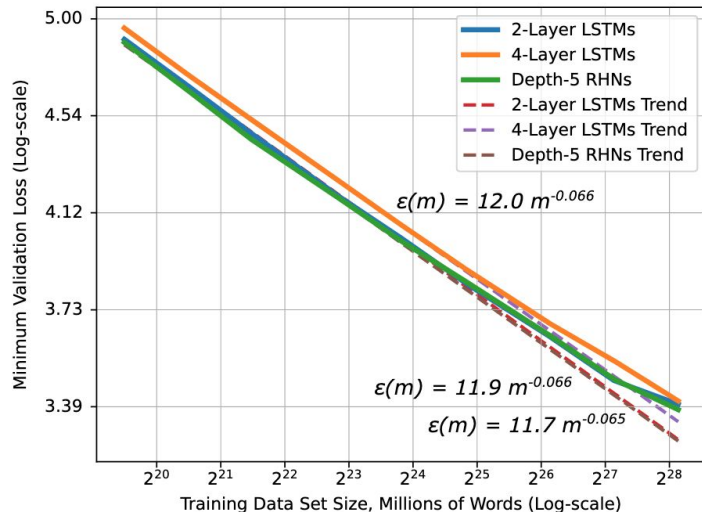
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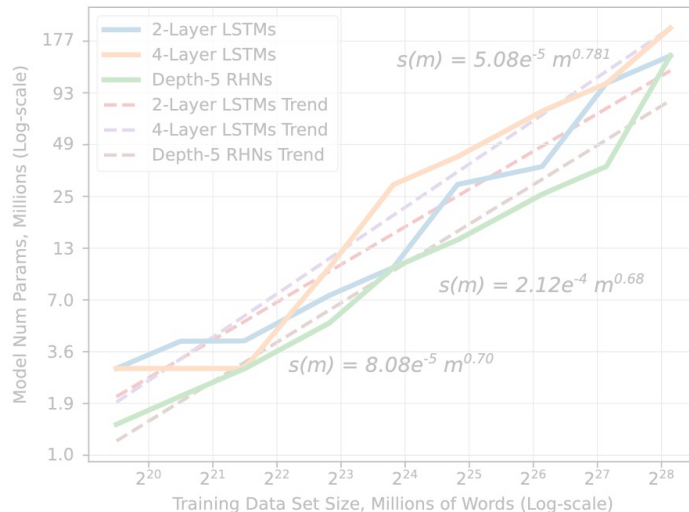
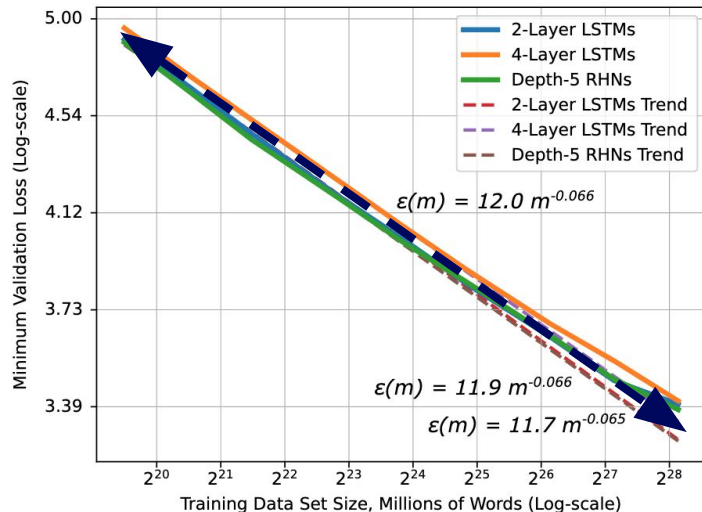
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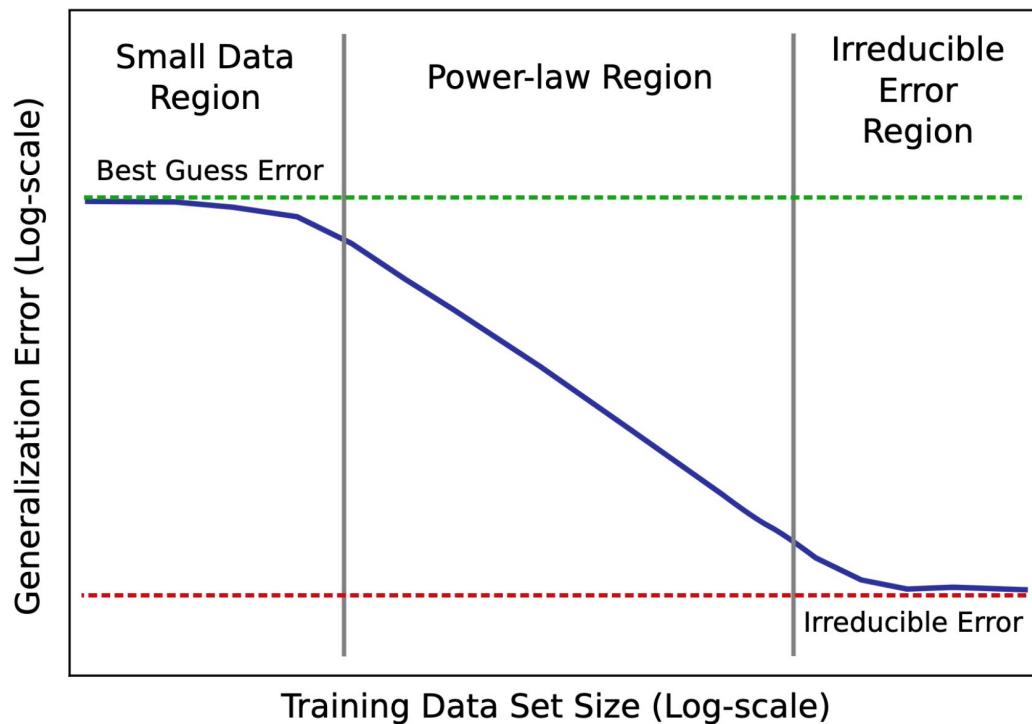
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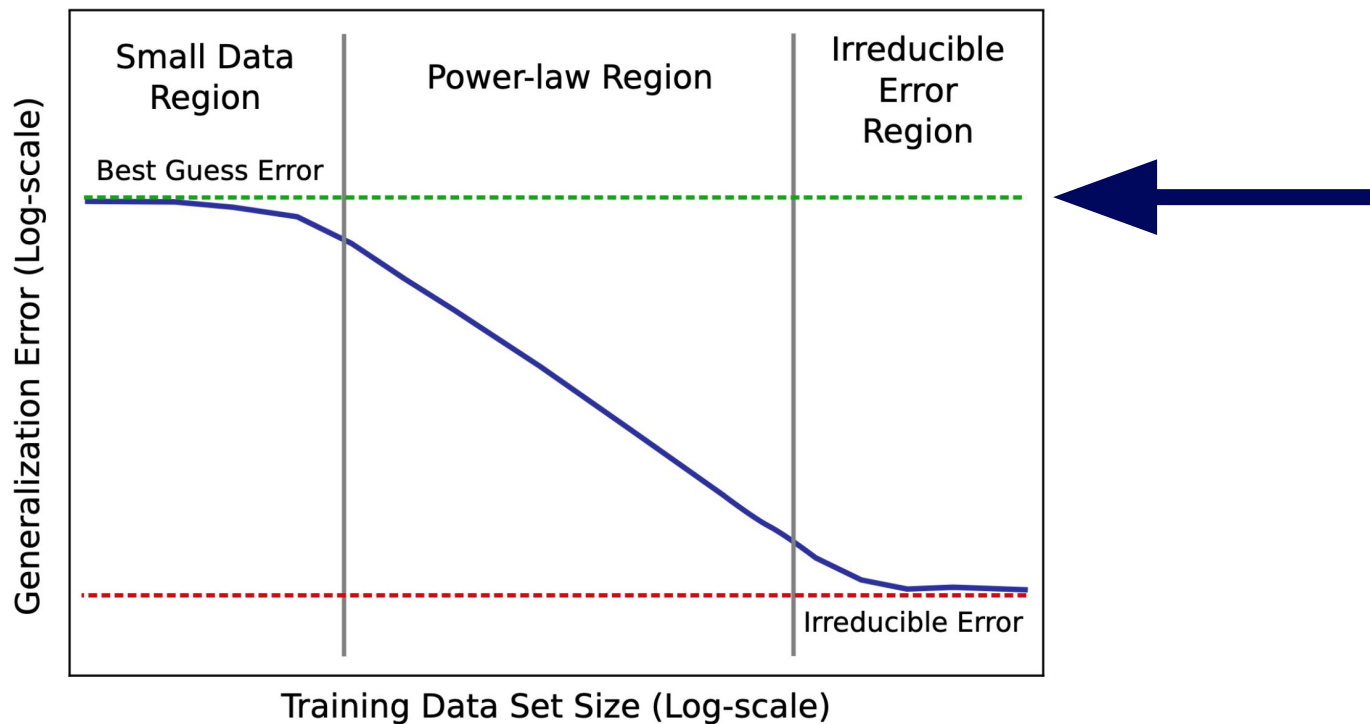


$$\varepsilon(m) \sim \alpha m^{\beta_g}$$

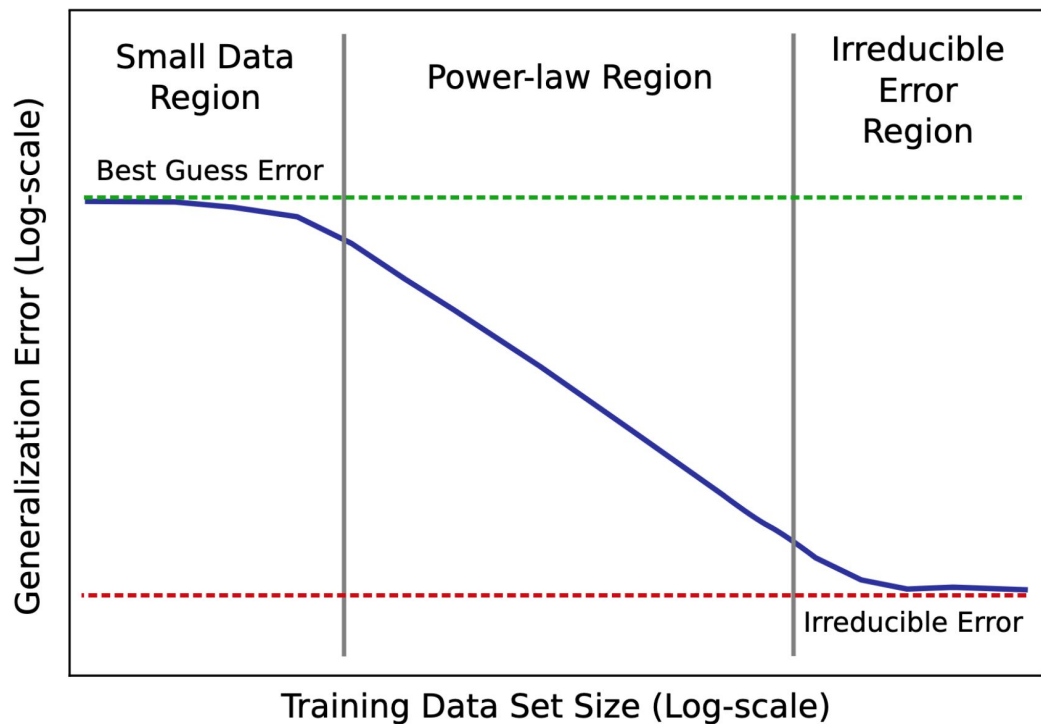
Power Law Scaling



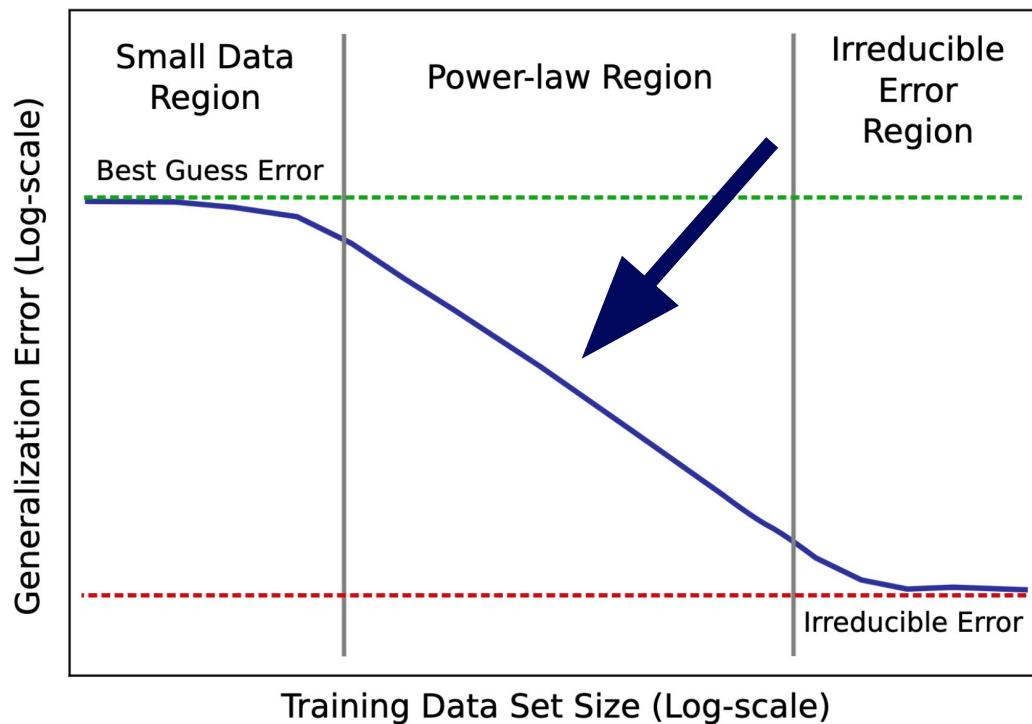
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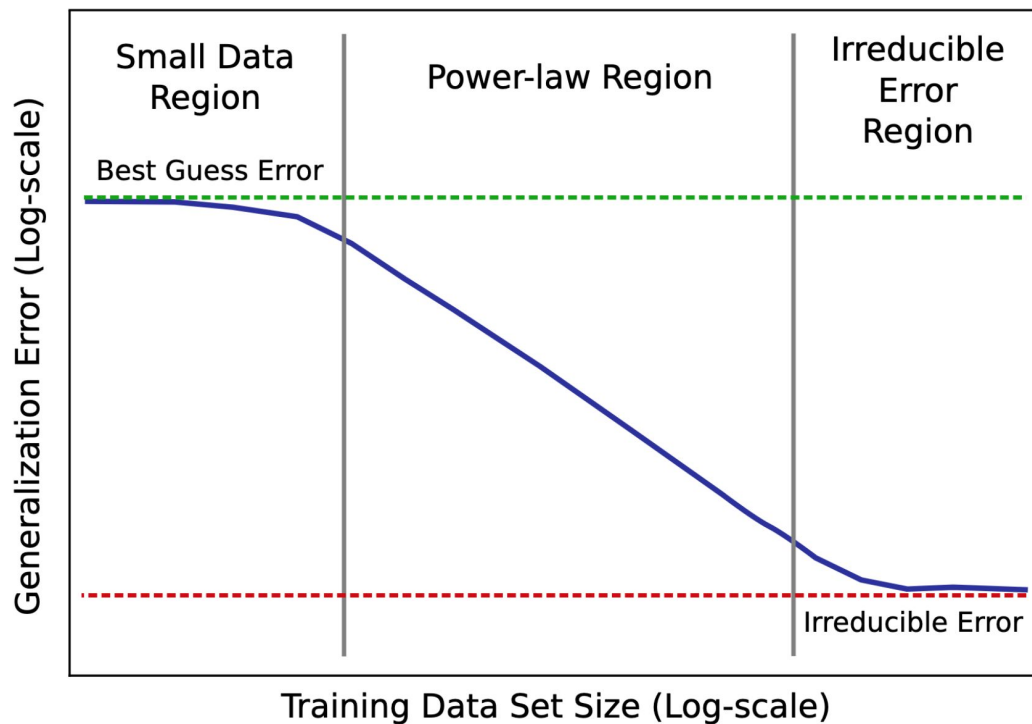
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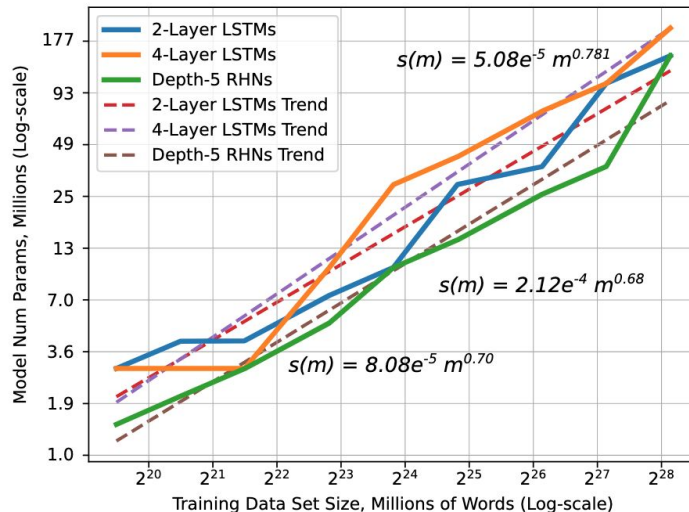
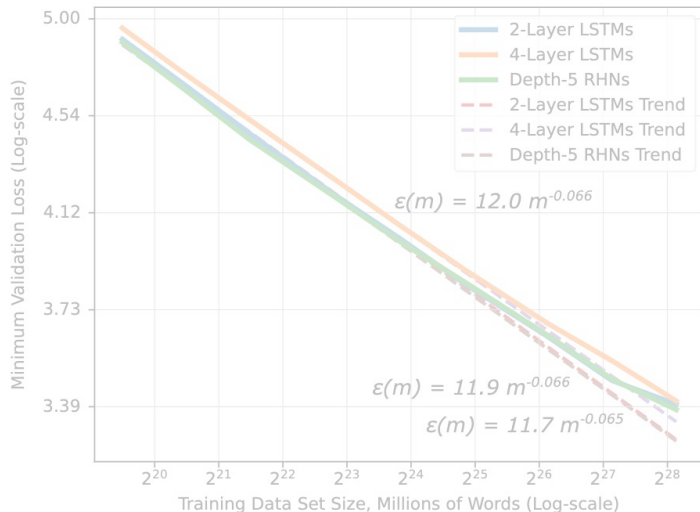
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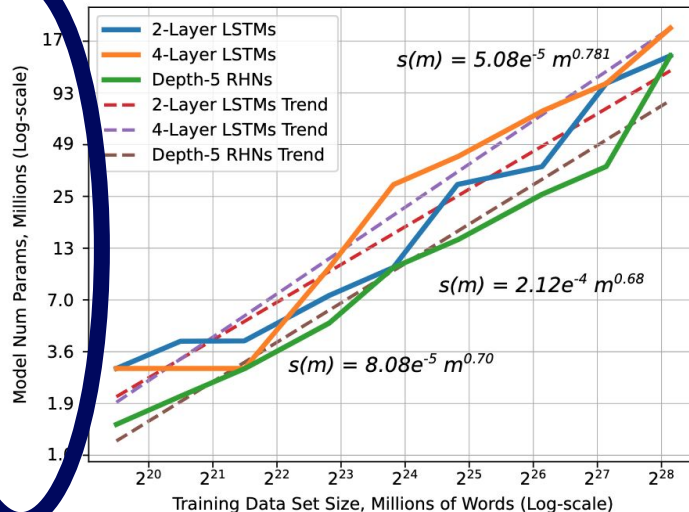
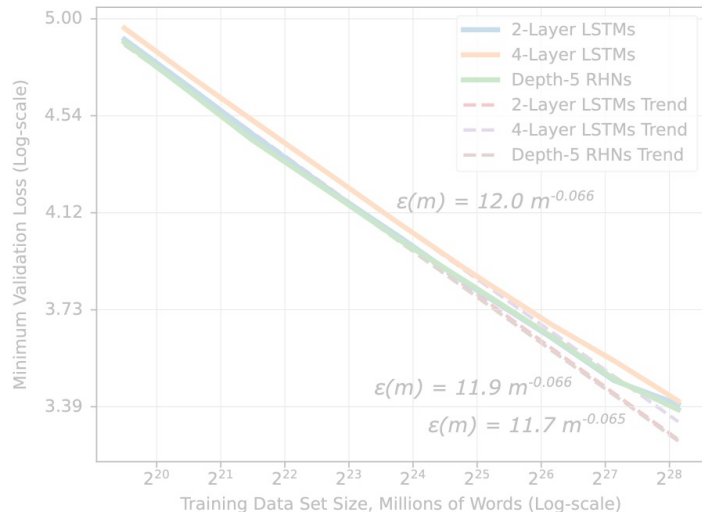
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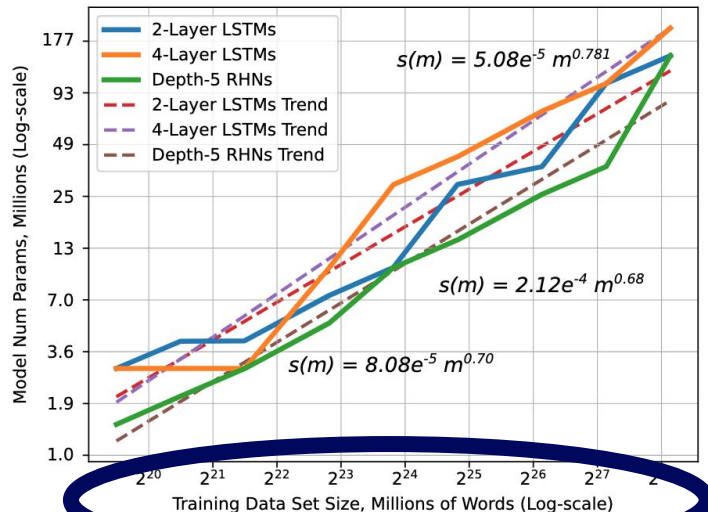
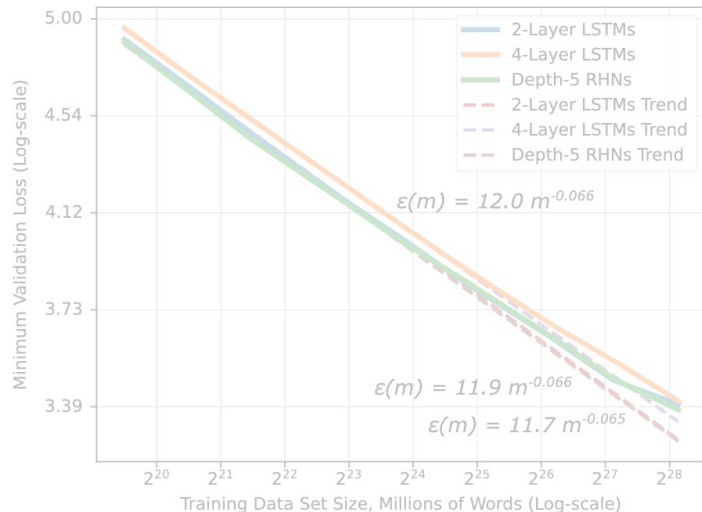
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Deep Learning Scaling is Predictable, Empirically (Hestness et al., 2017)



There's *more*
to learning
than **data**.

Old World

- **Supervised** learning *works*.
- Labeled **data** is *expensive*.
- Evaluate on **fixed** *benchmarks*.



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New World

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 - Unlabeled **data** is *cheap*.
 - Evaluate on **new** *capabilities*.
-

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Logical Reasoning

Question Answering

Search & Retrieval

Creative Writing

Paraphrasing

Old World

- **Supervised** learning *works*.
 - Labeled **data** is *expensive*.
 - Evaluate on **fixed** *benchmarks*.
-

New World

- **Unsupervised** learning *works*.
 - Unlabeled **data** is *cheap*.
 - Evaluate on **new** *capabilities*.
-

Scaling Laws for Pretraining

How do models improve as they *scale*?

What goes into
a foundation model?

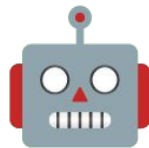
Intelligence =

Intelligence = Data

Intelligence = Data + Compute

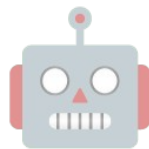


Intelligence = Data + Compute





Intelligence = Data + Compute



How do you
measure **data**?

Scaling Laws for Pretraining

$$(X_i, Y_i) \sim \mathcal{D}$$

Examples are sampled independently from a distribution.

Scaling Laws for Pretraining

$$(X_i, Y_i) \sim \mathcal{D}$$

$$(X_1, Y_1), \dots, (X_n, Y_n)$$

The number of examples is the dataset size.

Data **isn't** just *independent examples*.

Scaling Laws for Pretraining

How *much* data is a **document**?

Scaling Laws for Pretraining

How do you measure data?

Scaling Laws for Pretraining

How do you measure data?

- Documents

Scaling Laws for Pretraining

How do you measure data?

- Documents
- Tokens

Scaling Laws for Pretraining

How do you measure data?

- Documents
- Tokens
- Bytes

Scaling Laws for Pretraining

How do you measure data?

- Documents
- Tokens
- Bytes

Scaling laws are a **tool**.

Scaling laws are a **tool**.

Use whatever makes them **work best**.

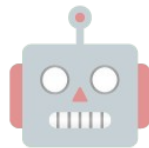
Scaling Laws for Pretraining

How do you measure data?

- Documents
- Tokens
- Bytes

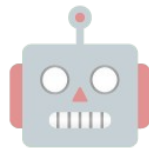


Intelligence = Data + Compute





Intelligence = Data + Compute



How do you
measure **compute**?

Scaling Laws for Pretraining

compute $\propto$ parameters $\times$ data

*Measure compute by counting **batches** and **parameters**.*

Scaling Laws for Pretraining

compute $\propto$ parameters $\times$ data

*Measure compute by counting **batches** and **parameters**.*

Scaling Laws for Pretraining

The gradient for *each* **batch of data**

Scaling Laws for Pretraining

The gradient for *each* **batch of data**
uses *each* **parameter** a few times.

Scaling Laws for Pretraining

The gradient for *each* **batch of data**
uses *each* **parameter** a few times.

Scaling Laws for Pretraining

How do you measure compute?

Scaling Laws for Pretraining

How do you measure compute?

- FLOPs

Scaling Laws for Pretraining

How do you measure compute?

- FLOPs
- GPU hours

Scaling Laws for Pretraining

How do you measure compute?

- FLOPs
- GPU hours
- Dollars

Scaling Laws for Pretraining

How do you measure compute?

- FLOPs
- GPU hours
- Dollars

Scaling Laws for Pretraining

How do you measure compute?

- FLOPs
- GPU hours
- Dollars

Scaling Laws for Pretraining

How do we convert parameters to FLOPs?

Scaling Laws for Pretraining

$$C \approx 6NBS$$

Multiply the FLOPs per parameter by parameters by batch size by steps.

Scaling Laws for Pretraining

$$C \approx 6NBS$$

Multiply the FLOPs per parameter by parameters by batch size by steps.

Scaling Laws for Pretraining

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Multiply the FLOPs per parameter by parameters by batch size by steps.

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Scaling Laws for Pretraining

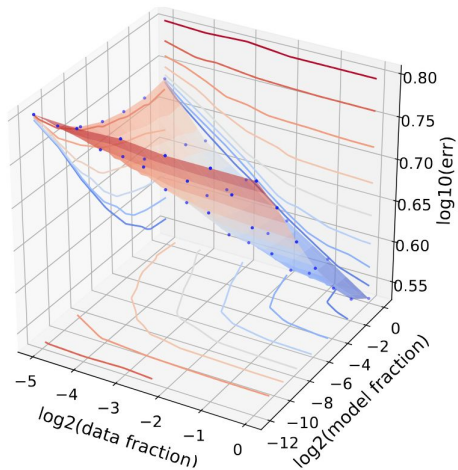
$$C \approx 6NBS$$

Scaling Laws for Neural Language Models (Kaplan et al., 2020)

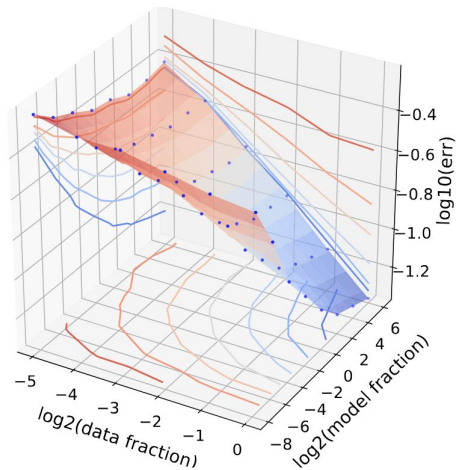
How does *scale*
improve *loss*?

Scaling Laws for Pretraining

A Constructive Prediction of the Generalization Error Across Scales (Rosenfeld et al., 2020)



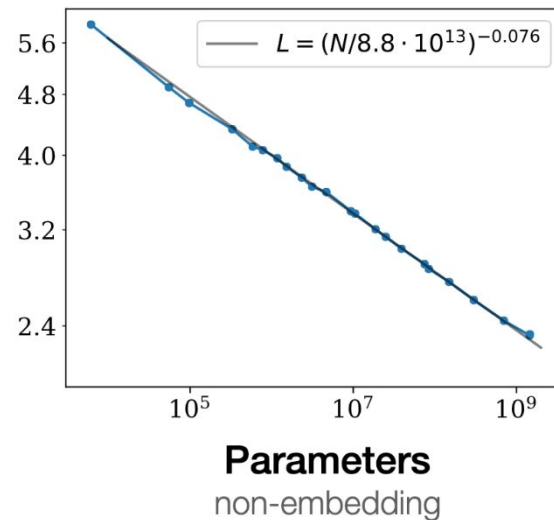
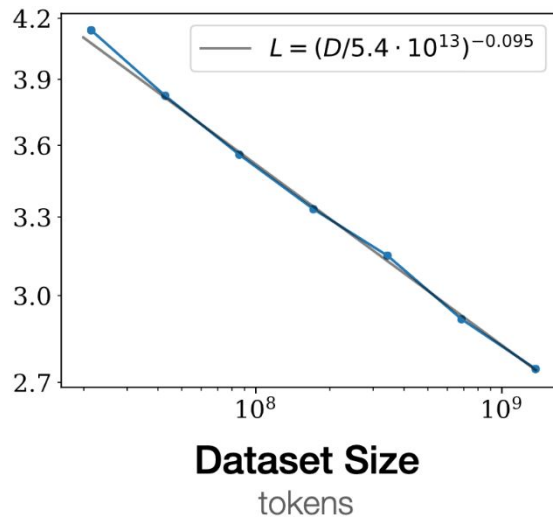
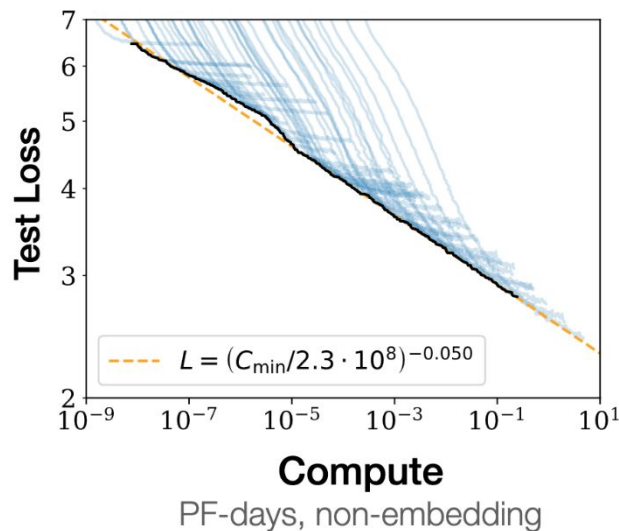
(a) Wiki103 error (cross entropy) landscape.



(b) CIFAR10 error (top1) landscape.

Scaling Laws for Pretraining

Scaling Laws for Neural Language Models (Kaplan et al., 2020)

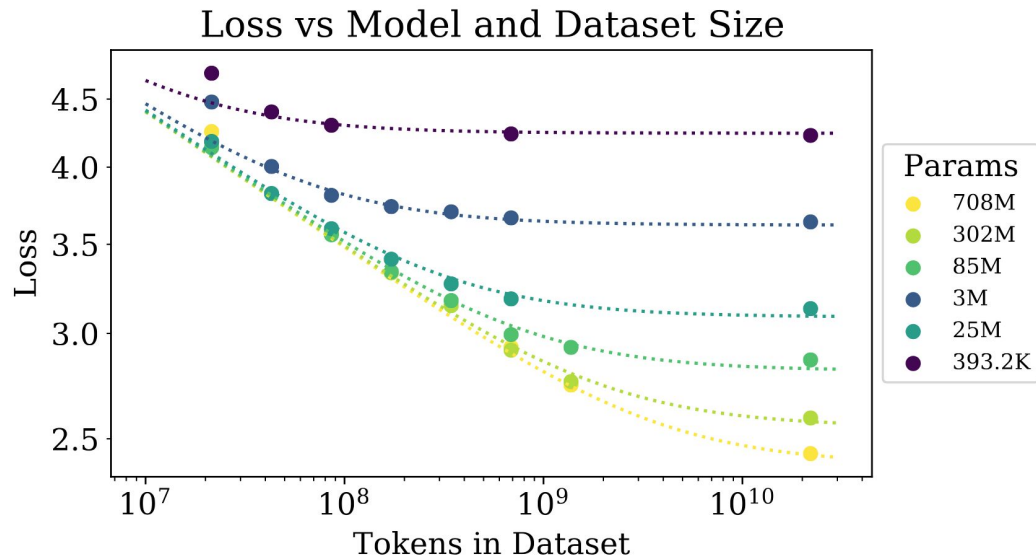


Scaling Laws for Pretraining

How does *scaling* **data** improve *loss*?

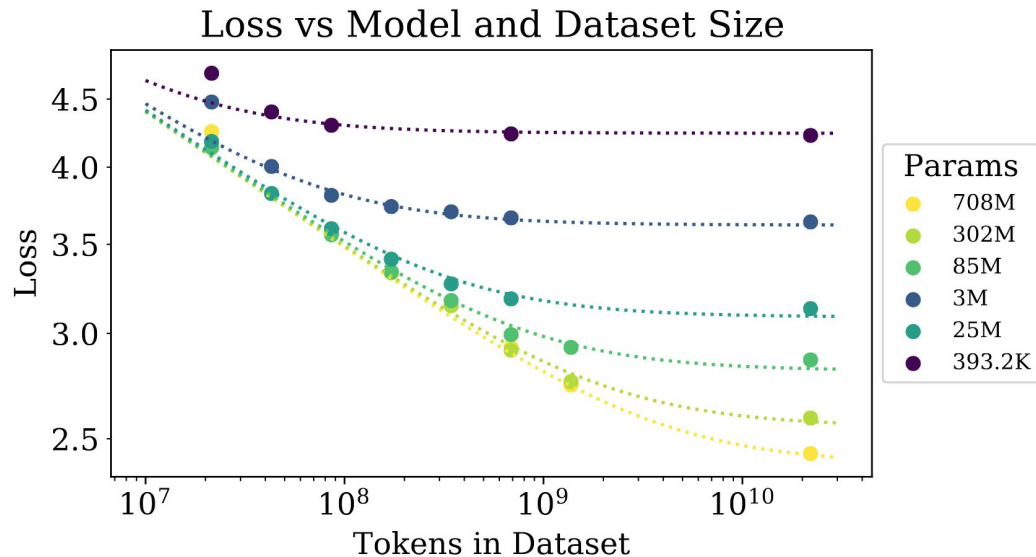
Scaling Laws for Pretraining

Scaling Laws for Neural Language Models (Kaplan et al., 2020)



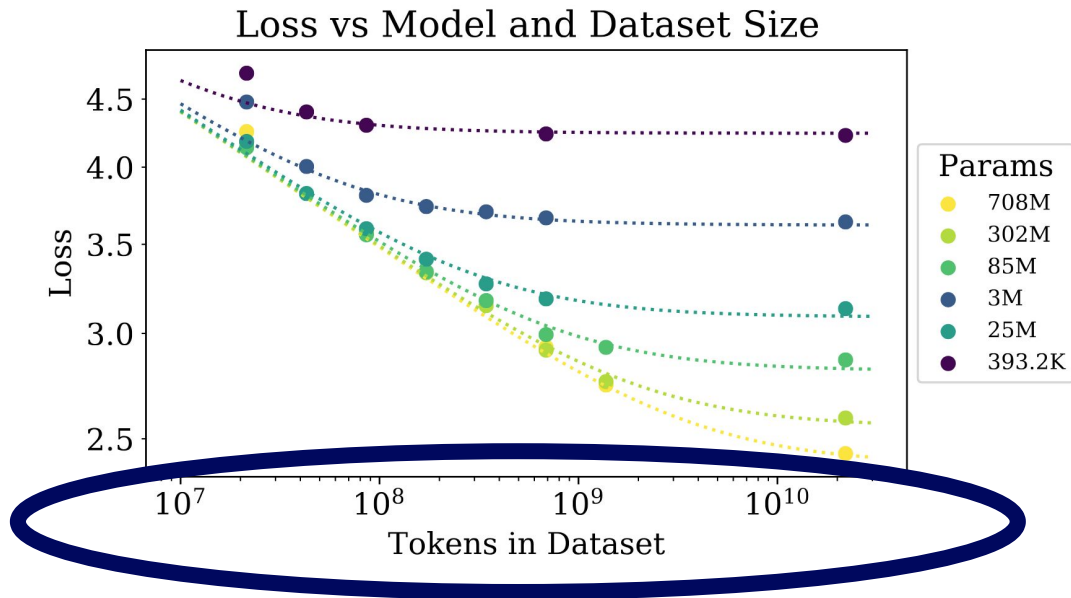
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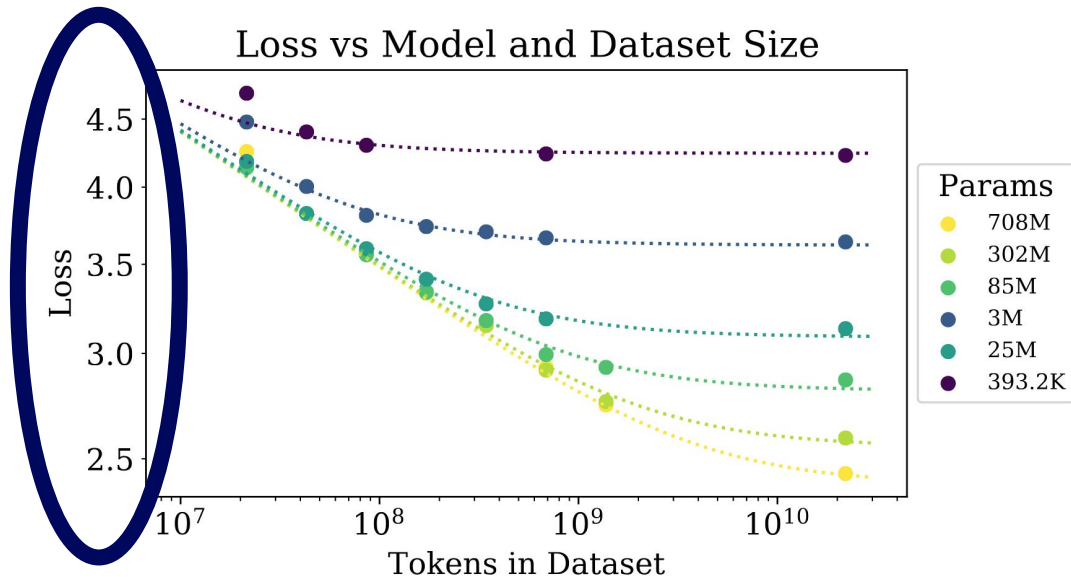
Scaling Laws for Pretraining

Scaling Laws for Neural Language Models (Kaplan et al., 2020)



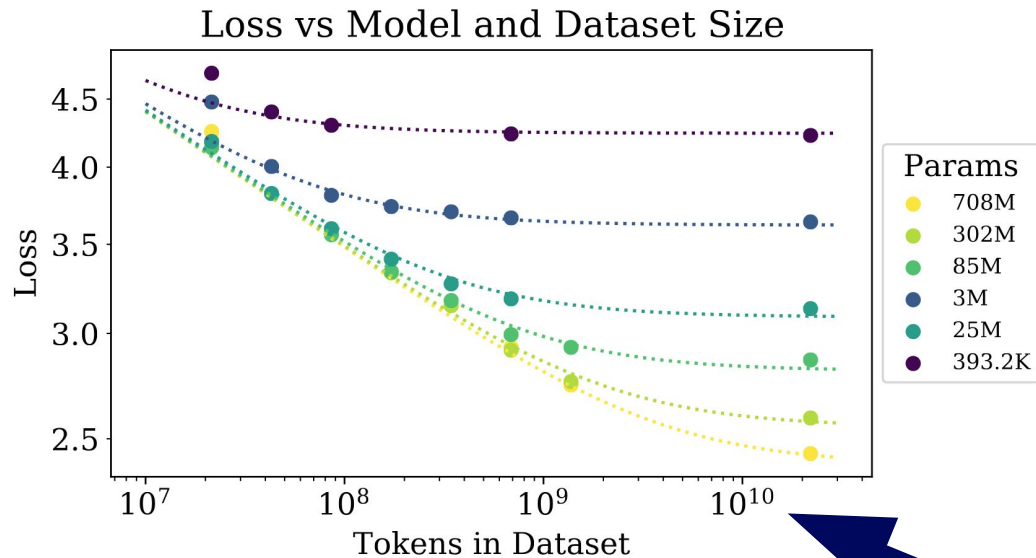
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Scaling Laws for Neural Language Models (Kaplan et al., 2020)



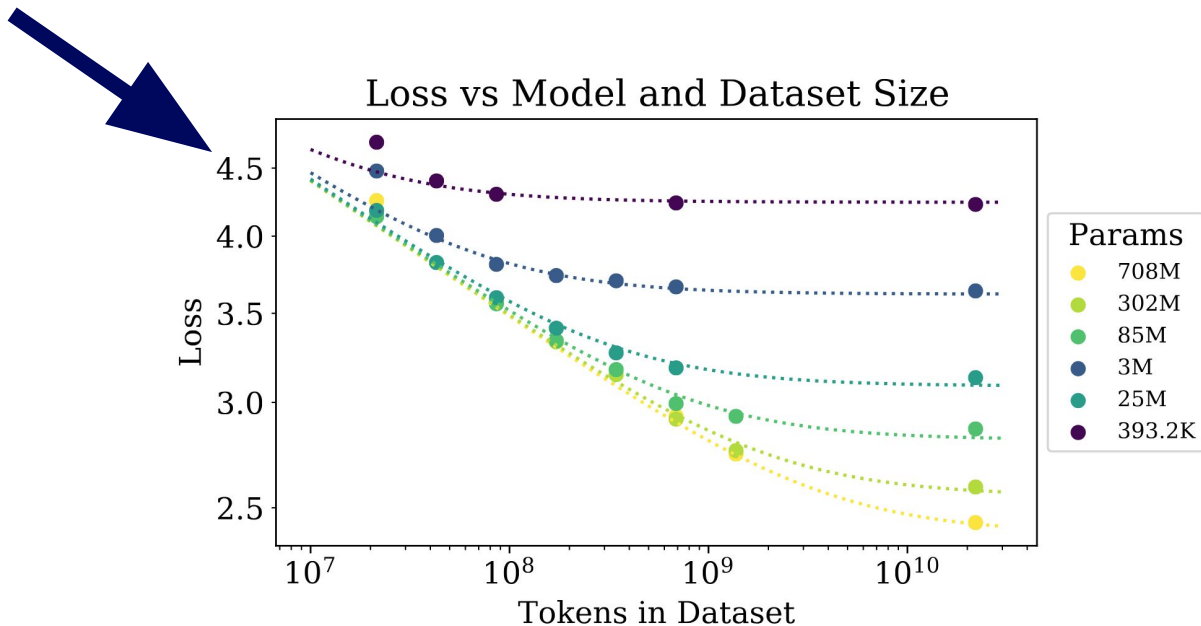
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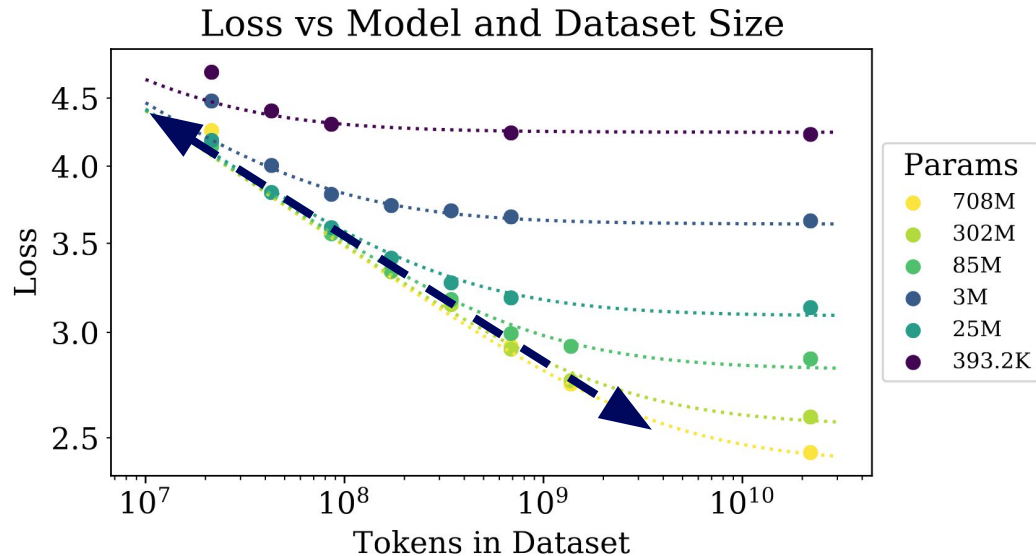
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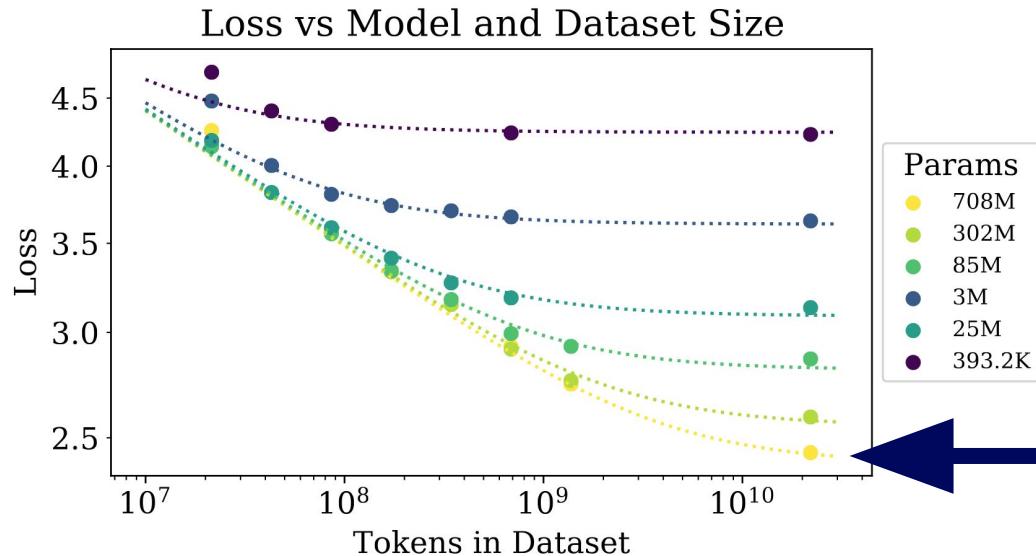
Scaling Laws for Pretraining

Scaling Laws for Neural Language Models (Kaplan et al., 2020)



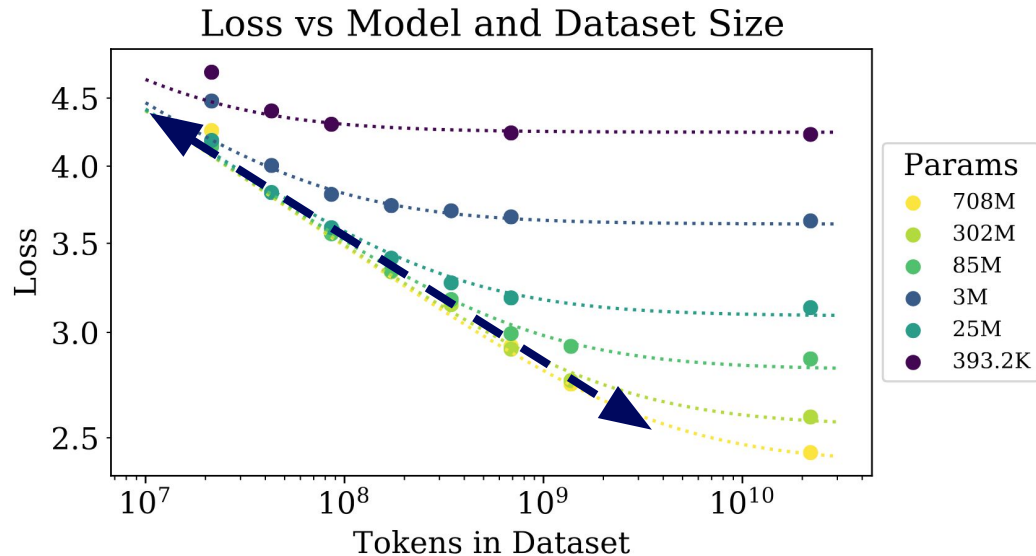
Scaling Laws for Pretraining

Scaling Laws for Neural Language Models (Kaplan et al., 2020)



Scaling Laws for Pretraining

Scaling Laws for Neural Language Models (Kaplan et al., 2020)



Scaling Laws for Pretraining

$$Y = \alpha X^{\beta}$$

*A linear trend on a log-log plot is a **power law**.*

Scaling Laws for Pretraining

$$Y = \alpha X^{\beta}$$

$$\log Y = \log \alpha + \beta \log X$$

*A linear trend on a log-log plot is a **power law**.*

Scaling Laws for Pretraining

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$$\log Y = \log \alpha + \beta \log X$$

*A linear trend on a log-log plot is a **power law**.*

Scaling Laws for Pretraining

$$L(D) \approx \left(\frac{D_c}{D} \right)^{\alpha_D}$$

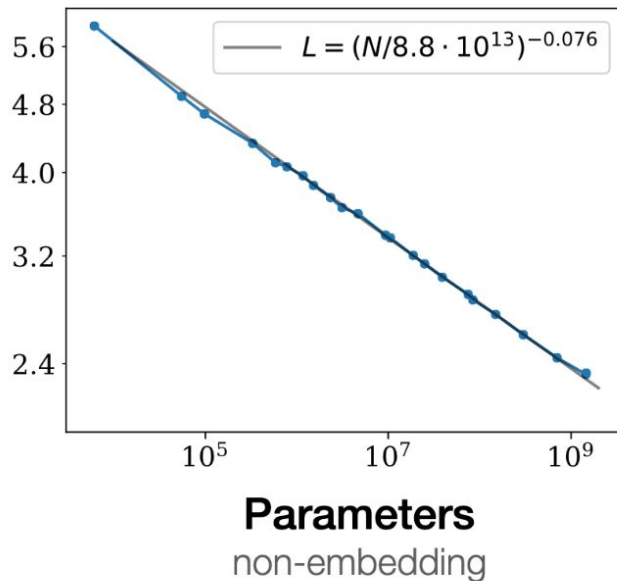
*Language modeling loss is a **power law** of data.*

Scaling Laws for Pretraining

How does *scaling* **compute** improve *loss*?

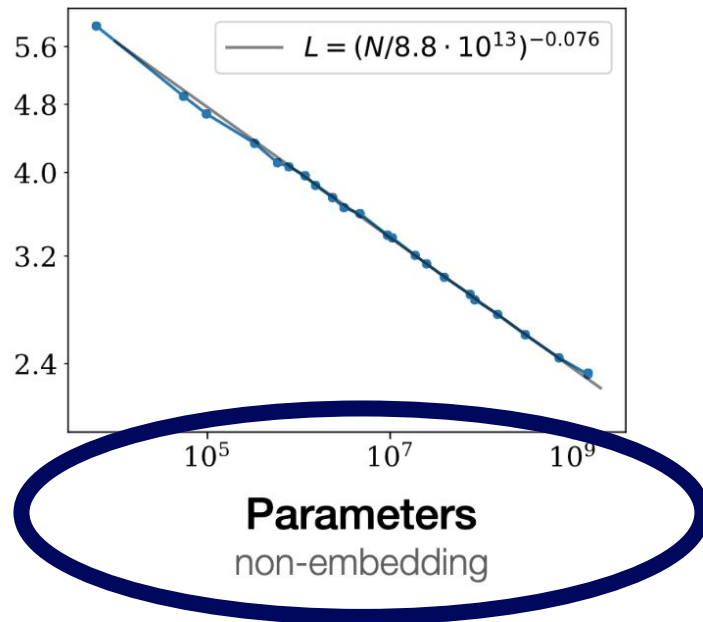
Scaling Laws for Pretraining

Scaling Laws for Neural Language Models (Kaplan et al., 2020)



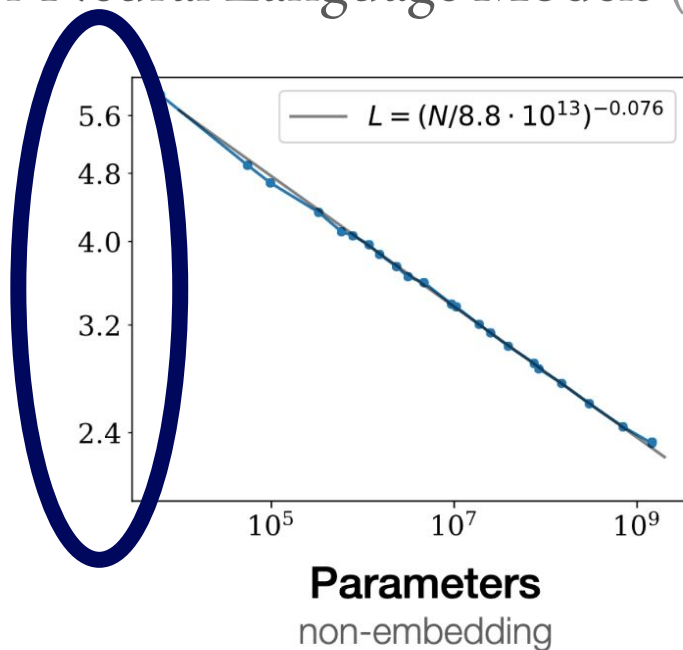
Scaling Laws for Pretraining

Scaling Laws for Neural Language Models (Kaplan et al., 2020)



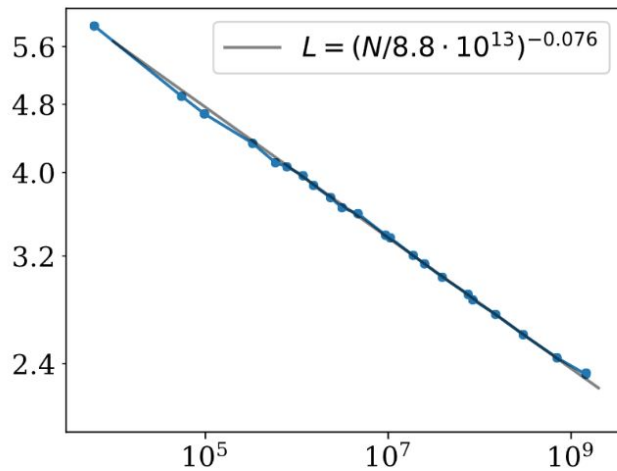
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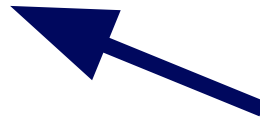


Scaling Laws for Pretraining

Scaling Laws for Neural Language Models (Kaplan et al., 2020)

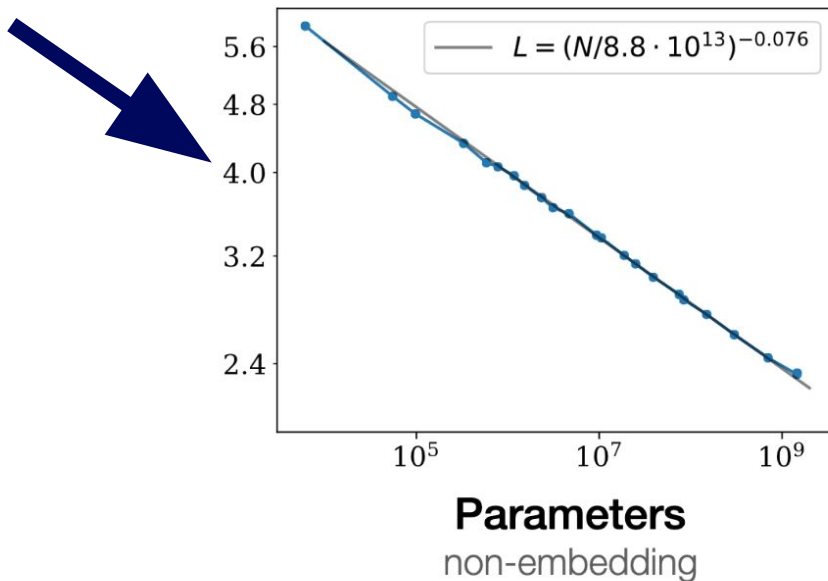


Parameters
non-embedding



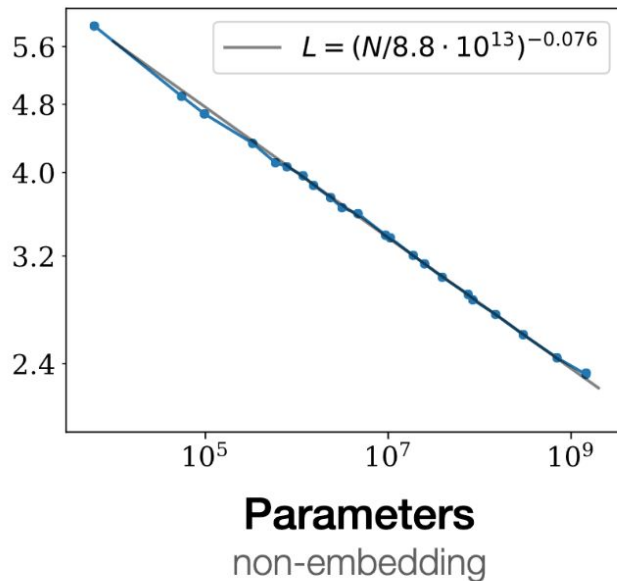
Scaling Laws for Pretraining

Scaling Laws for Neural Language Models (Kaplan et al., 2020)



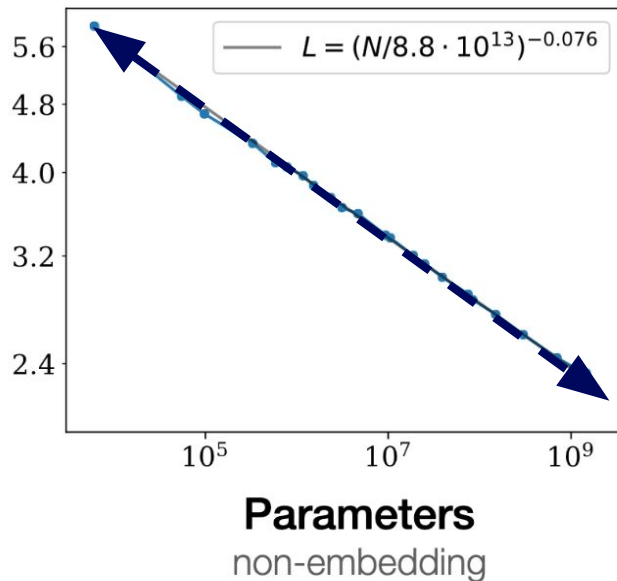
Scaling Laws for Pretraining

Scaling Laws for Neural Language Models (Kaplan et al., 2020)



Scaling Laws for Pretraining

Scaling Laws for Neural Language Models (Kaplan et al., 2020)



Scaling Laws for Pretraining

$$L(N) \approx \left(\frac{N_c}{N} \right)^{\alpha_N}$$

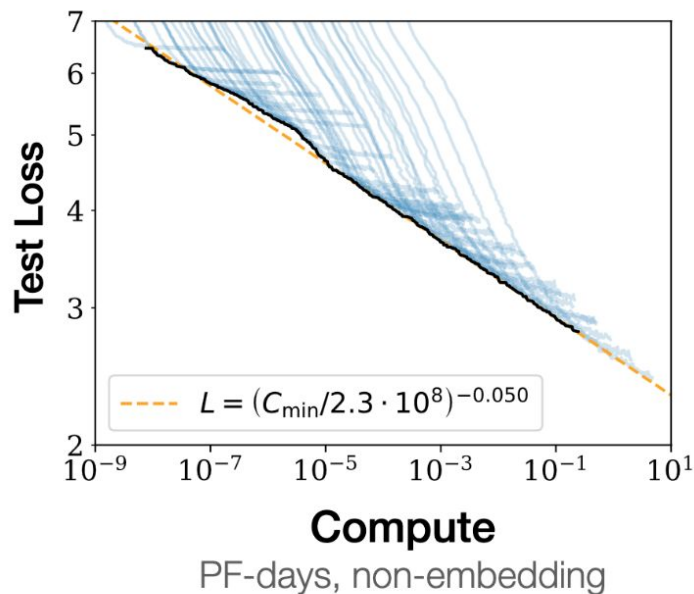
*Language modeling loss is a **power law** of parameters.*

Scaling Laws for Pretraining

Parameters $\neq$ Compute

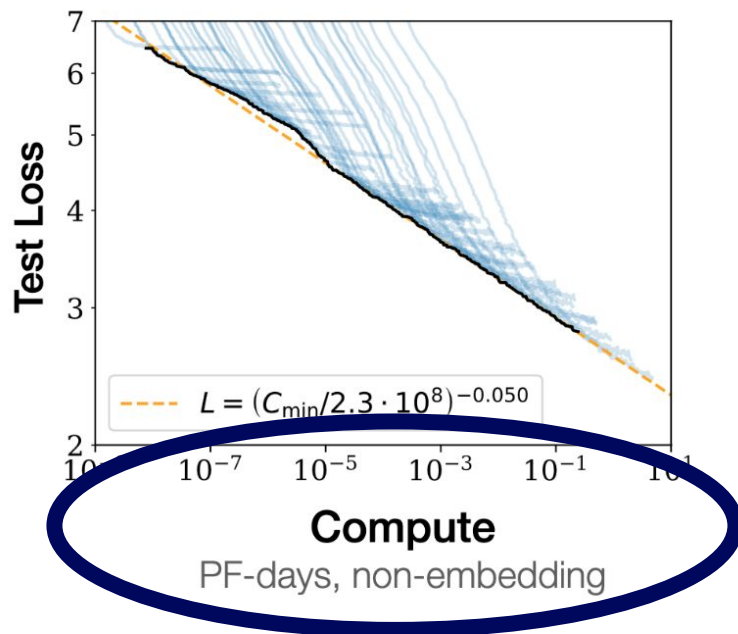
Scaling Laws for Pretraining

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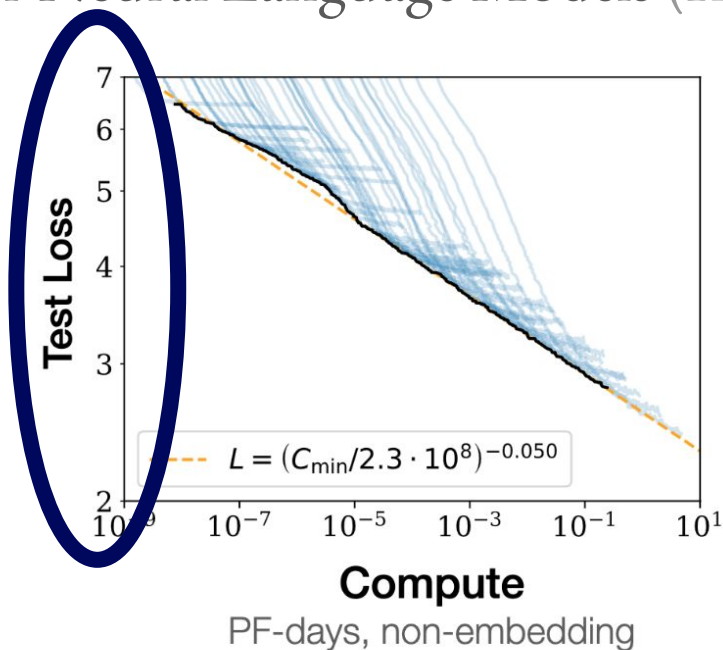
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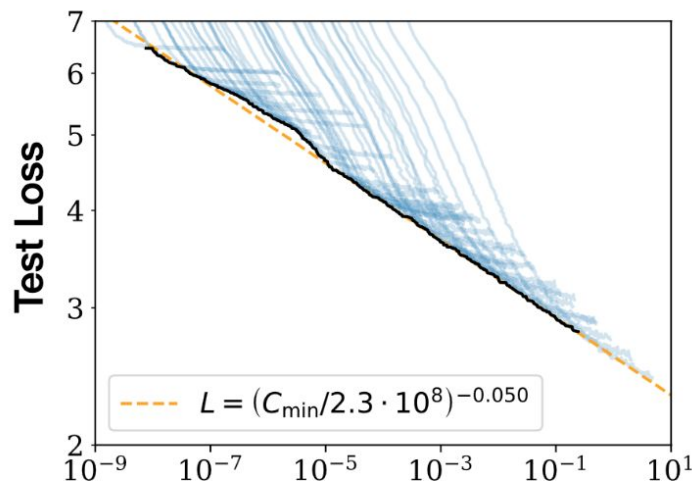
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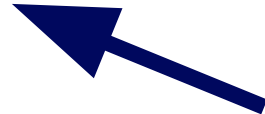
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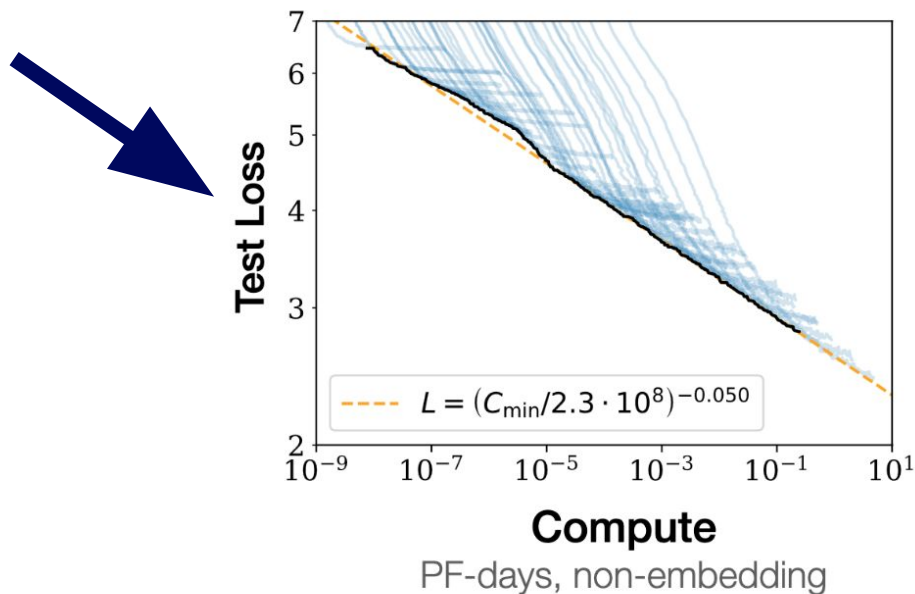
Compute

PF-days, non-embedding



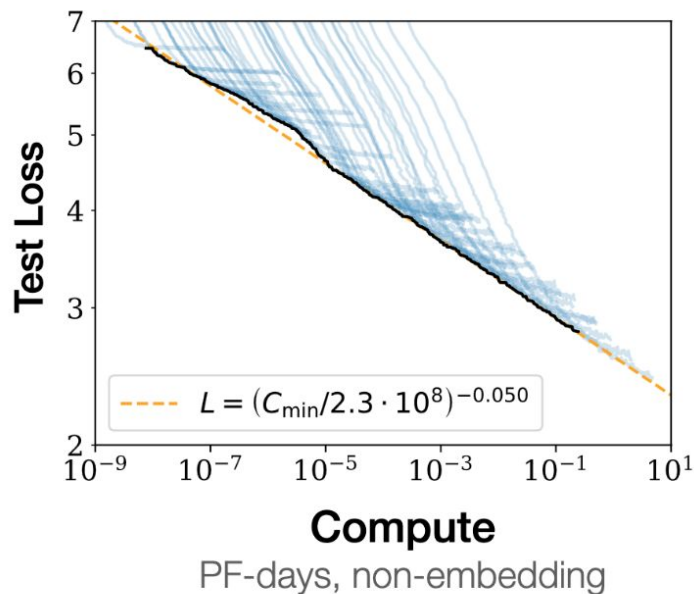
Scaling Laws for Pretraining

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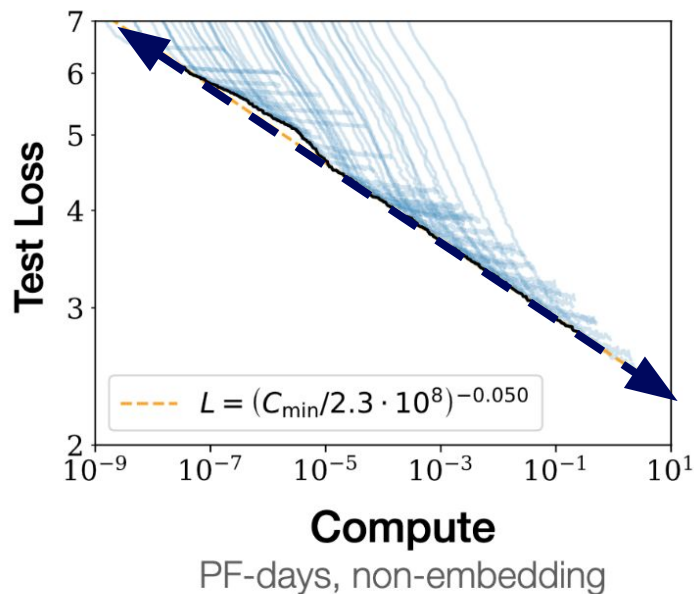
Scaling Laws for Pretraining

Scaling Laws for Neural Language Models (Kaplan et al., 2020)



Scaling Laws for Pretraining

Scaling Laws for Neural Language Models (Kaplan et al., 2020)



Scaling Laws for Pretraining

$$L(C) \approx \left(\frac{C_c}{C} \right)^{\alpha_C}$$

*Language modeling loss is a **power law** of **optimal** compute.*

We can use *scaling laws* for **optimal scaling**.

Scaling Laws for Pretraining

How does *scaling* **data & compute** improve *loss*?

Scaling Laws for Pretraining

How do the individual laws **constrain** the joint law?

Scaling Laws for Pretraining

How do the individual laws **constrain** the joint law?

- The data scaling law must be recovered as $N \rightarrow \infty$.

Scaling Laws for Pretraining

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- The parameters scaling law must be recovered as $D \rightarrow \infty$.

Scaling Laws for Pretraining

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- The data scaling law must be recovered as $N \rightarrow \infty$.
- The parameters scaling law must be recovered as $D \rightarrow \infty$.
- Scaling shouldn't *hurt* performance.

Scaling Laws for Pretraining

How do the individual laws **constrain** the joint law?

- The data scaling law must be recovered as $N \rightarrow \infty$.
- The parameters scaling law must be recovered as $D \rightarrow \infty$.
- Scaling shouldn't *hurt* performance.

Scaling Laws for Pretraining

$$\hat{\epsilon}(m, n) = \epsilon_0 \left\| \frac{\tilde{\epsilon}(m, n)}{\tilde{\epsilon}(m, n) - i\eta} \right\| = \epsilon_0 \left\| \frac{an^{-\alpha} + bm^{-\beta} + c_{\infty}}{an^{-\alpha} + bm^{-\beta} + c_{\infty} - i\eta} \right\|$$

A Constructive Prediction of the Generalization Error Across Scales (Rosenfeld et al., 2020)

Scaling Laws for Pretraining

$$L(N, D) = \left[\left(\frac{N_c}{N} \right)^{\frac{\alpha_N}{\alpha_D}} + \frac{D_c}{D} \right]^{\alpha_D}$$

Scaling Laws for Neural Language Models (Kaplan et al., 2020)

Scaling Laws for Pretraining

$$\hat{L}(N, D) \triangleq E + \frac{A}{N^\alpha} + \frac{B}{D^\beta}.$$

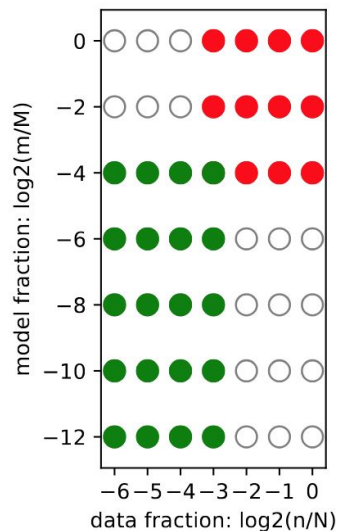
Training Compute-Optimal Large Language Models (Hoffman et al., 2022)

Without a theory, a scaling law's

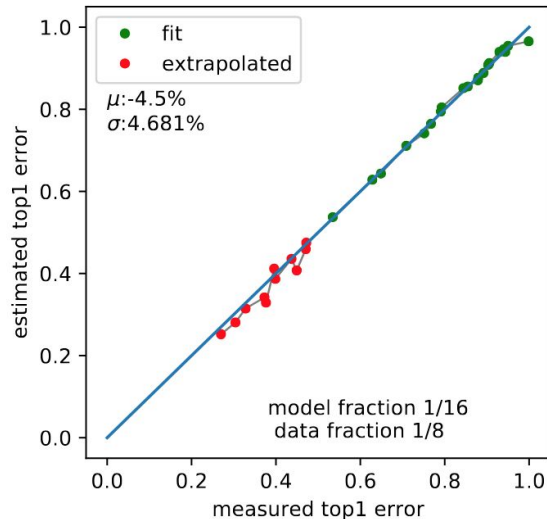
Without a theory, a scaling law's
functional form is only an **approximation**.

Scaling Laws for Pretraining

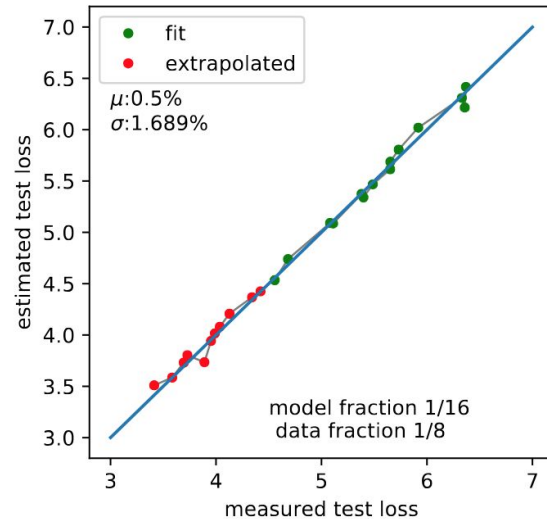
A Constructive Prediction for the Generalization Error Across Scales (Rosenfeld et al., 2020)



(a) Illustration.



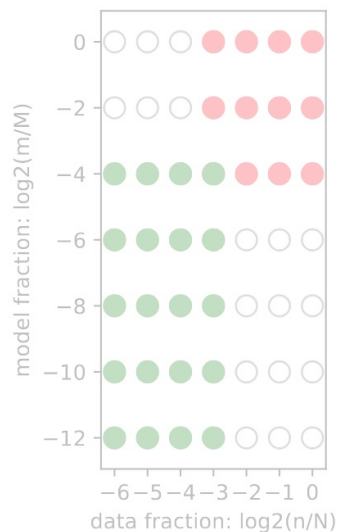
(b) Extrapolation on ImageNet



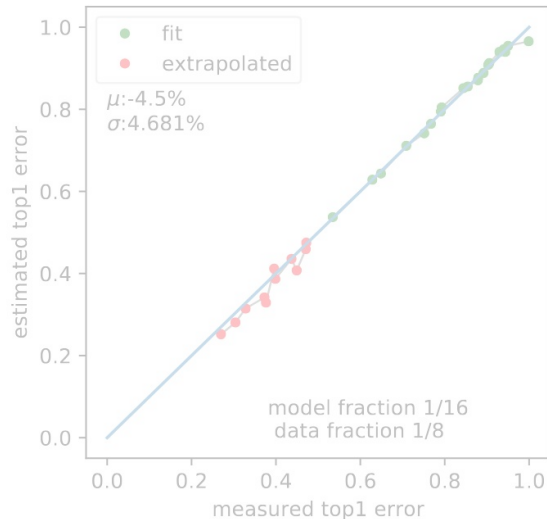
(c) Extrapolation on WikiText-103.

Scaling Laws for Pretraining

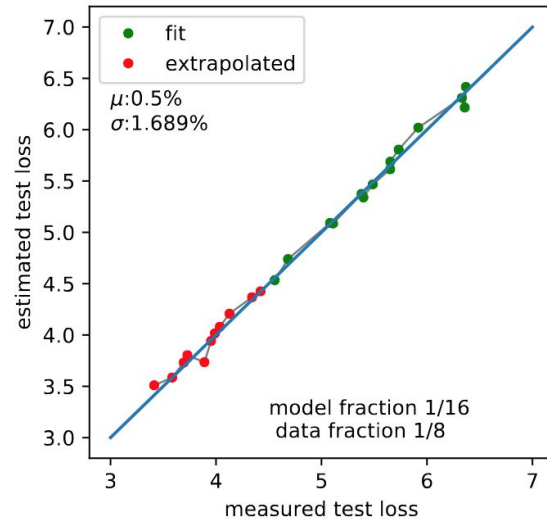
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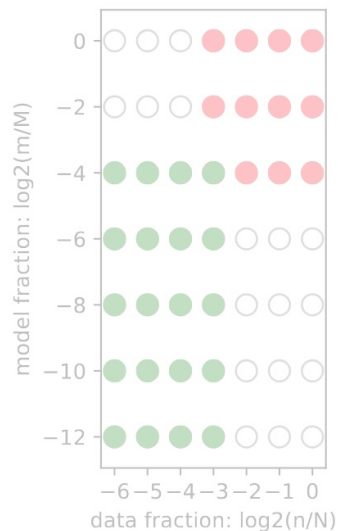
(b) Extrapolation on ImageNet



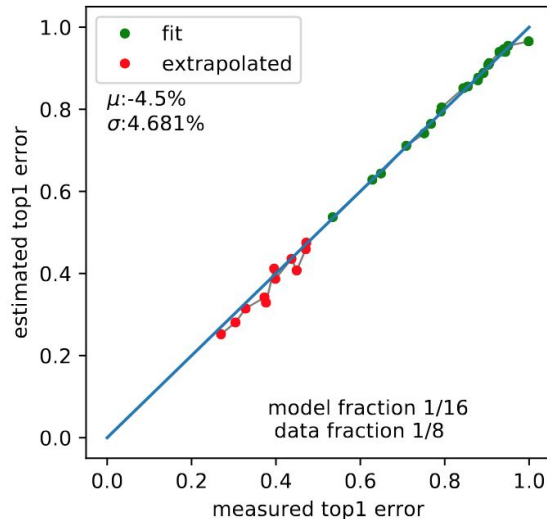
(c) Extrapolation on WikiText-103.

Scaling Laws for Pretraining

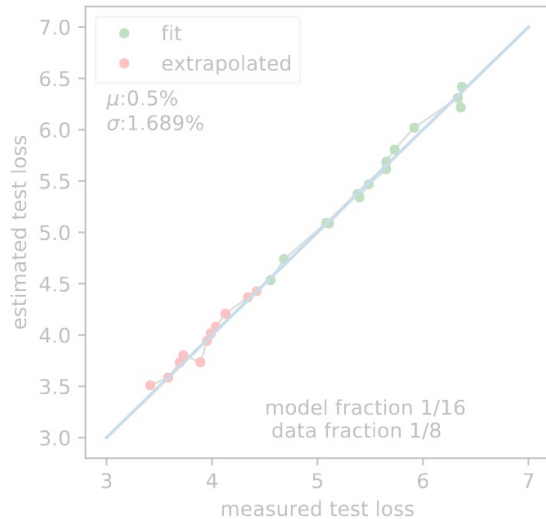
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(a) Illustration.



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Scaling Laws for Pretraining

$$\hat{L}(N, D) \triangleq E + \frac{A}{N^\alpha} + \frac{B}{D^\beta}.$$

Training Compute-Optimal Large Language Models (Hoffman et al., 2022)

Details matter,

Details matter, and can differ!

Scaling Laws for Pretraining

When using scaling laws:

Scaling Laws for Pretraining

When using scaling laws:

- Choose or fit one appropriate for your situation.

Scaling Laws for Pretraining

When using scaling laws:

- Choose or fit one appropriate for your situation.
- Check the law against your data.

Scaling Laws for Pretraining

When using scaling laws:

- Choose or fit one appropriate for your situation.
- Check the law against your data.
- Go with what works best.

Scaling Laws for Downstream Tasks

How does scale improve *downstream tasks*?

We value
foundation models
for their **capabilities**.

Scaling Laws for Downstream Tasks

Do **downstream tasks** have *scaling laws*?

$$\{\text{Prompt, Finetune}\} \times \{\text{Pretrain, Posttrain}\}$$

adapt scale

$$\{\text{Prompt, Finetune}\} \times \{\text{Pretrain, Posttrain}\}$$

adapt scale

$$\{\text{Prompt, Finetune}\} \times \{\text{Pretrain, Posttrain}\}$$

adapt scale

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$$\{\text{Prompt, Finetune}\} \times \{\text{Pretrain, Posttrain}\}$$

adapt scale

Scaling Laws for Downstream Tasks

Do downstream tasks have scaling laws for...

Scaling Laws for Downstream Tasks

Do downstream tasks have scaling laws for...

- Data

Scaling Laws for Downstream Tasks

Do downstream tasks have scaling laws for...

- Data
- Parameters

Scaling Laws for Downstream Tasks

Do downstream tasks have scaling laws for...

- Data
- Parameters
- Compute

Scaling Laws for Downstream Tasks

Do downstream tasks have scaling laws for...

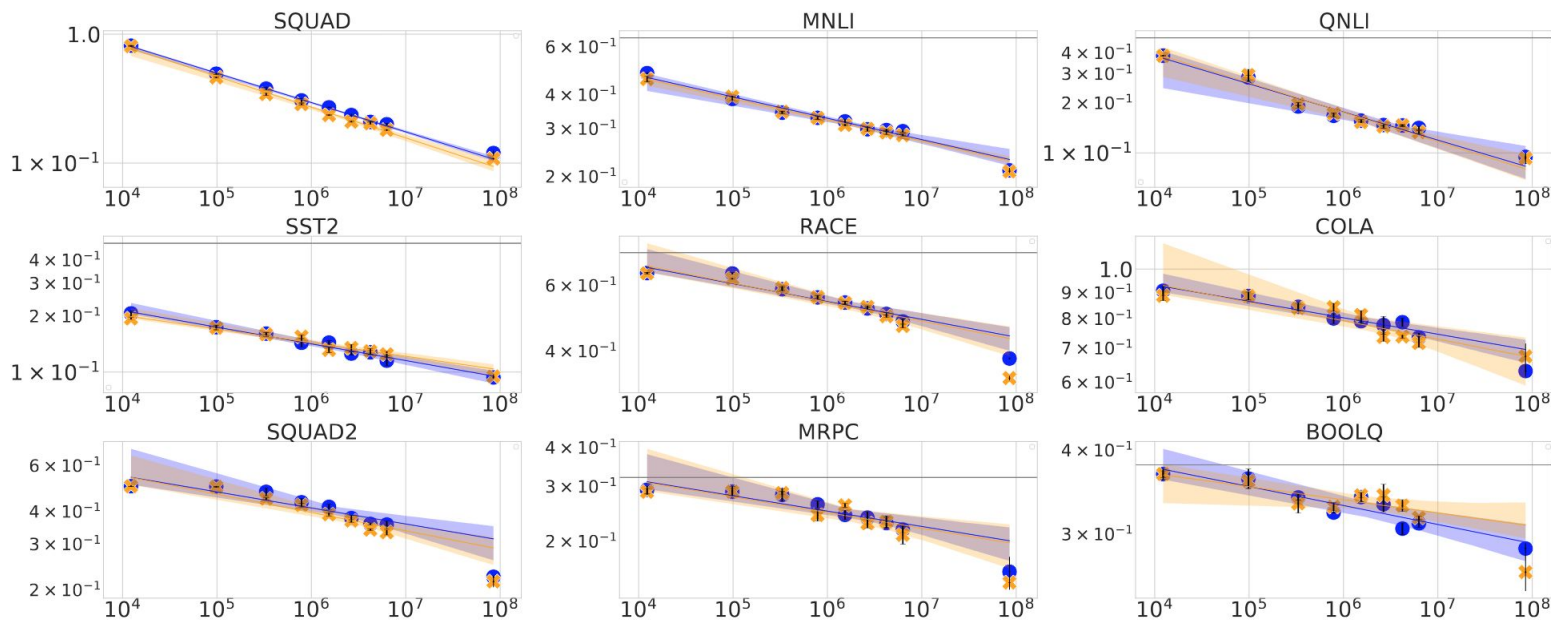
- Data
- Parameters
- Compute

Scaling Laws for Downstream Tasks

Does **finetuning** exhibit stable scaling laws?

Scaling Laws for Downstream Tasks

Scaling Laws Under the Microscope (Ivgi et al., 2022)



Scaling Laws for Downstream Tasks

Does **finetuning** exhibit stable scaling laws?

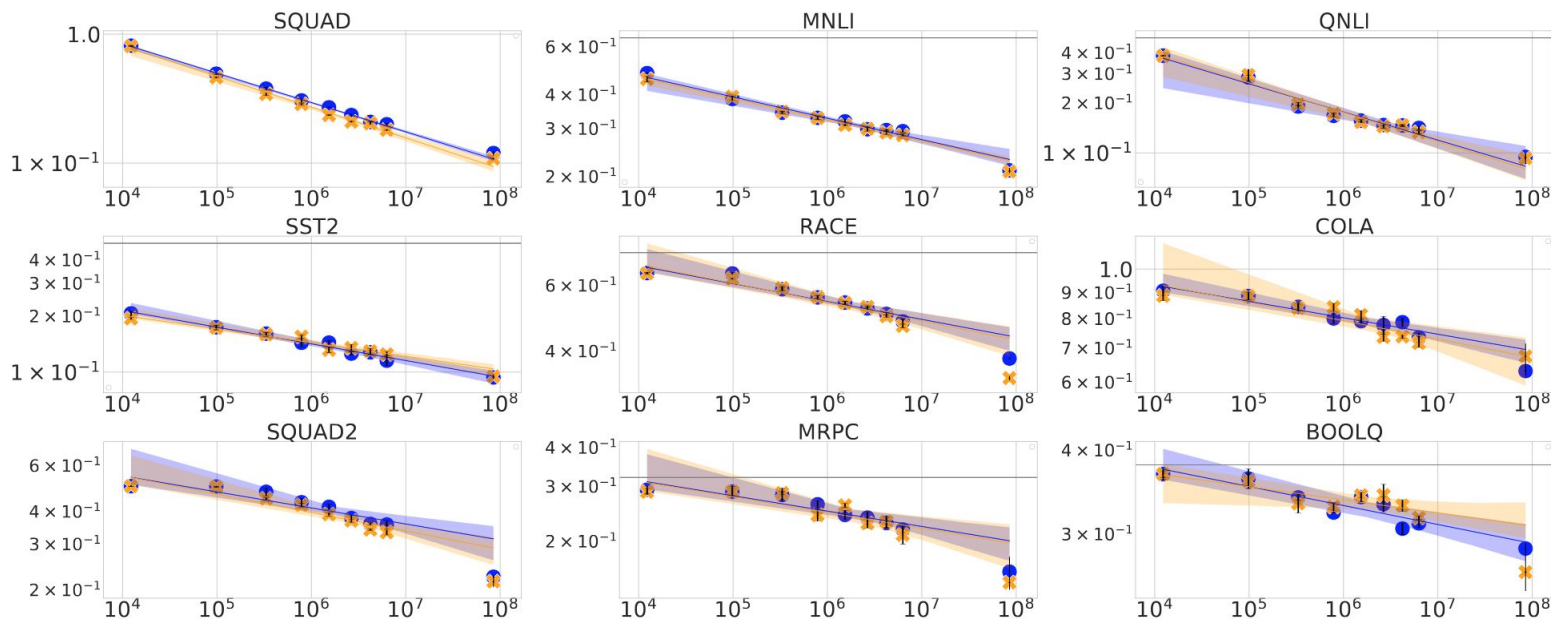
Scaling Laws for Downstream Tasks

Does **finetuning** exhibit stable scaling laws?

Trends exist, but can be **hard** to extrapolate.

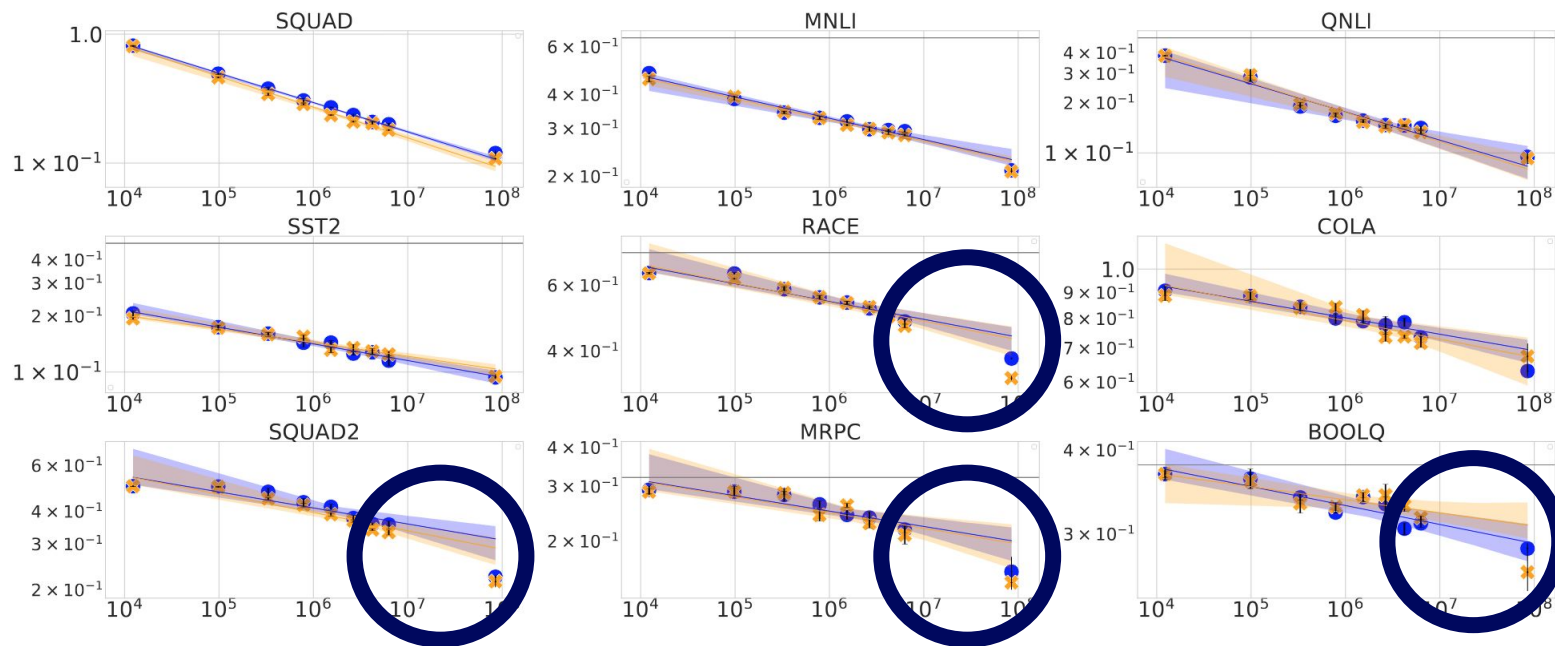
Scaling Laws for Downstream Tasks

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Scaling Laws for Downstream Tasks

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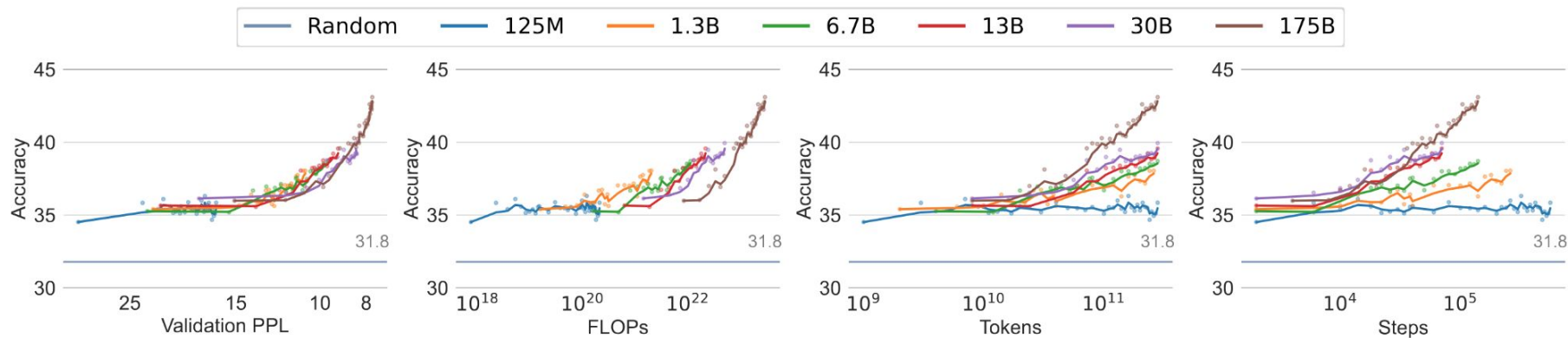


Scaling Laws for Downstream Tasks

Does **prompting** exhibit stable scaling laws?

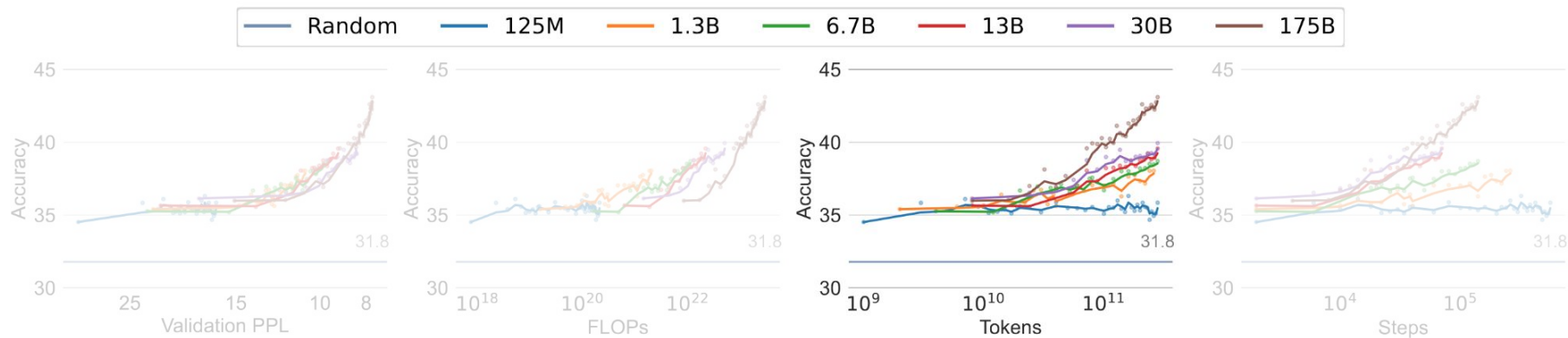
Scaling Laws for Downstream Tasks

Training Trajectories of Language Models Across Scales (Xia et al., 2023)



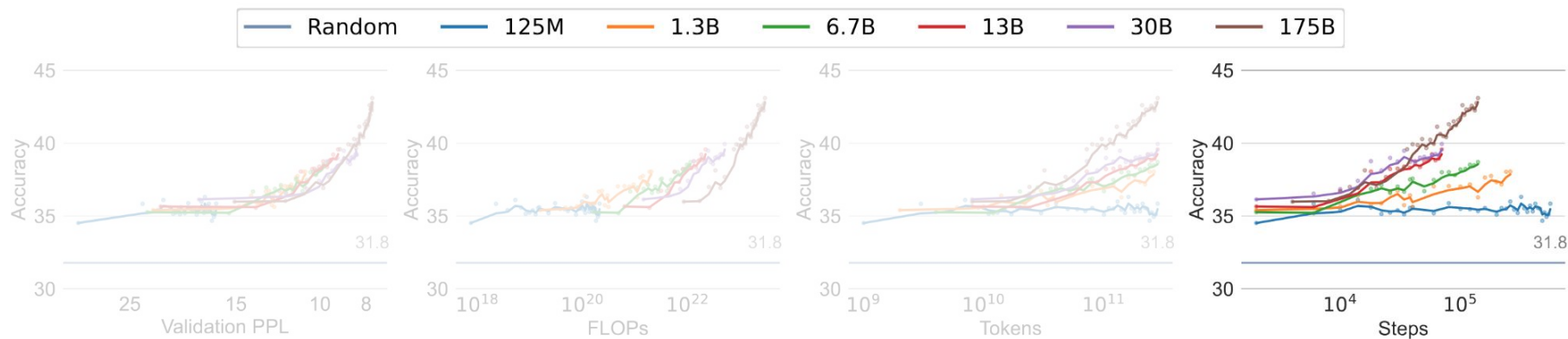
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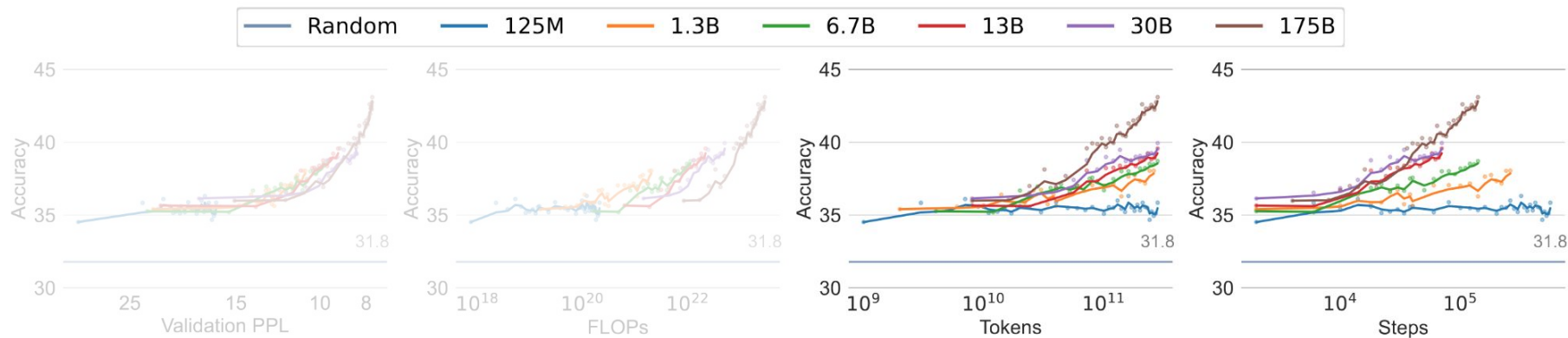
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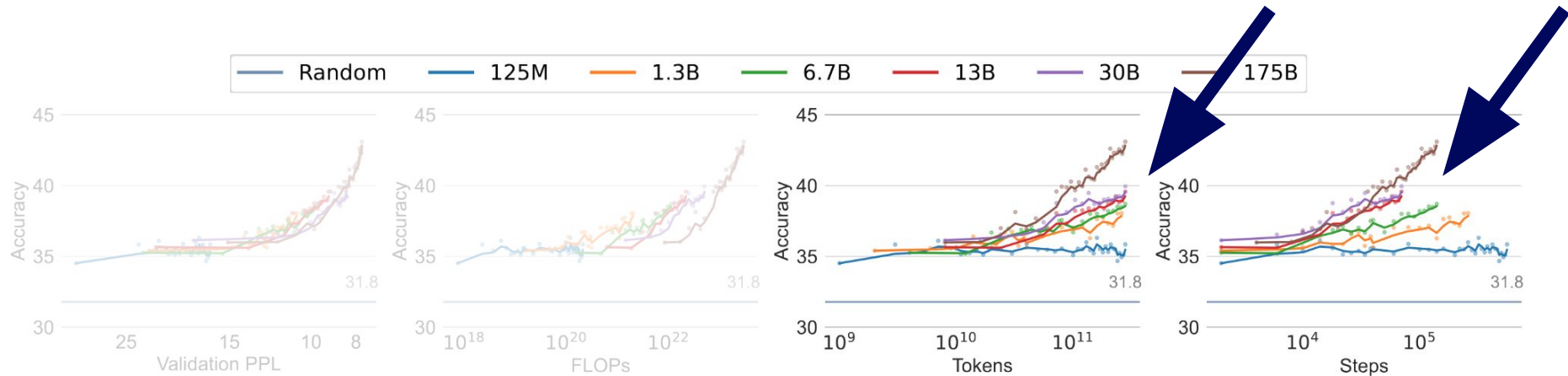
Scaling Laws for Downstream Tasks

Training Trajectories of Language Models Across Scales (Xia et al., 2023)



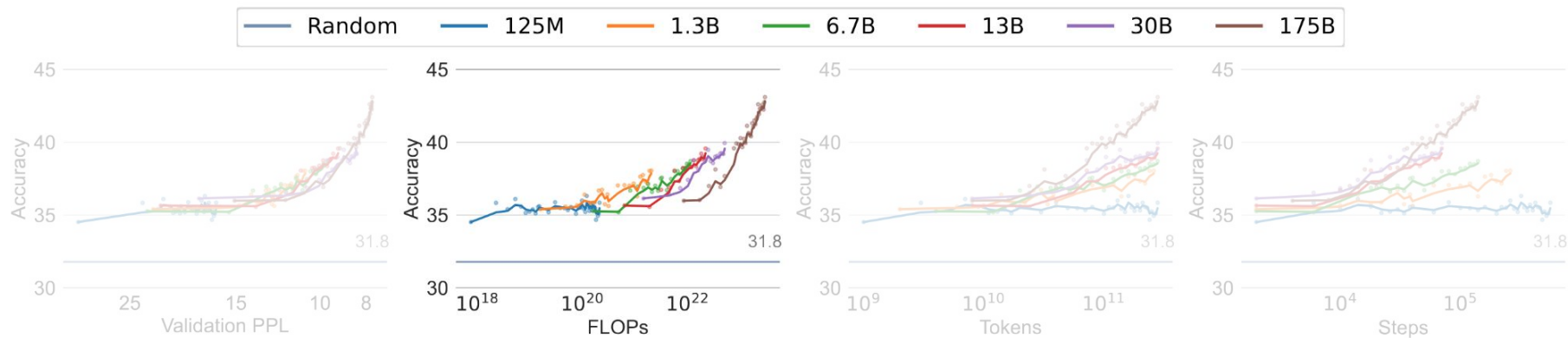
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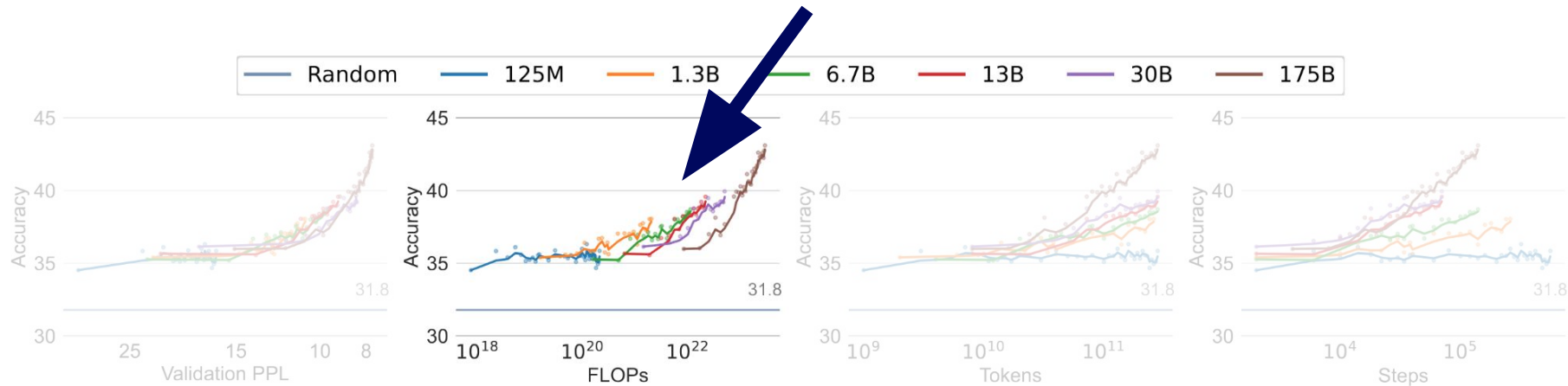
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Scaling Laws for Downstream Tasks

Training Trajectories of Language Models Across Scales (Xia et al., 2023)



Scaling Laws for Downstream Tasks

Does **prompting** exhibit stable scaling laws?

Scaling Laws for Downstream Tasks

Does **prompting** exhibit stable scaling laws?

Scaling laws *change* with the number of **parameters**.

Scaling Laws for Downstream Tasks

Do downstream tasks have scaling laws for...

- Data
- Parameters
- Compute

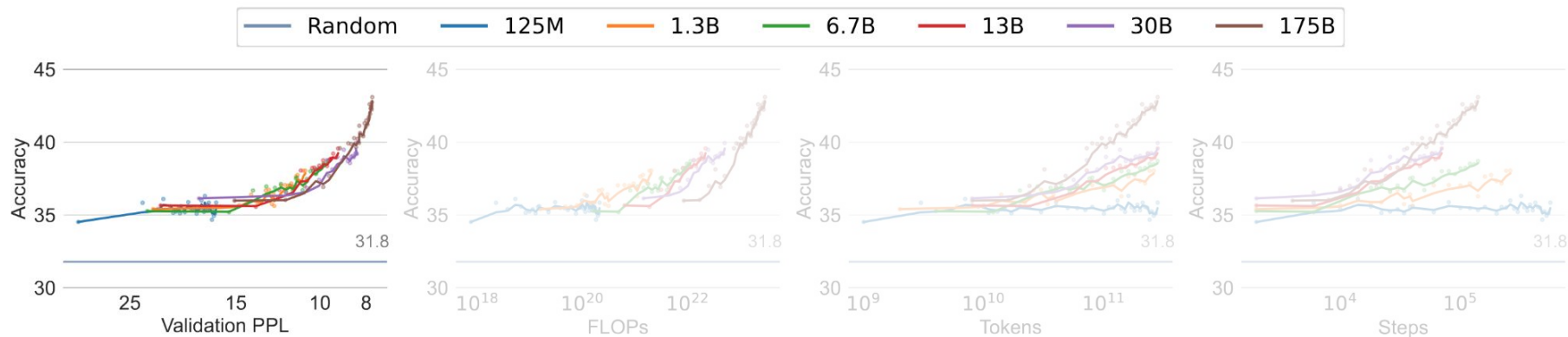
Scaling Laws for Downstream Tasks

Do downstream tasks have scaling laws for...

- Data
- Parameters
- Compute
- Cross-entropy

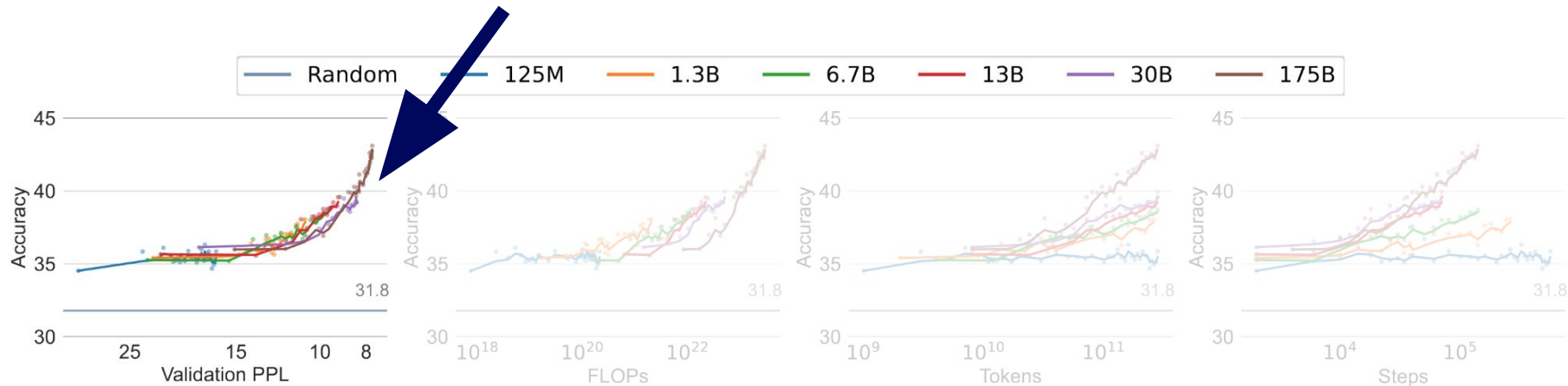
Scaling Laws for Downstream Tasks

Training Trajectories of Language Models Across Scales (Xia et al., 2023)



Scaling Laws for Downstream Tasks

Training Trajectories of Language Models Across Scales (Xia et al., 2023)



Use *pretraining loss*
for **downstream** *scaling laws*.

Scaling Laws for Downstream Tasks

Scaling laws for downstream tasks...

Scaling Laws for Downstream Tasks

Scaling laws for downstream tasks...

- depend on your performance metric,

Scaling Laws for Downstream Tasks

Scaling laws for downstream tasks...

- depend on your performance metric,
- might change for different (pre)training setups,

Scaling Laws for Downstream Tasks

Scaling laws for downstream tasks...

- depend on your performance metric,
- might change for different (pre)training setups,
- are often based on pretraining loss.

Scaling Laws for Downstream Tasks

data

parameters

compute

Scaling Laws for Downstream Tasks

data

parameters

compute



Scaling Laws for Downstream Tasks

data

parameters

compute



pretraining loss

Scaling Laws for Downstream Tasks

data

parameters

compute



pretraining loss



Scaling Laws for Downstream Tasks



Are scaling laws a
panacea?

Scaling Laws for Downstream Tasks

The Problem

Scaling Laws for Downstream Tasks

 The Problem of

Scaling Laws for Downstream Tasks

 The Problem of **Emergence** 

Emergence

Emergence

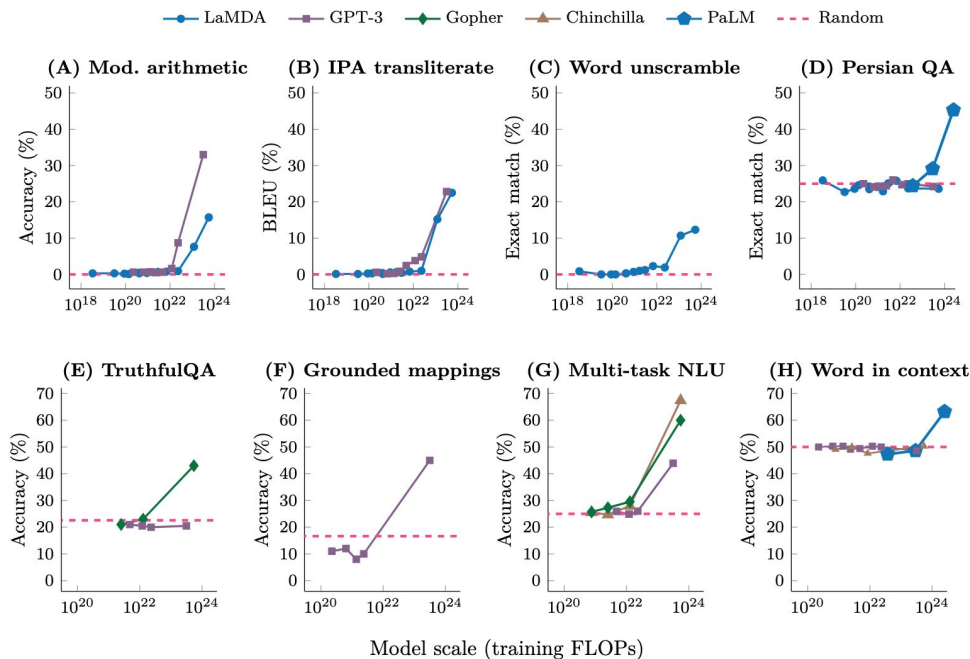
Some capabilities only appear at a given scale.

Scaling Laws for Downstream Tasks

Emergence occurs in **task performance**.

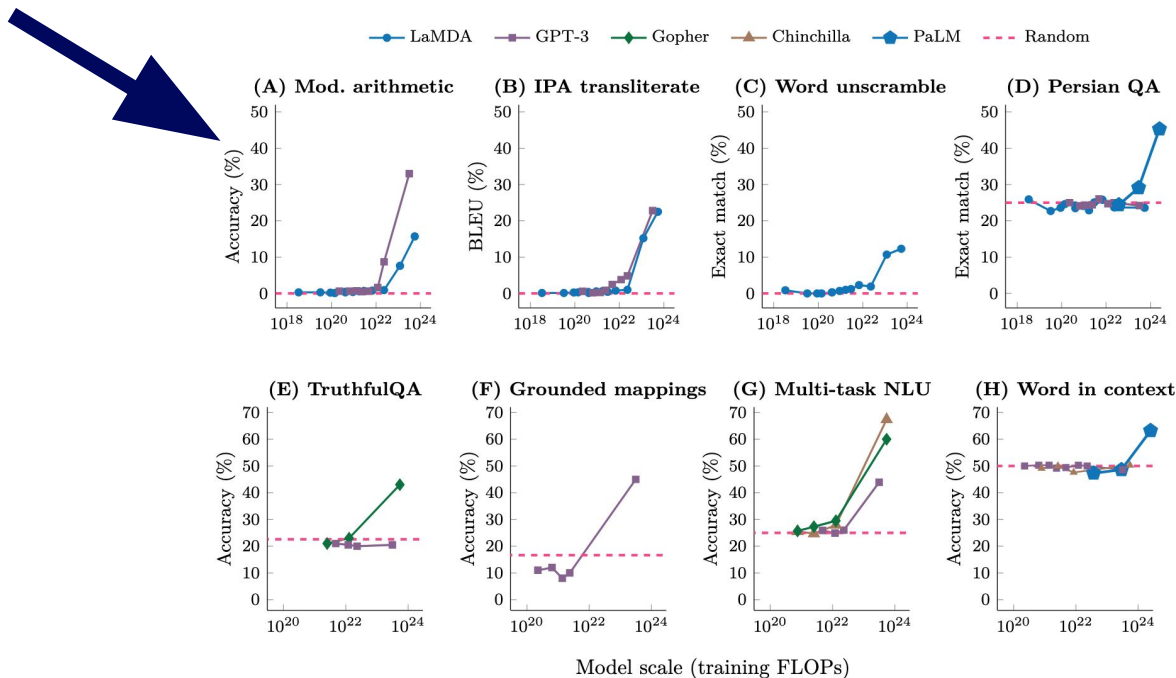
Scaling Laws for Downstream Tasks

Emergent Abilities of Large Language Models (Wei et al., 2022)



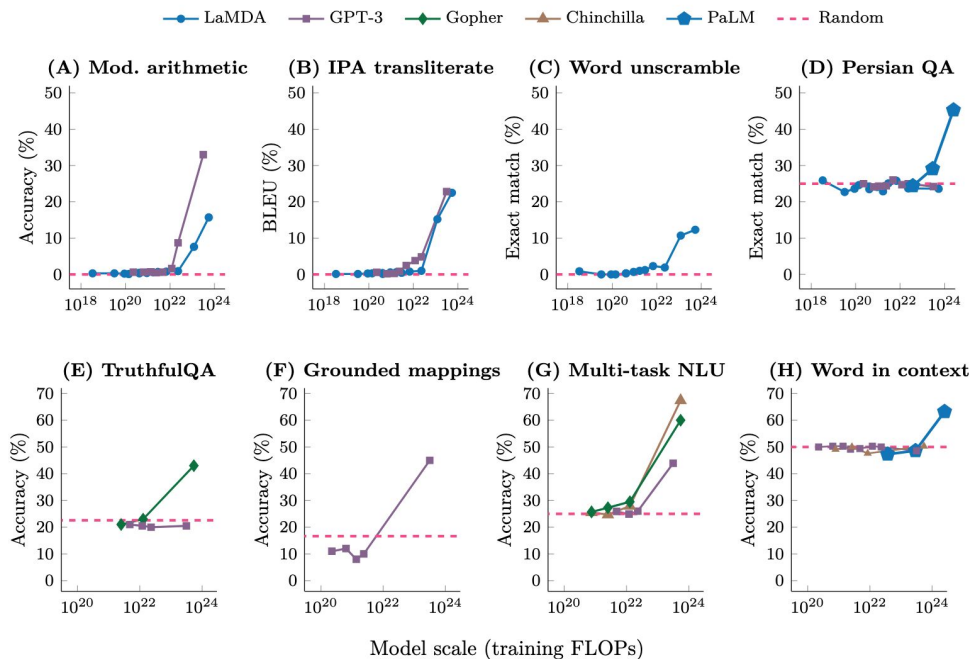
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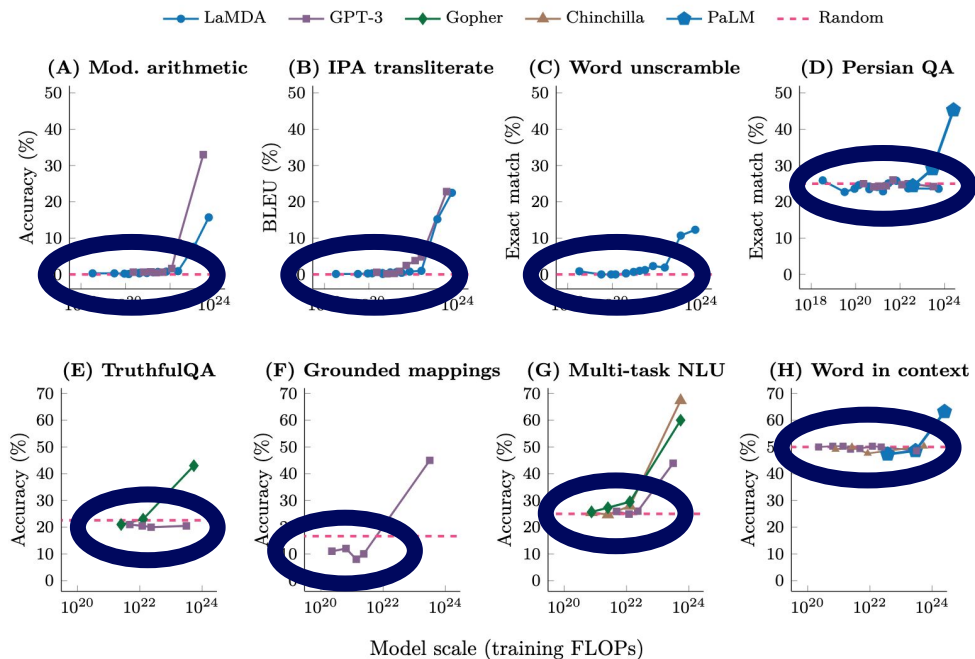
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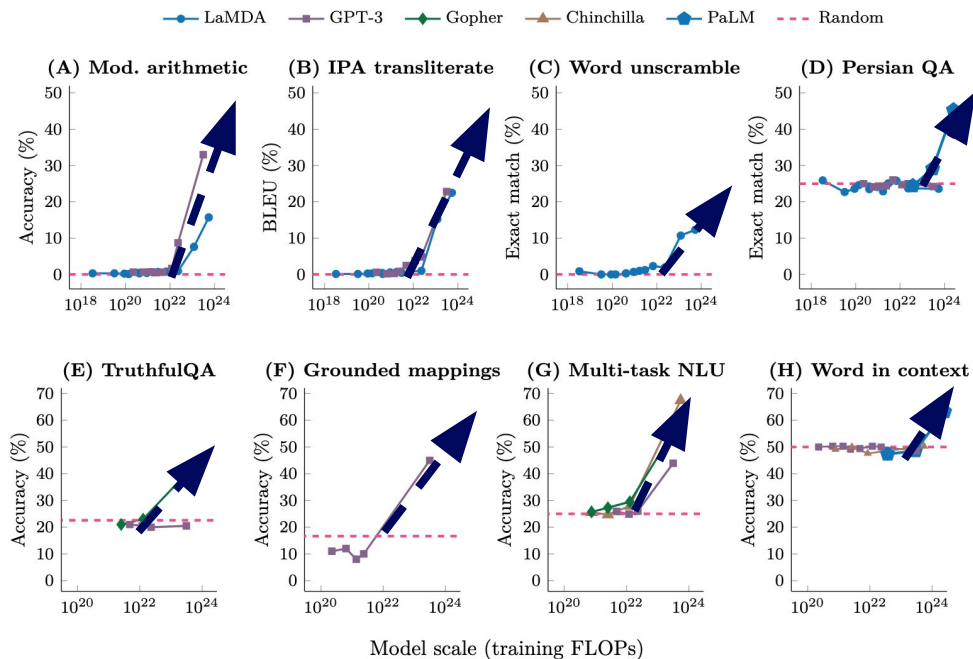
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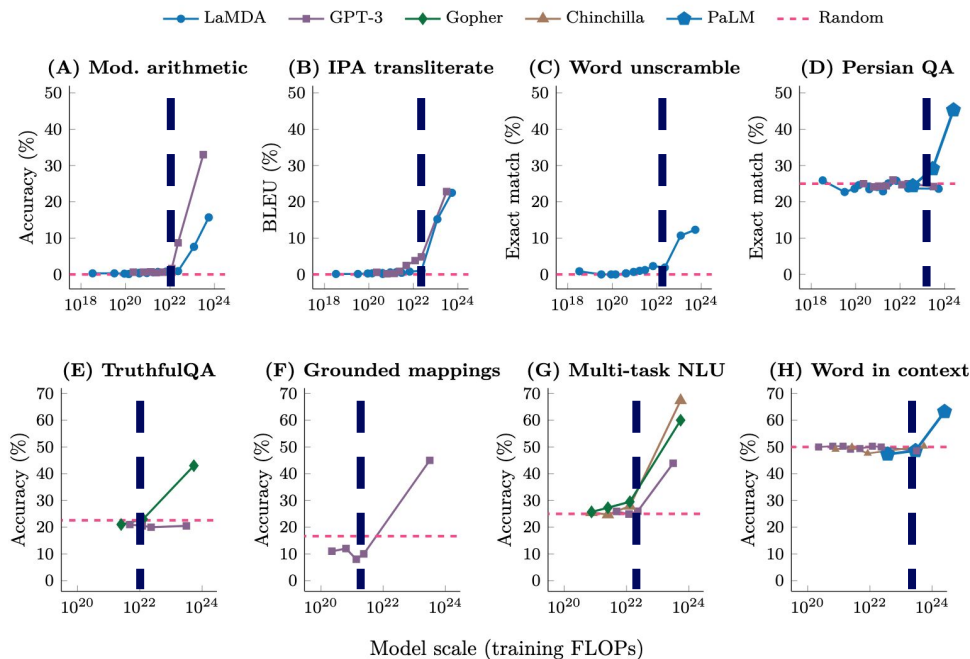
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Scaling Laws for Downstream Tasks

Emergent Abilities of Large Language Models (Wei et al., 2022)



You can't *extrapolate* a line

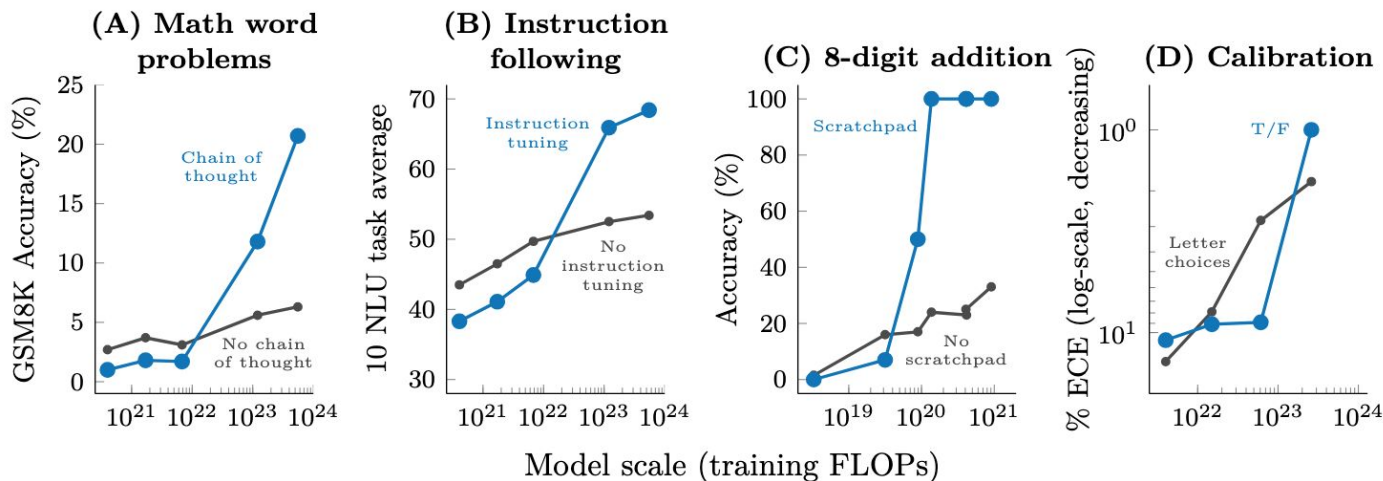
You can't *extrapolate* a line
across a **structural break**.

Scaling Laws for Downstream Tasks

Emergence occurs in **techniques' usefulness**.

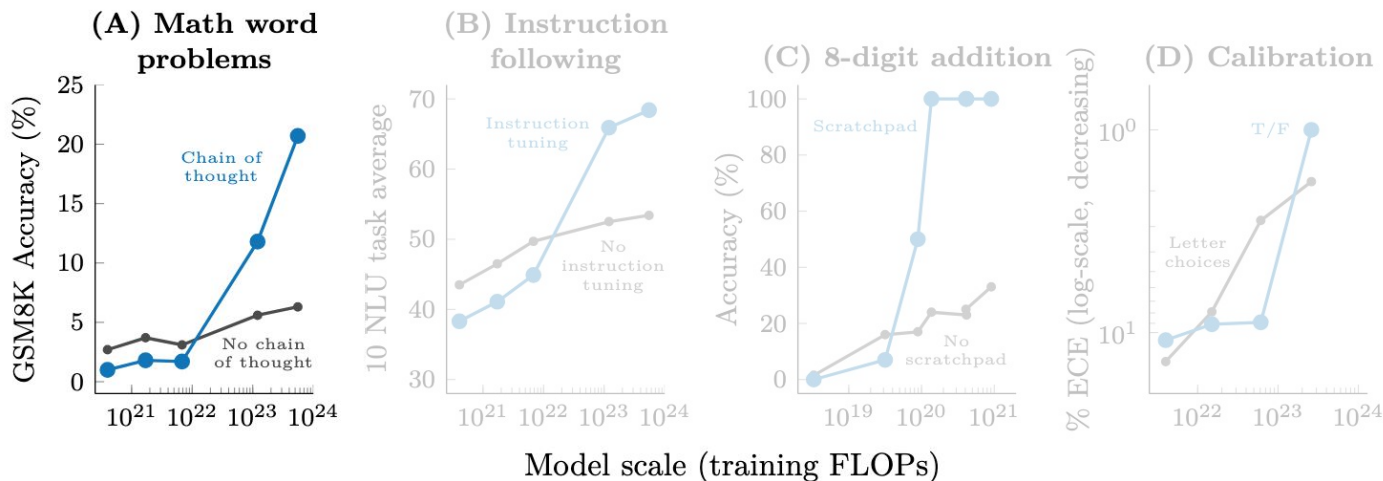
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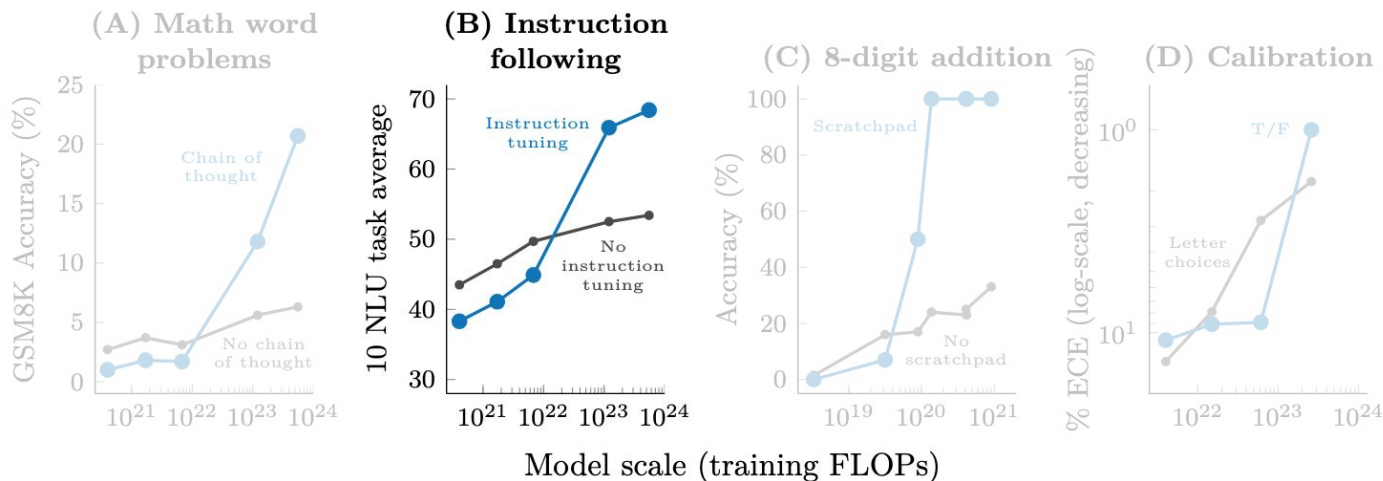
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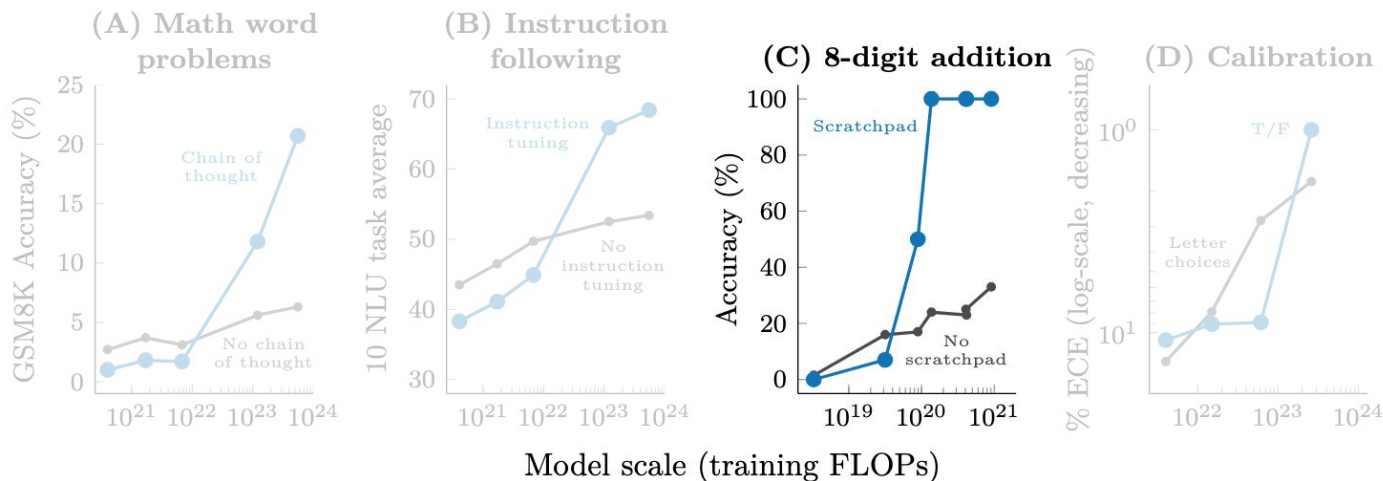
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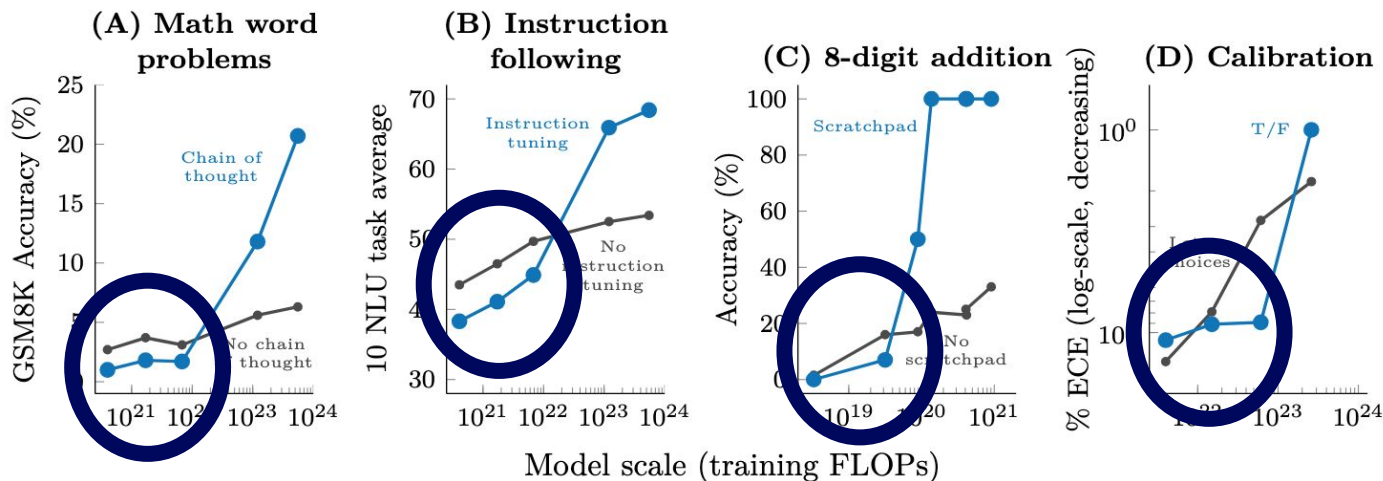
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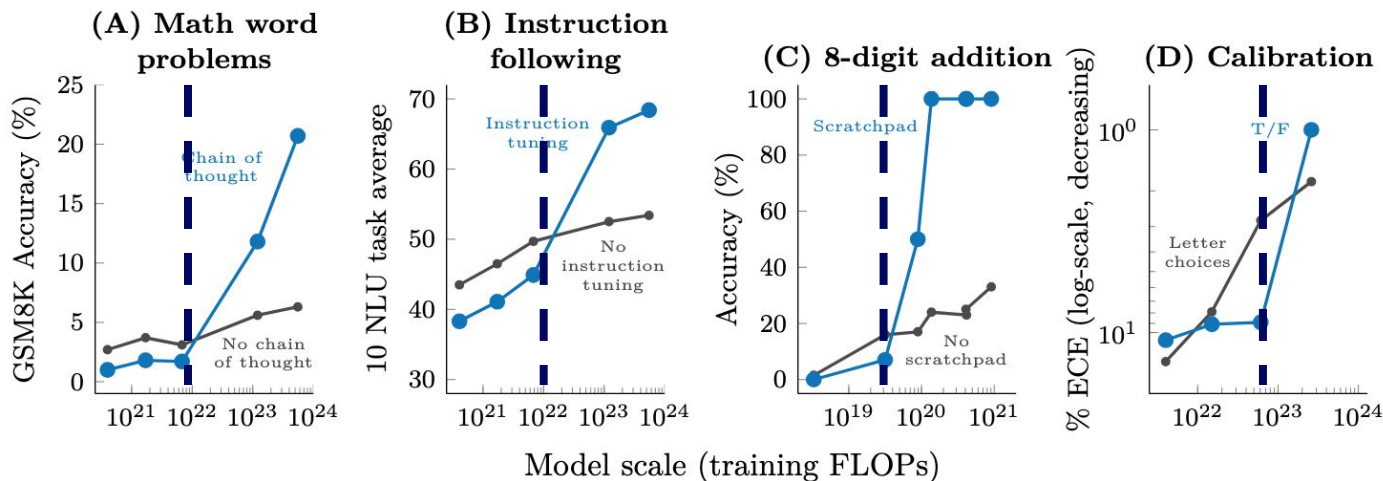
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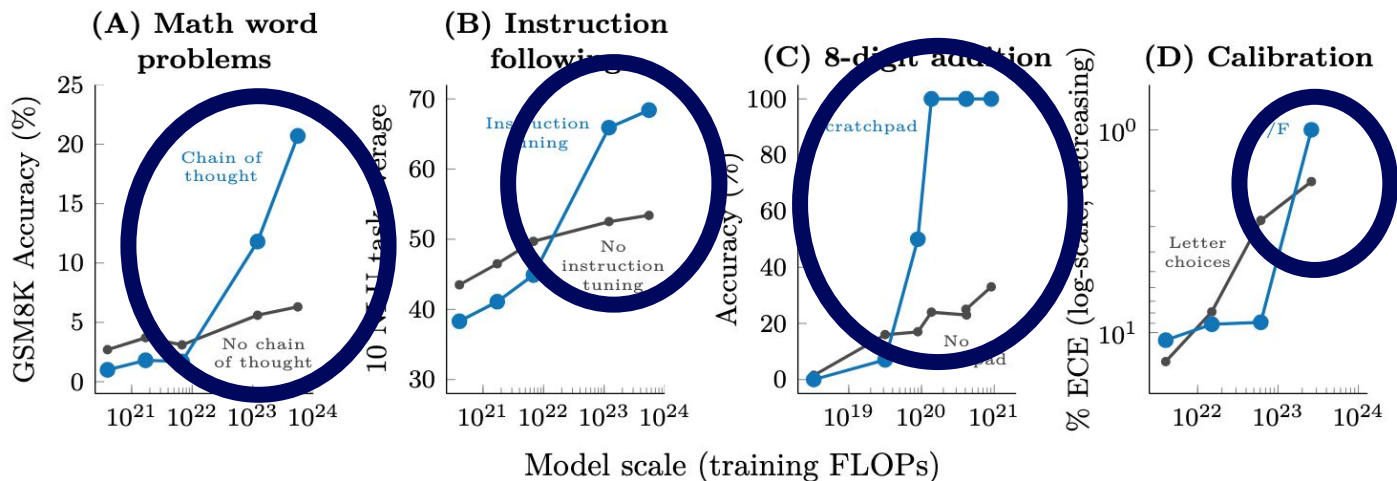
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Scaling Laws for Downstream Tasks

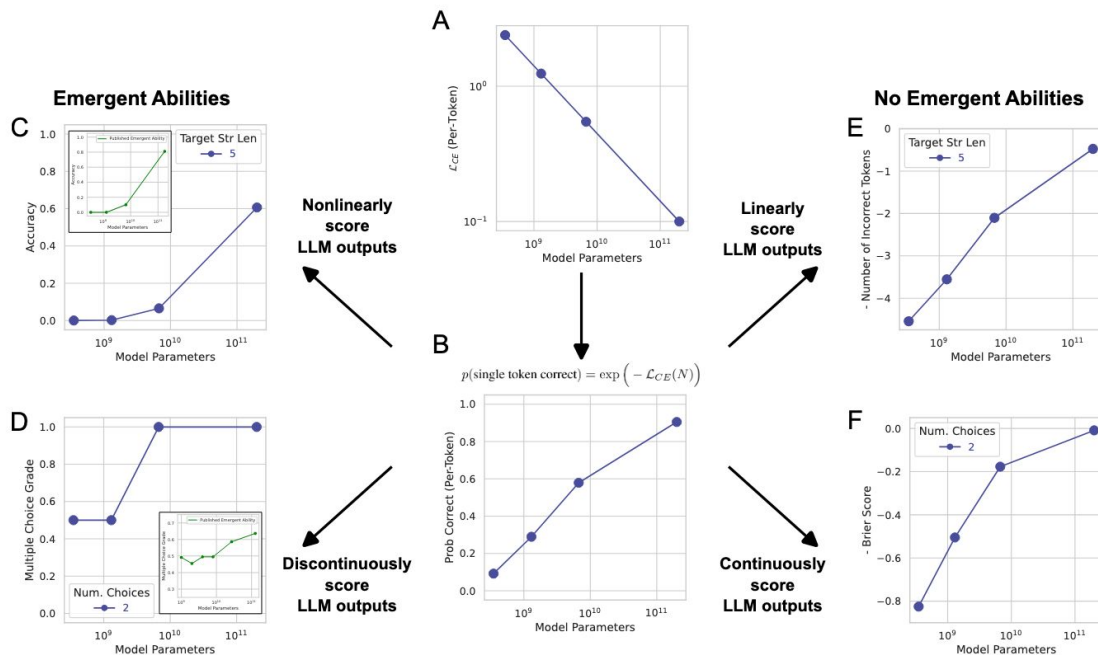
Emergent Abilities of Large Language Models (Wei et al., 2022)



Do we create
emergence?

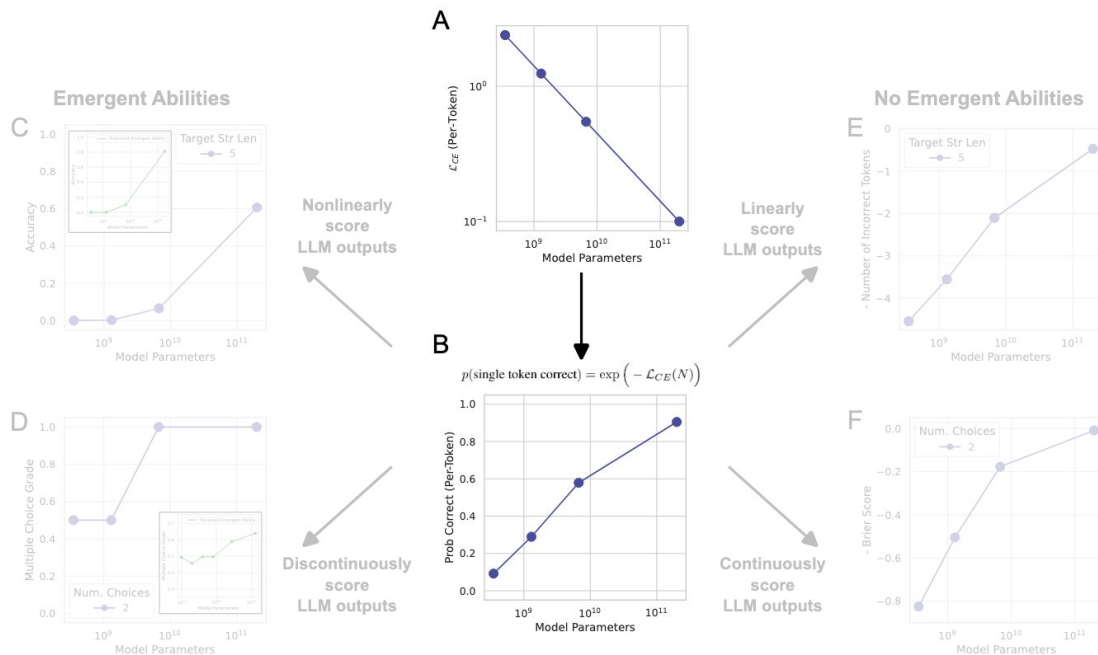
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Are Emergent Abilities of Large Language Models a Mirage (Schaeffer et al., 2023)



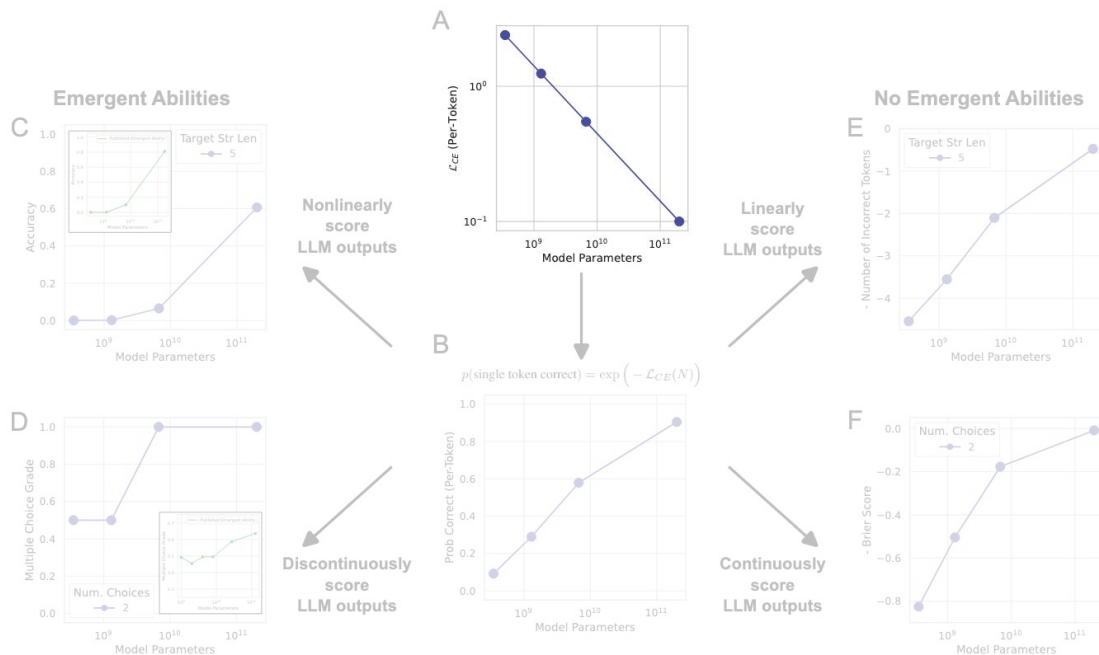
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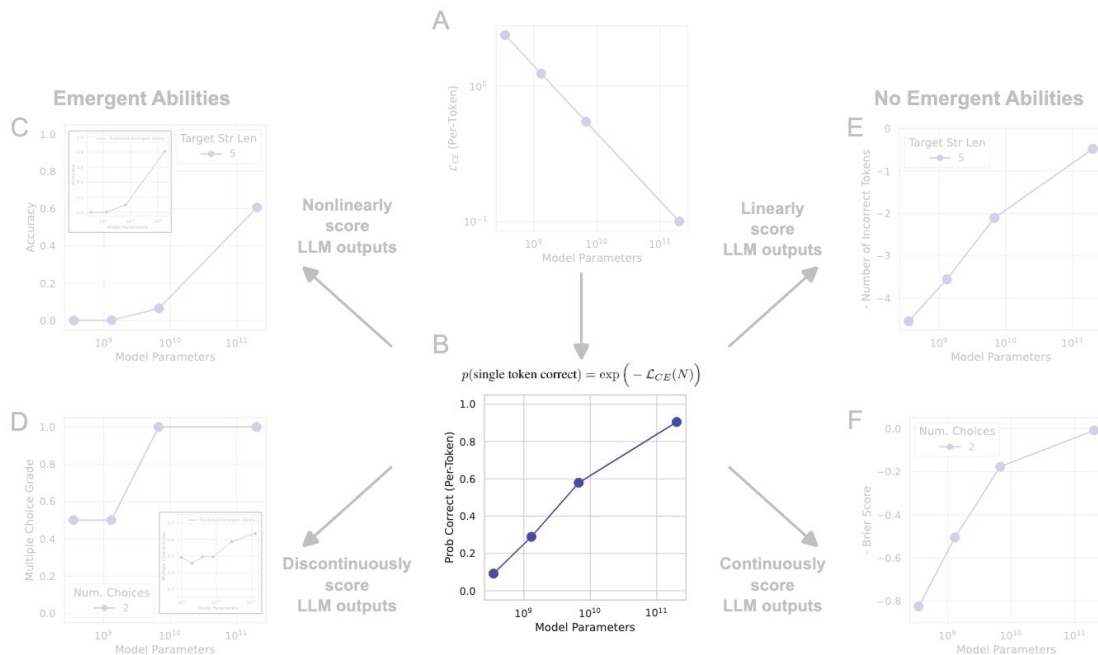
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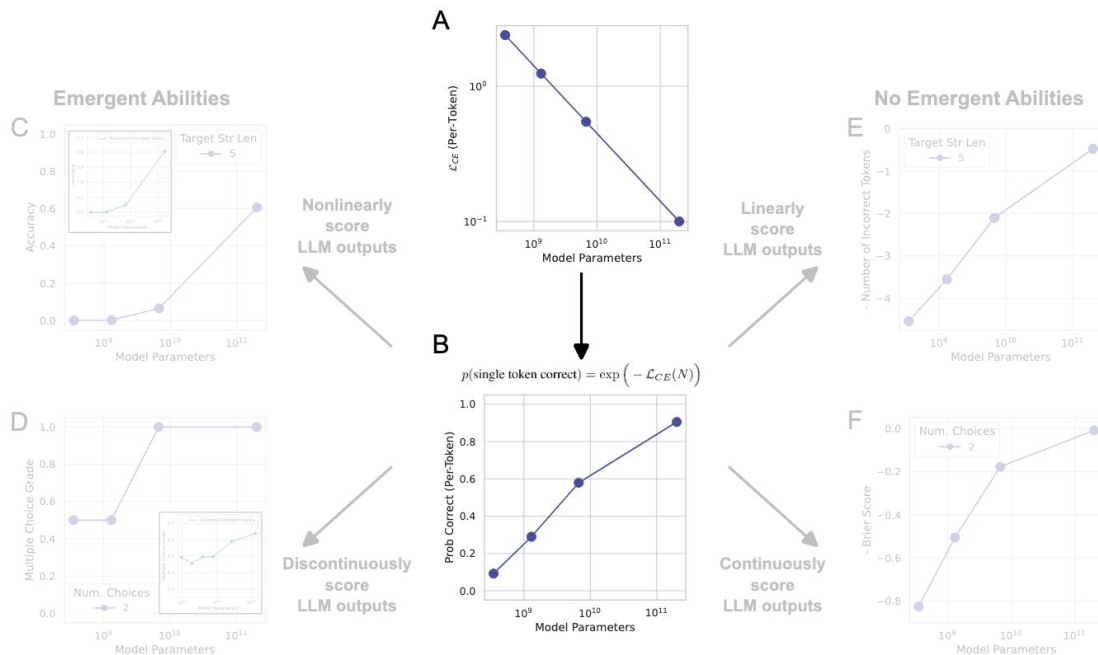
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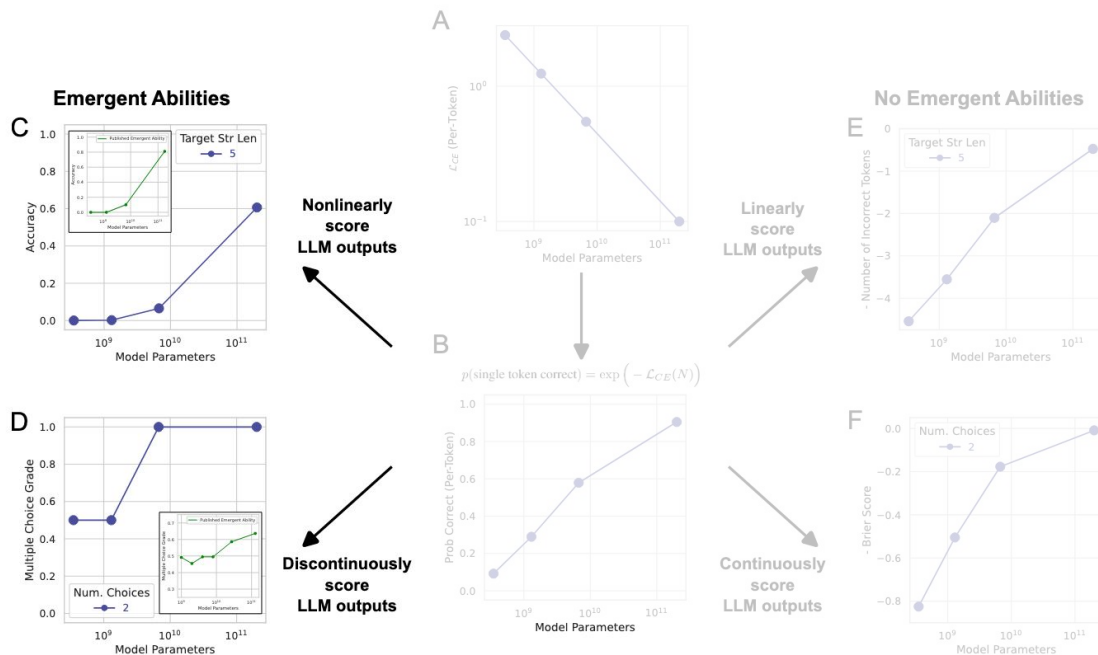
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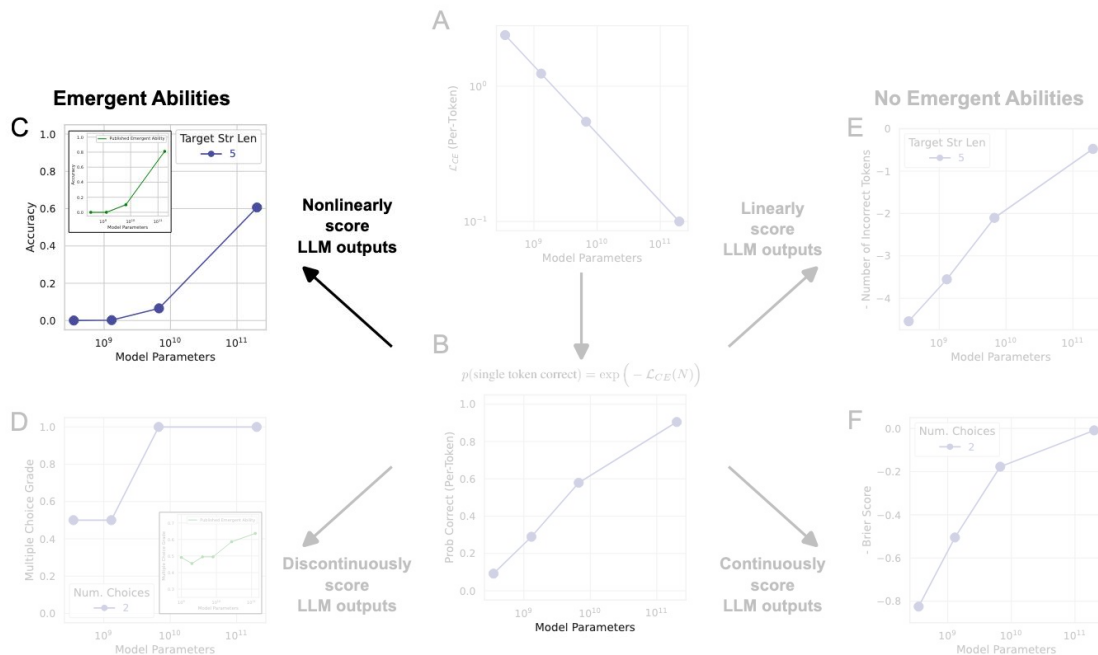
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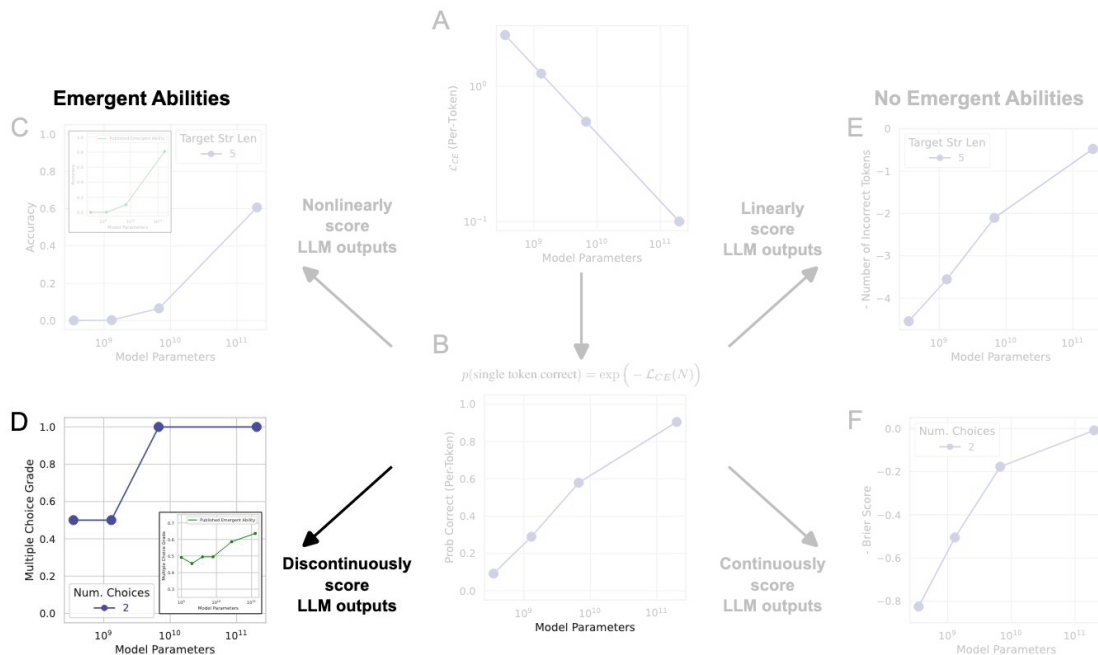
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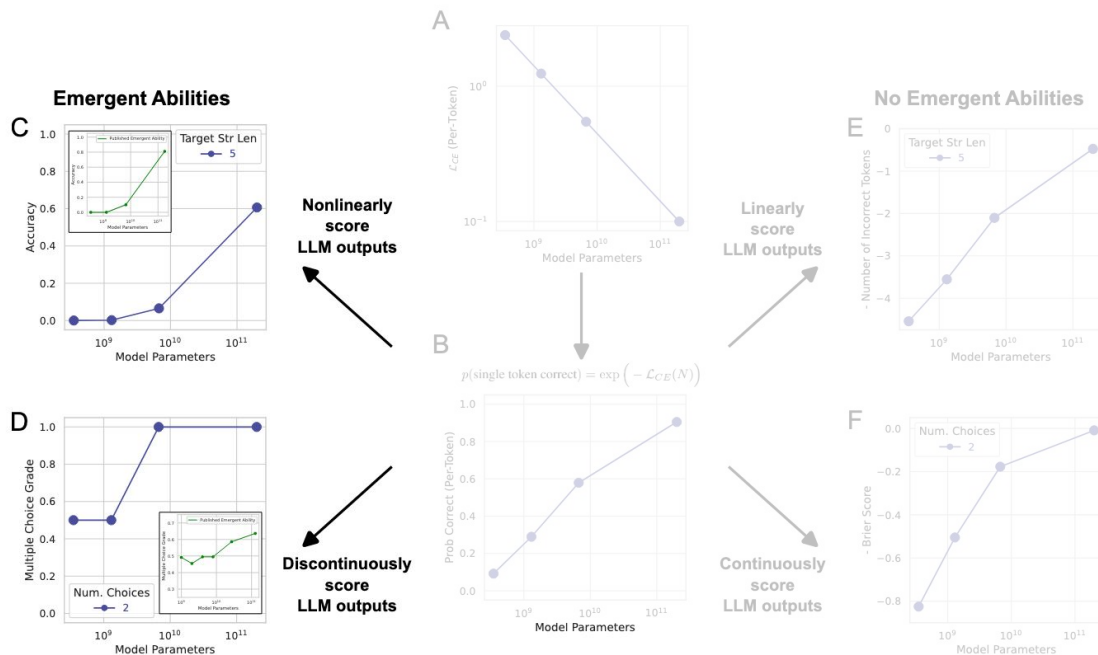
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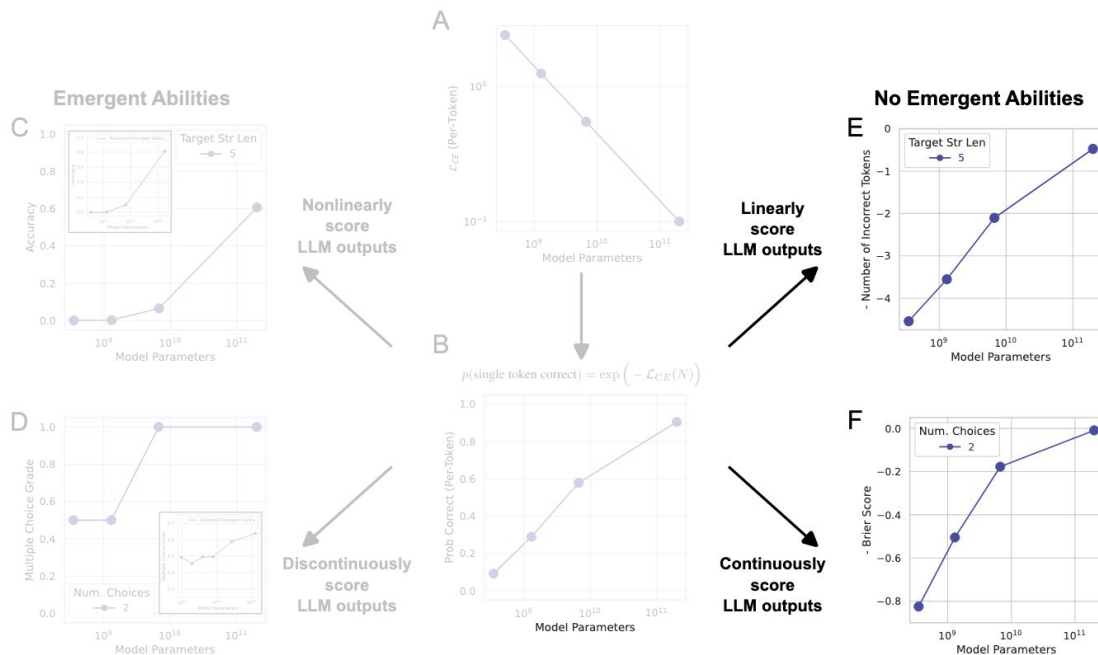
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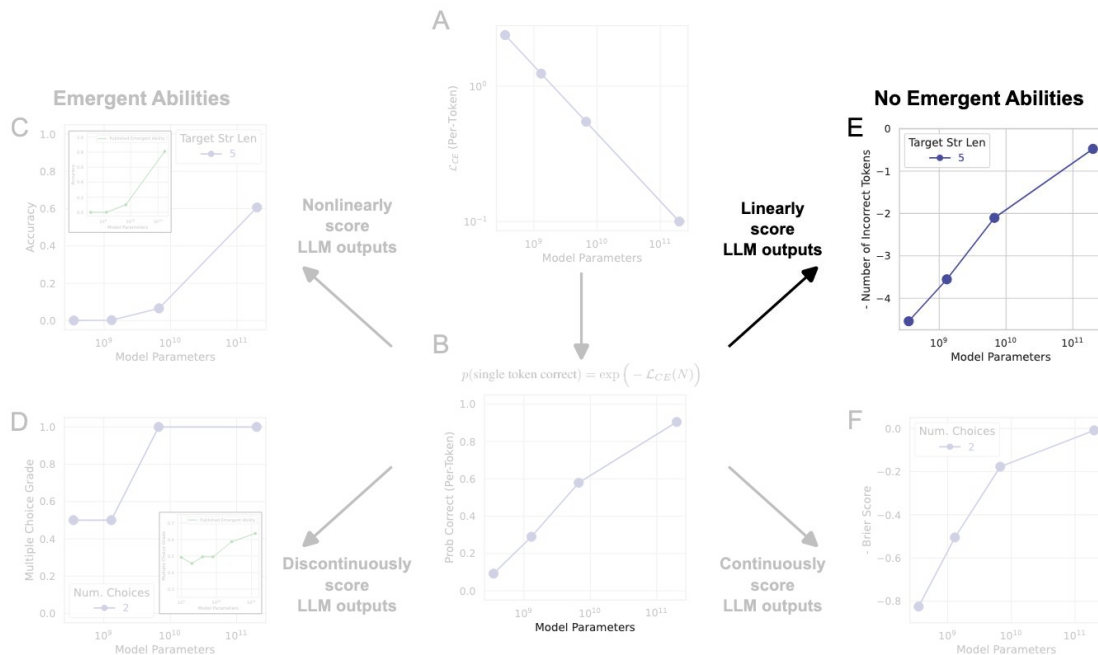
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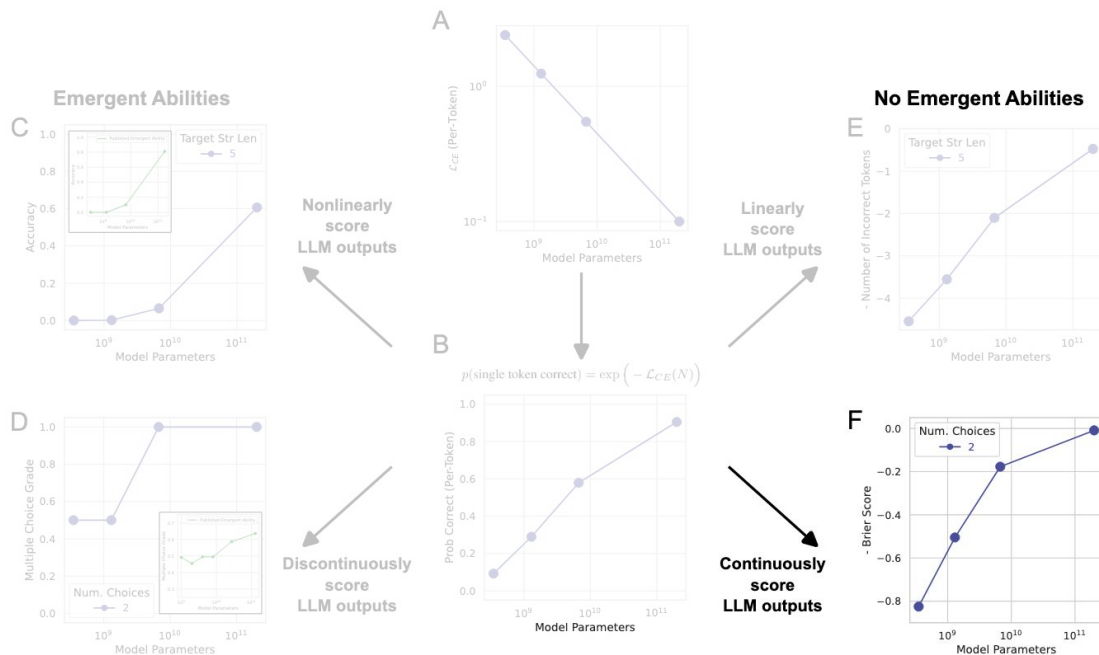
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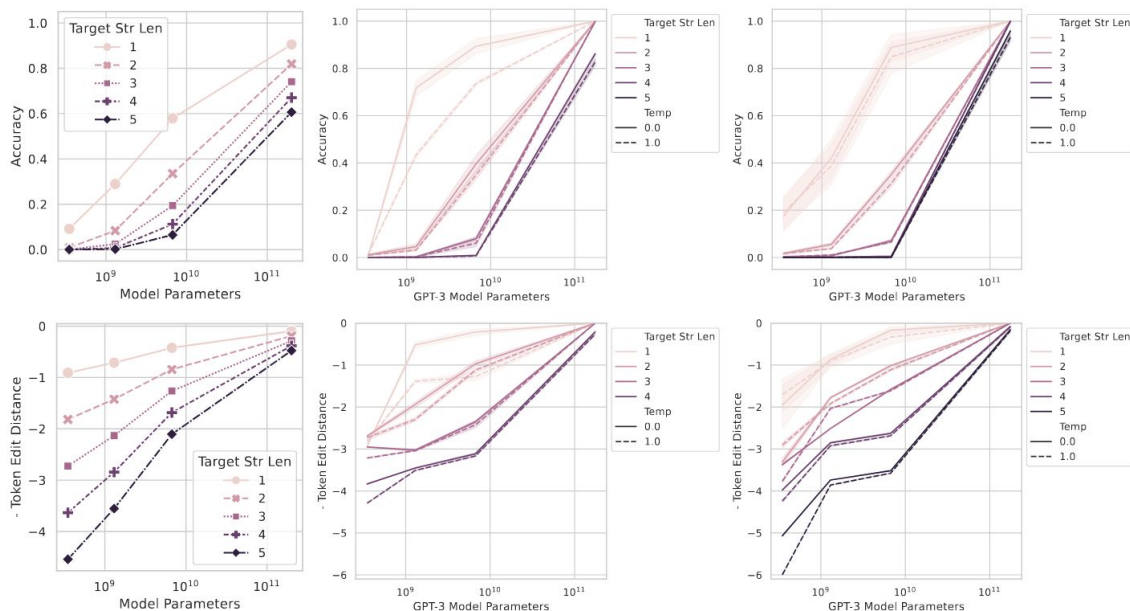


We should **design** *metrics*

We should **design** *metrics*
that are **compatible** with *scaling laws*.

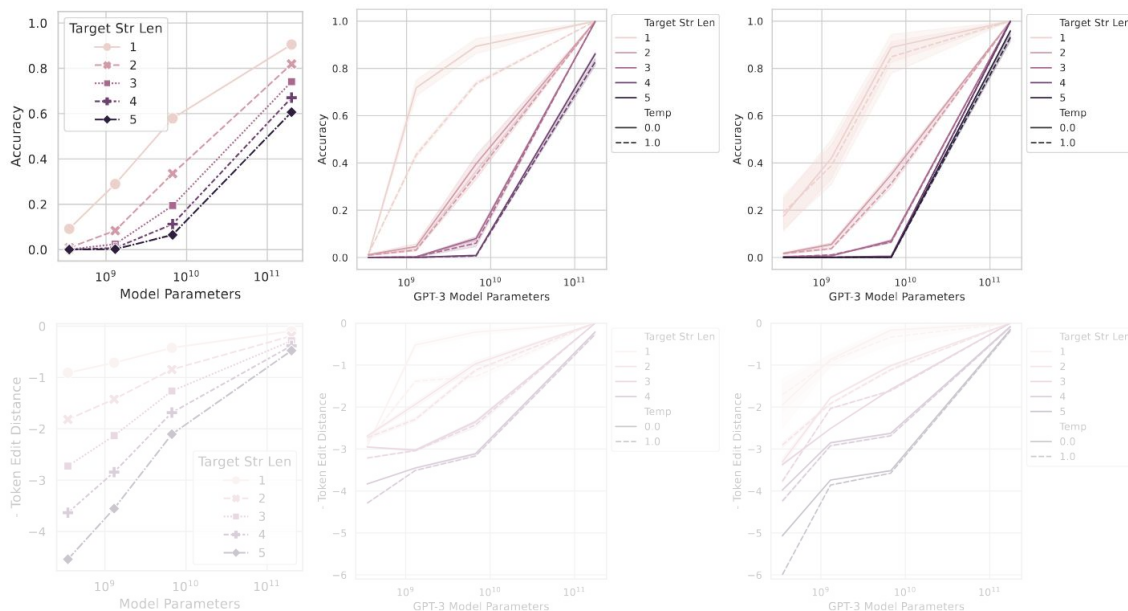
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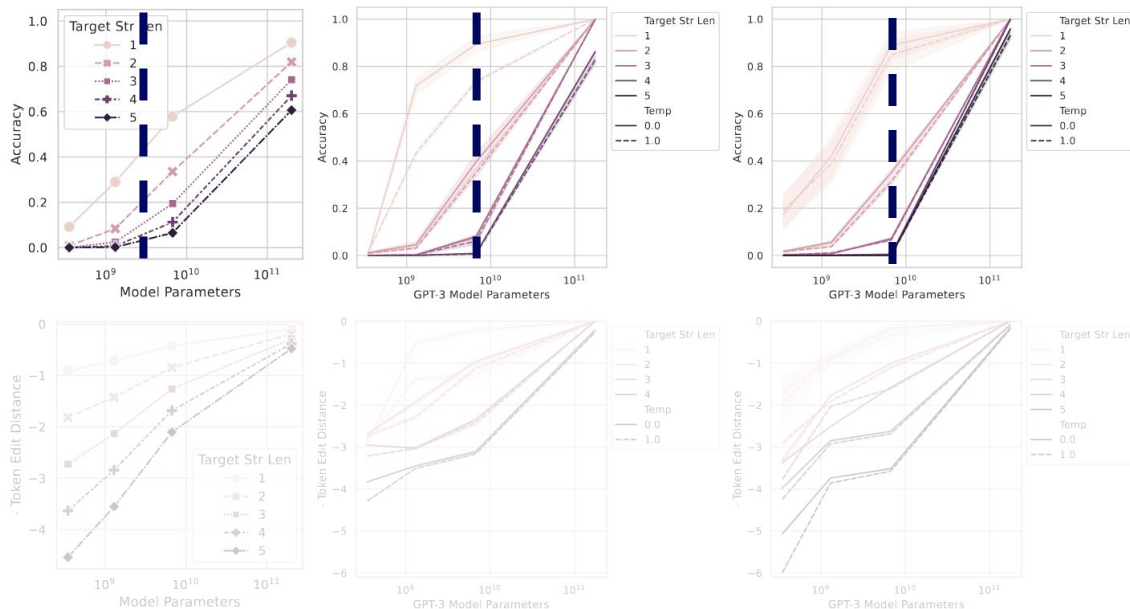
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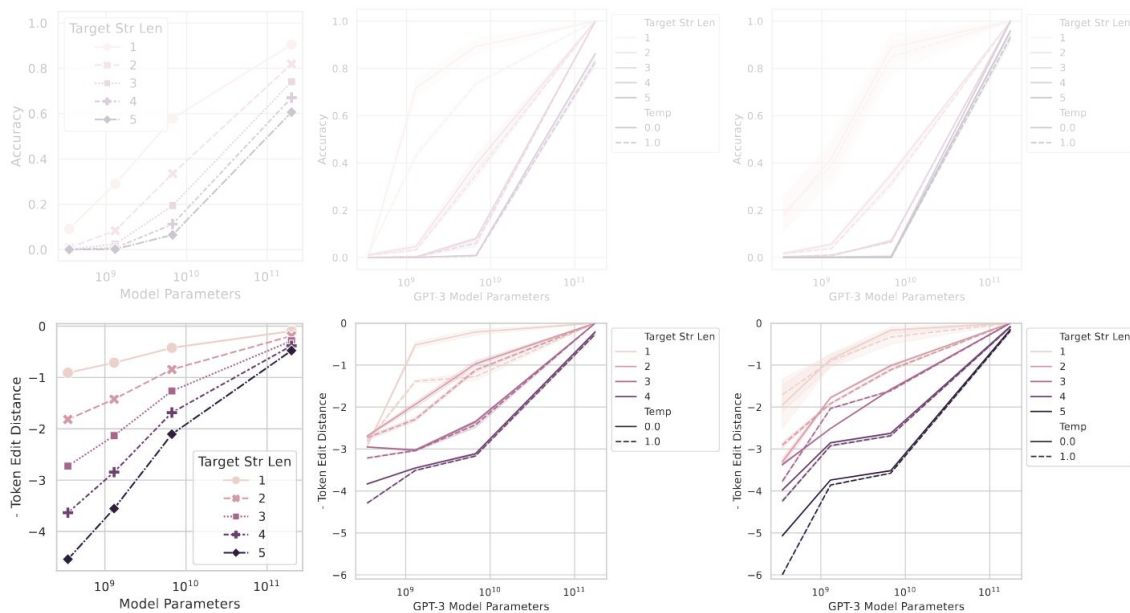
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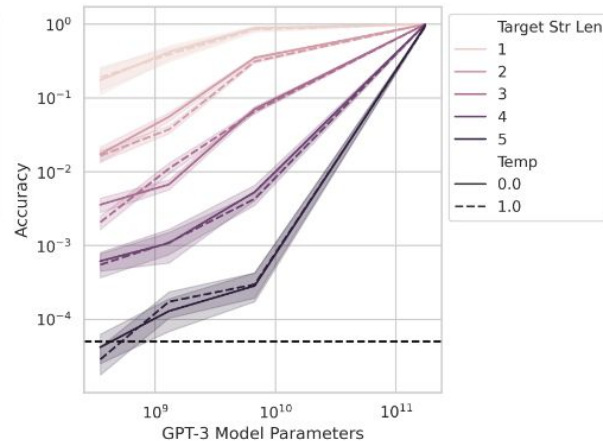
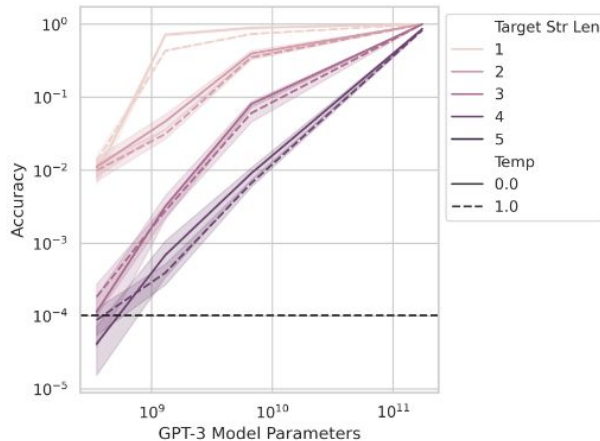
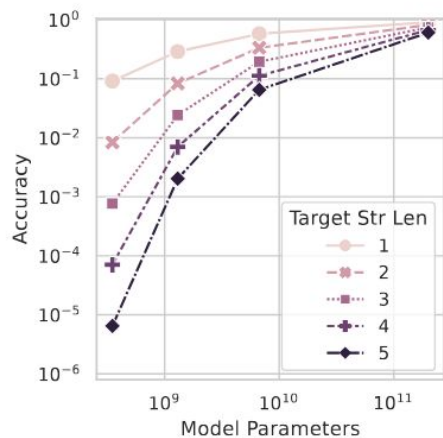


Emergence is *mitigated*

Emergence is *mitigated*
by **higher resolution** measurements.

Scaling Laws for Downstream Tasks

Are Emergent Abilities of Large Language Models a Mirage (Schaeffer et al., 2023)



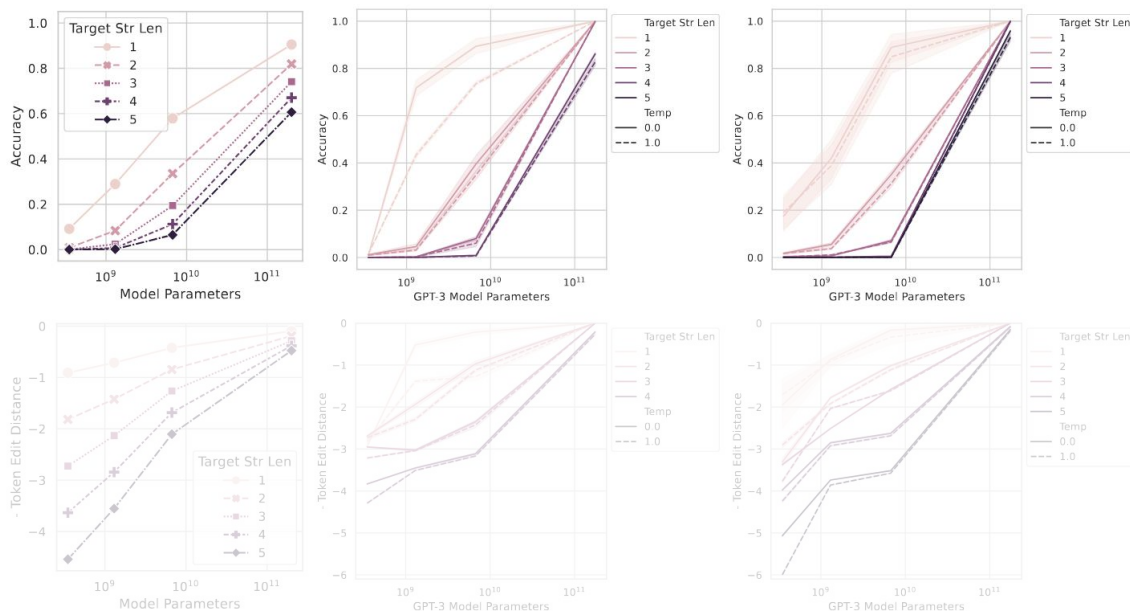
How can we
fix *emergence*?

Scaling Laws for Downstream Tasks

Consider **continuous, linear** metrics.

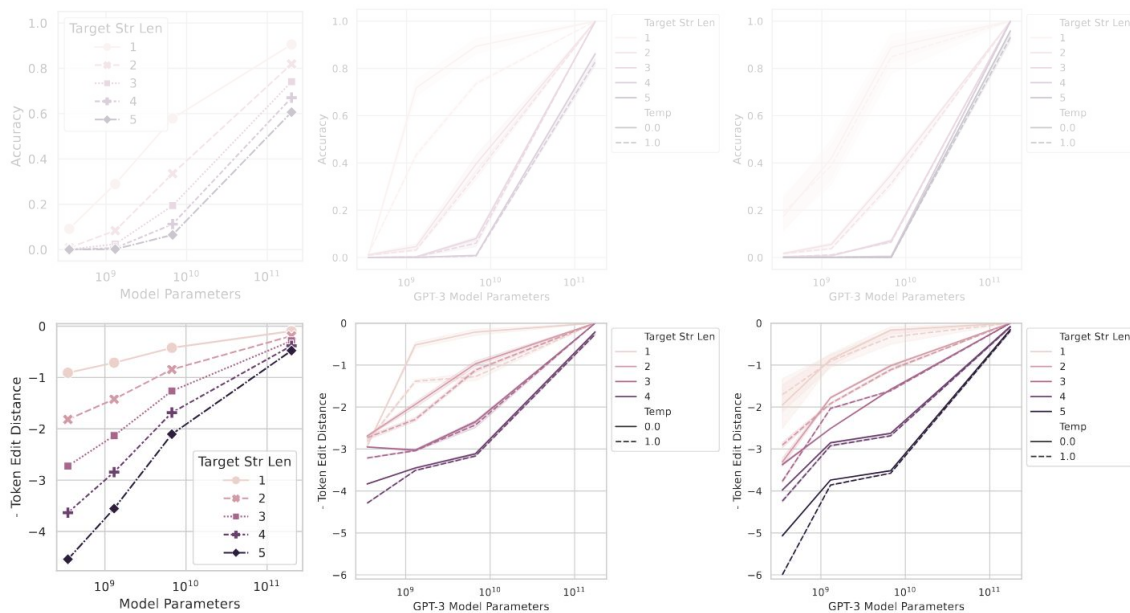
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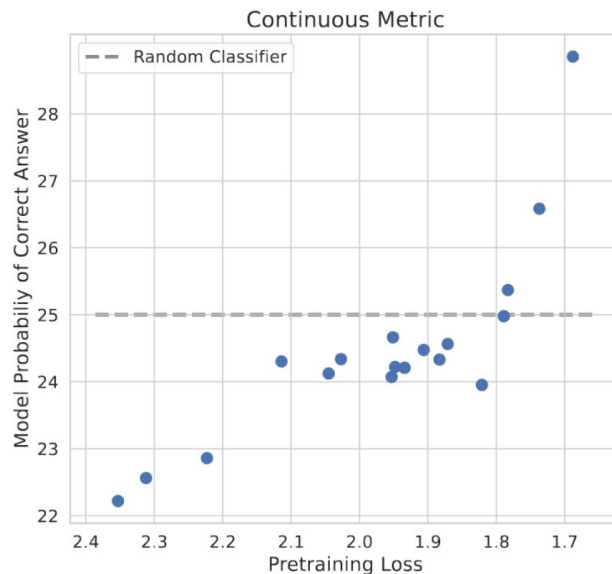
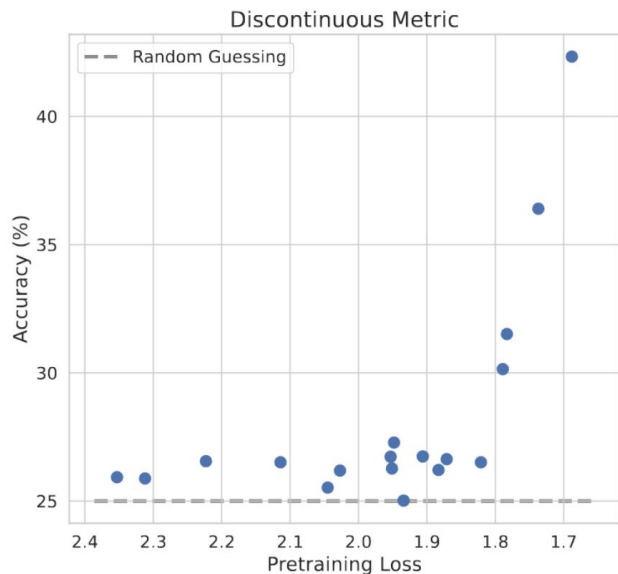
Scaling Laws for Downstream Tasks

But **scale** might *still* be necessary.

Scaling Laws for Downstream Tasks

Predicting Emergent Capabilities by Finetuning (Snell et al., 2024)

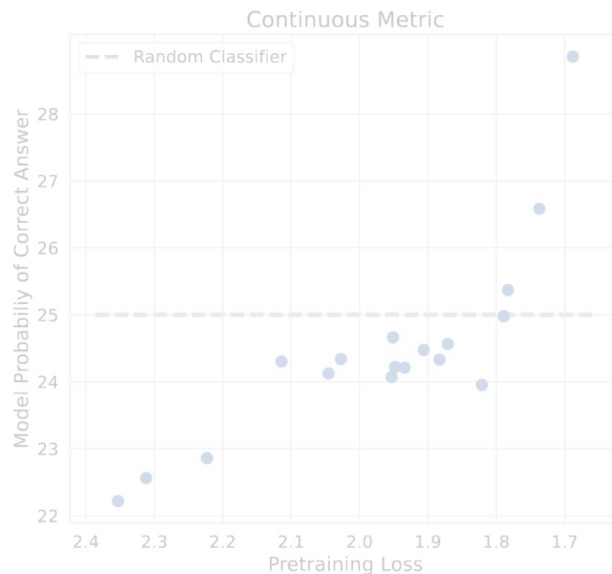
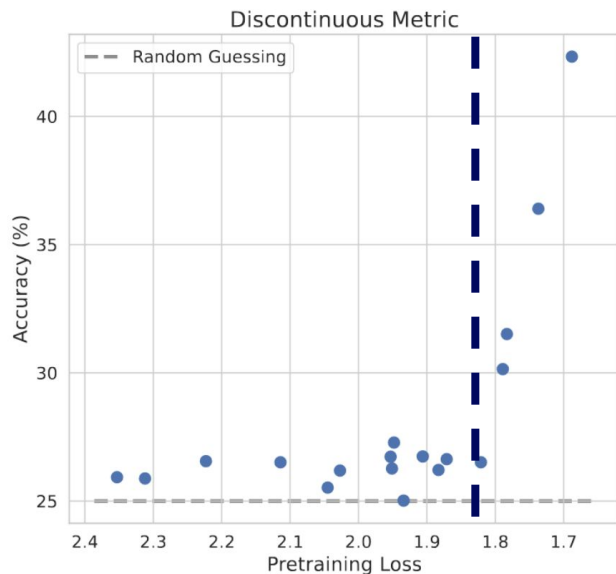
MMLU Emergence with Discontinuous and Continuous Metrics



Scaling Laws for Downstream Tasks

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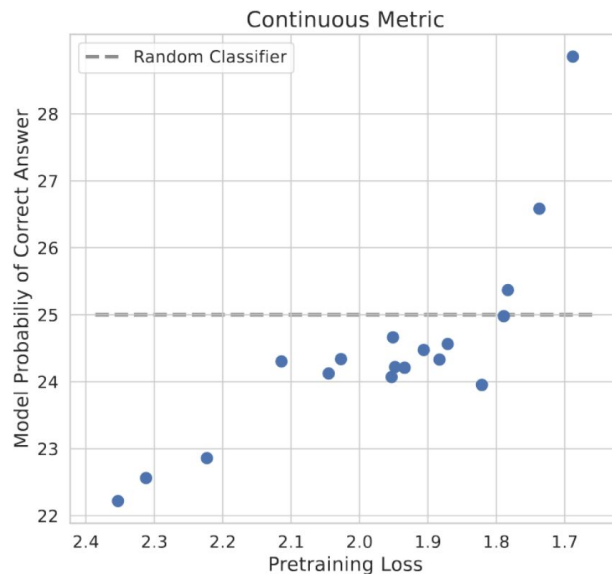
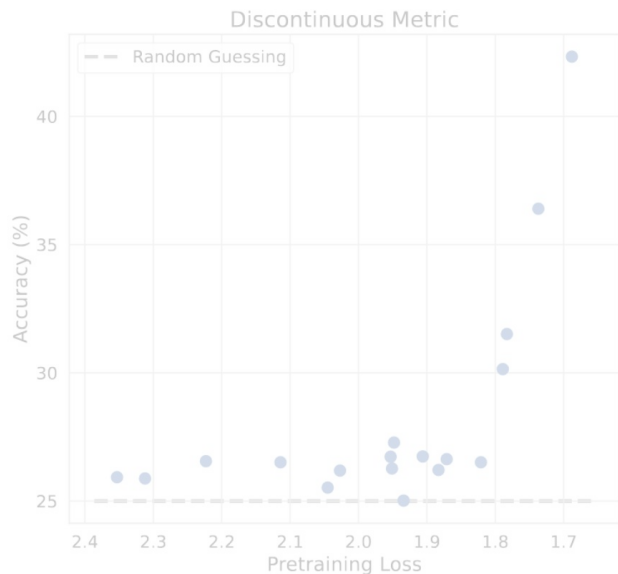
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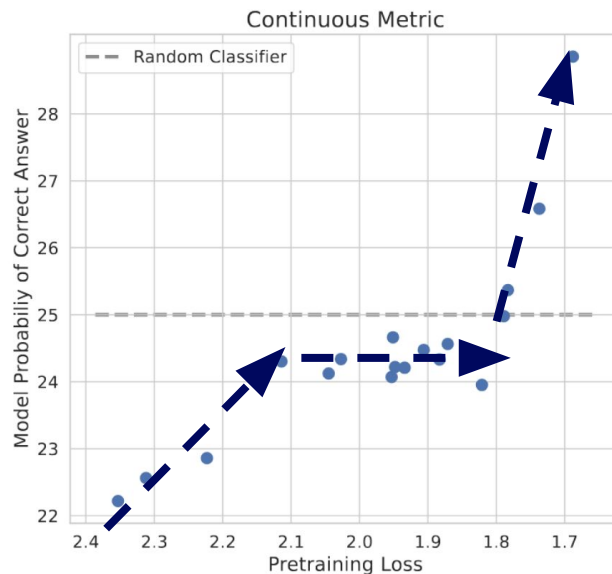
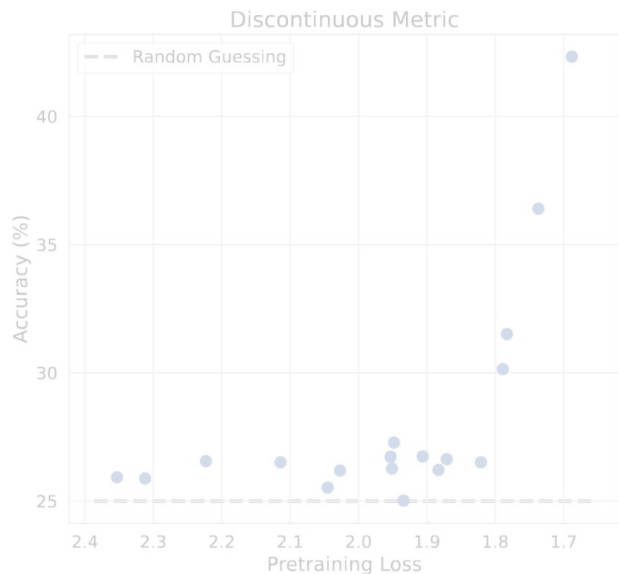
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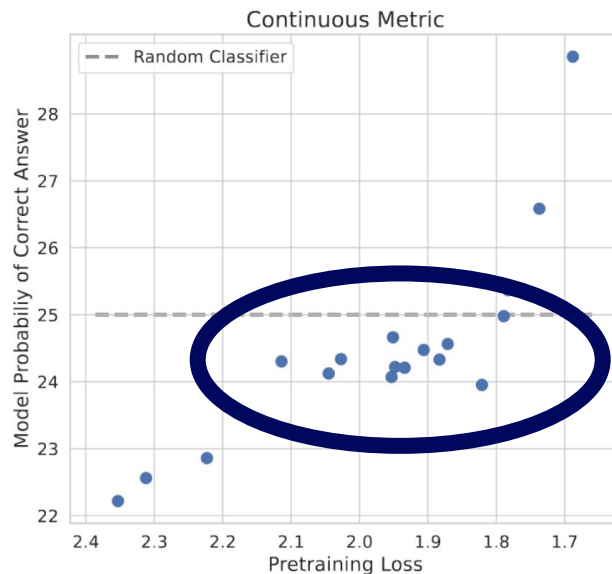
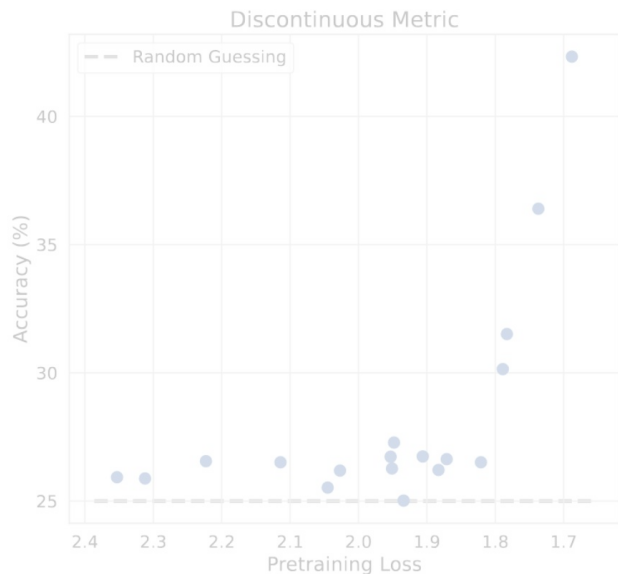
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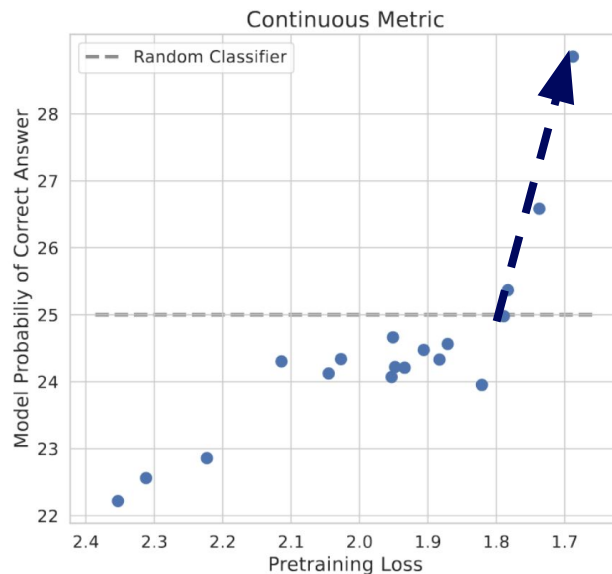
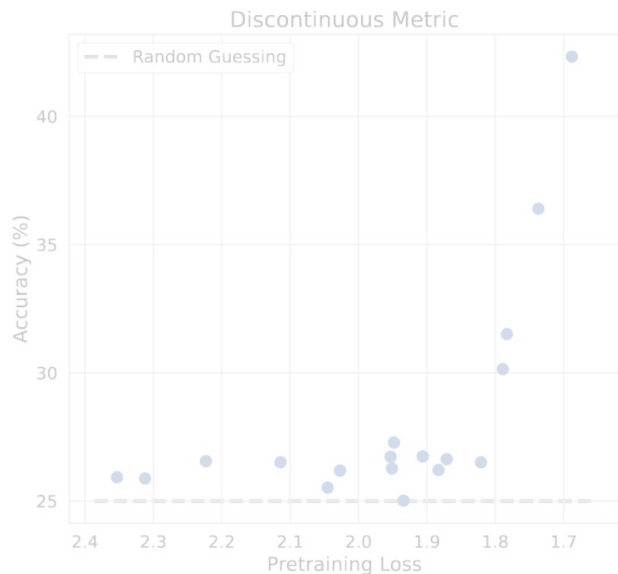
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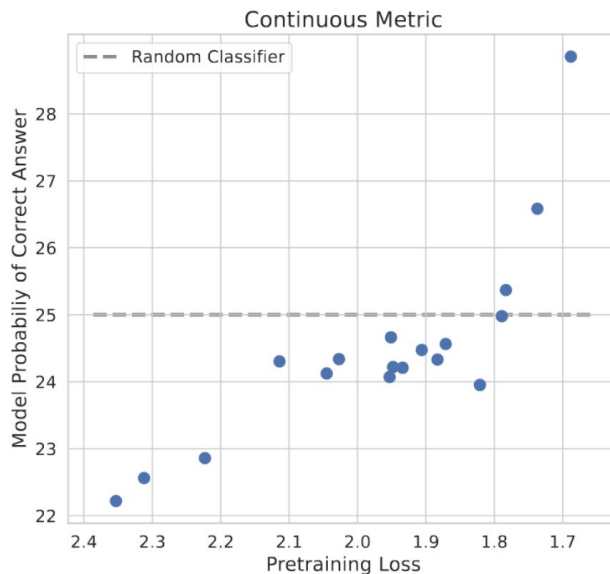
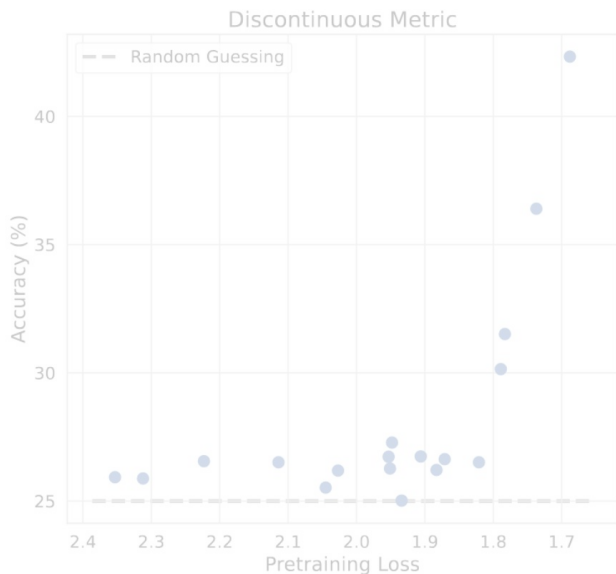
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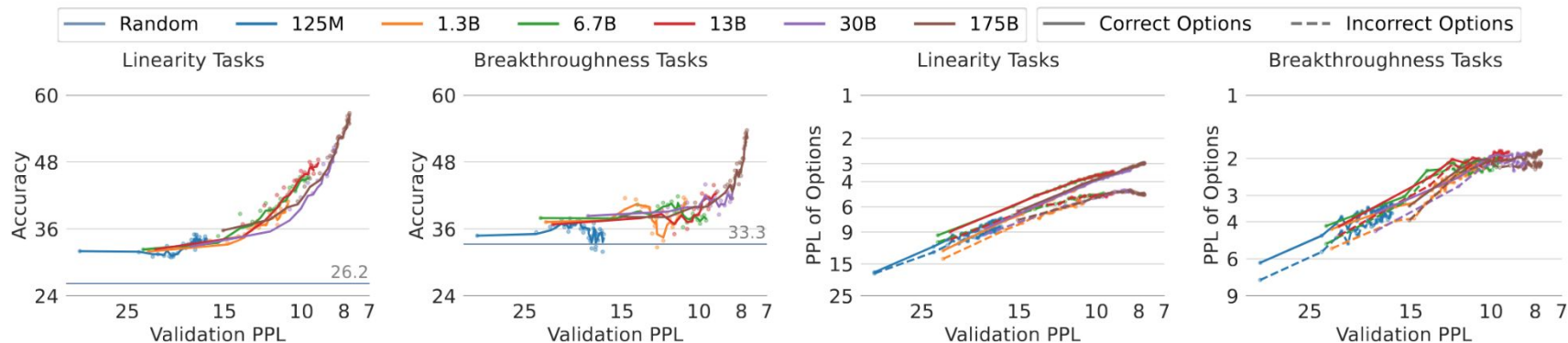
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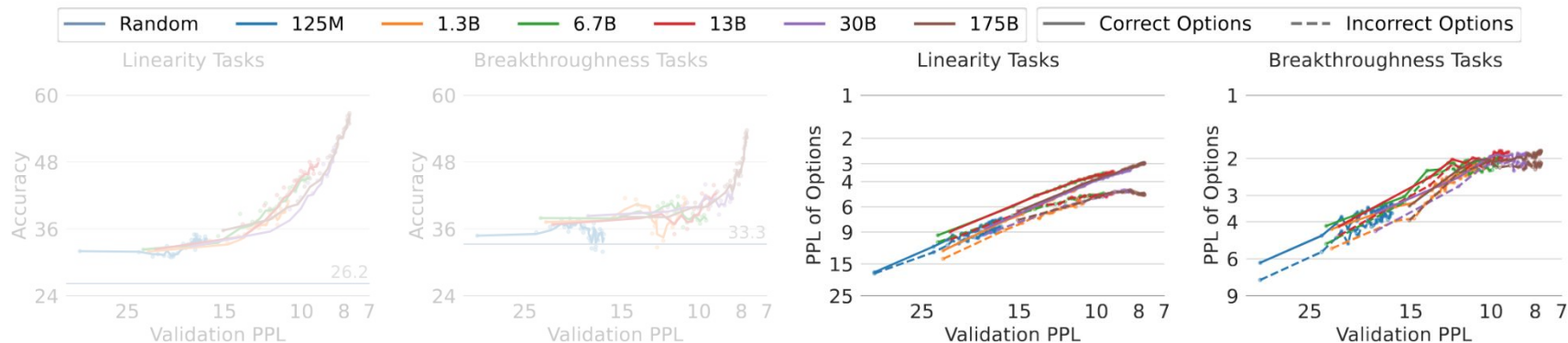
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Scaling Laws for Downstream Tasks

Training Trajectories of Language Models Across Scales (Xia et al., 2023)



We **create** emergence.

We ~~create~~ emergence.

We **worsen** emergence.

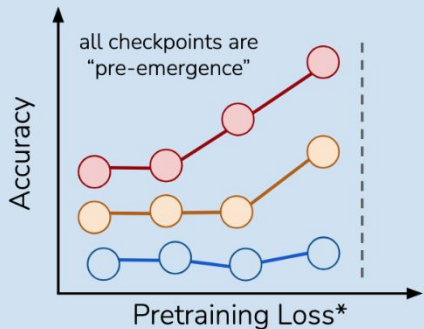
Scaling Laws for Downstream Tasks

Predicting Emergent Capabilities by Finetuning (Snell et al., 2024)

Observation

Finetuning shifts the point of emergence

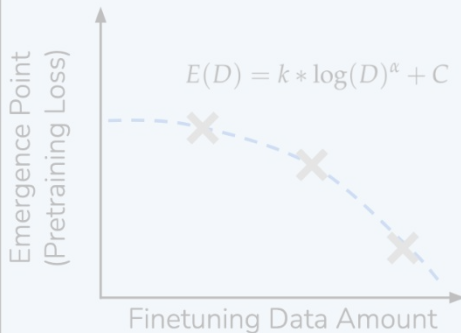
few-shot finetune big finetune



Our Method

1) Fit an "emergence law", which models how the point of emergence shifts with data

2) To predict few-shot emergence, take the emergence law to the "low data limit"



*Pretraining loss is decreasing as the x-axis increases

Scaling Laws for Downstream Tasks

The more data you **finetune** on,

Scaling Laws for Downstream Tasks

The more data you **finetune** on,
the **sooner** emergence occurs.

Scaling Laws for Downstream Tasks

Predicting Emergent Capabilities by Finetuning (Snell et al., 2024)

Observation

Finetuning shifts the point of emergence

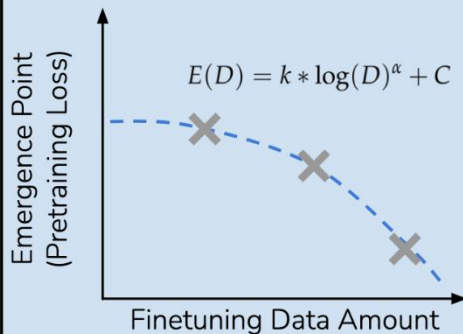
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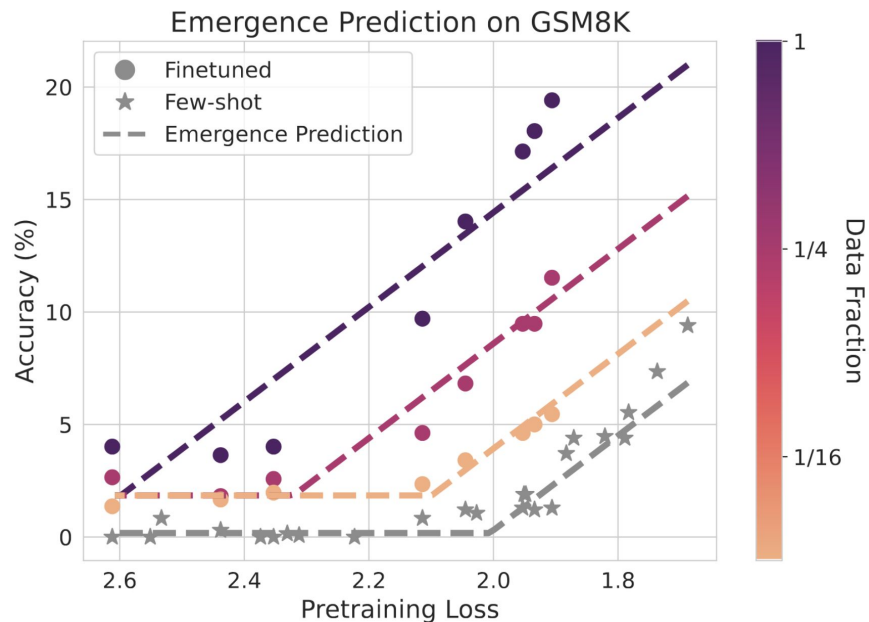
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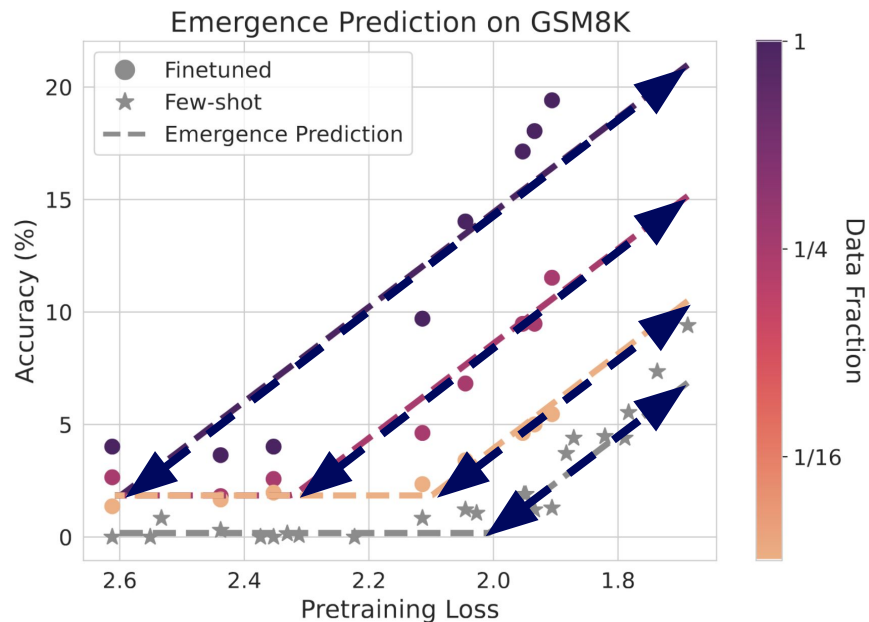
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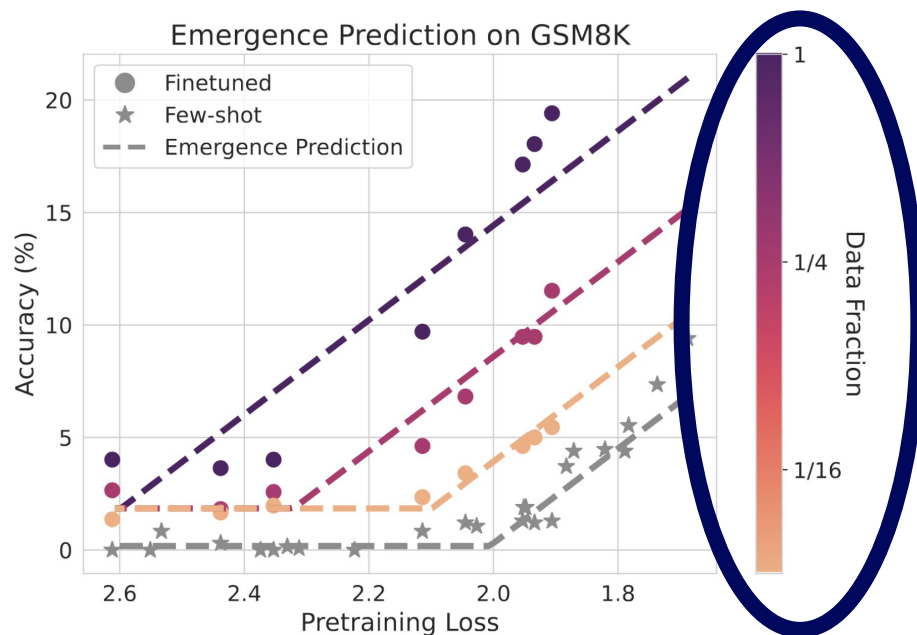
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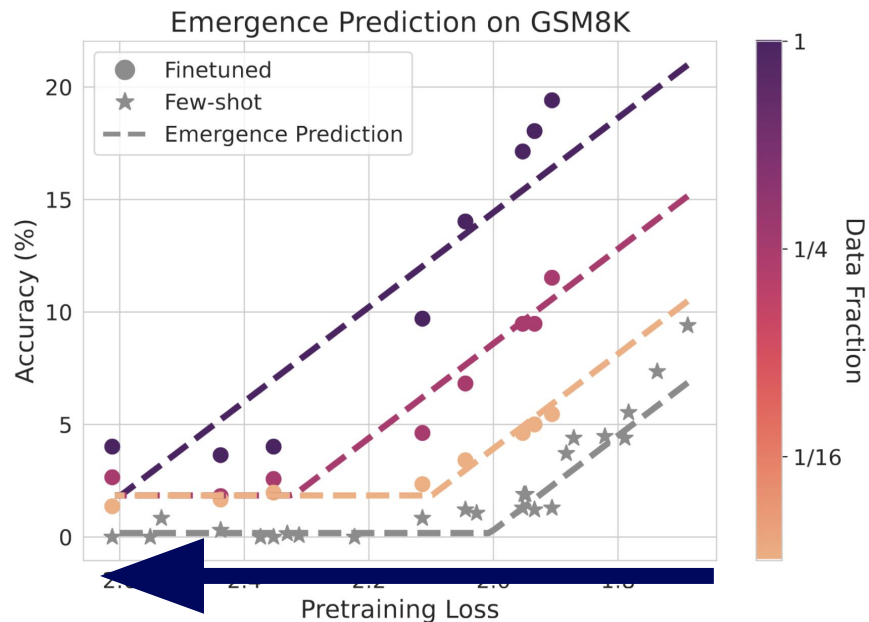
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Using Scaling Laws

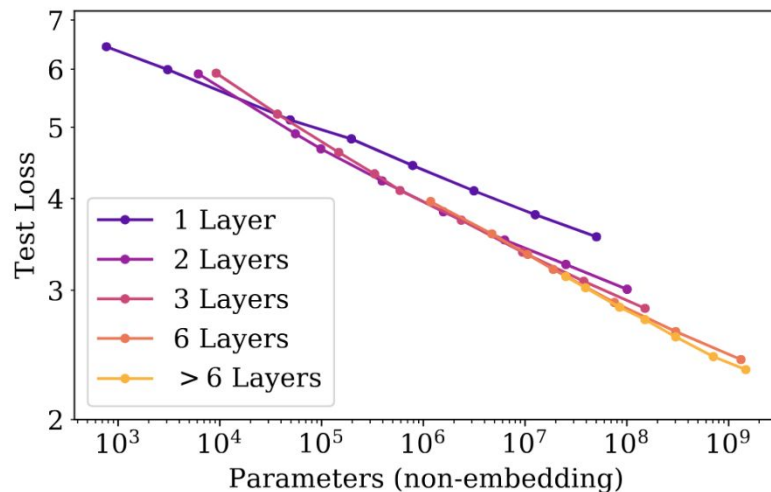
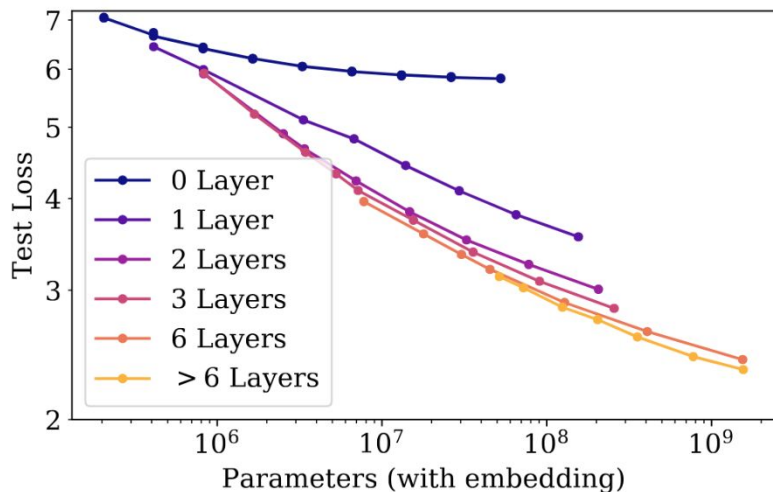
What are *scaling laws* **good for**?

Use scaling laws
to **design**
the *network*.

You can't *tune* hyperparameters **at scale**.

Using Scaling Laws

Scaling Laws for Neural Language Models (Kaplan et al., 2020)



Use scaling laws
to **balance**
data & compute.

Optimal Scaling

Optimal Scaling

Choose the model size

Optimal Scaling

Choose the model size
that achieves a given loss

Optimal Scaling

Choose the model size
that achieves a given loss
using the least possible compute.

Using Scaling Laws

$$L(C) \approx \left(\frac{C_c}{C} \right)^{\alpha_C}$$

*Language modeling loss is a **power law** of **optimal** compute.*

Using Scaling Laws

$$L(N, D) = \left[\left(\frac{N_c}{N} \right)^{\frac{\alpha_N}{\alpha_D}} + \frac{D_c}{D} \right]^{\alpha_D}$$

Scaling Laws for Neural Language Models (Kaplan et al., 2020)

Using Scaling Laws

Scale *parameters* **faster** than *data*.

Using Scaling Laws

$$\hat{L}(N, D) \triangleq E + \frac{A}{N^\alpha} + \frac{B}{D^\beta}.$$

Training Compute-Optimal Large Language Models (Hoffman et al., 2022)

Using Scaling Laws

Scale *parameters* **faster** than *data*.

Using Scaling Laws

~~Scale *parameters* **faster** than *data*.~~

Scale *parameters* **the same** as *data*.

But,
what *is* **optimal**
anyway?

Using Scaling Laws

Optimal scaling *minimizes* the compute for **training**.

Using Scaling Laws

We need to *consider* the compute for **inference**.

Overtraining

Overtraining

Training a smaller model

Overtraining

Training a smaller model
beyond the optimal compute.

What if data
isn't *infinite*?

Infinite Data

Finite Data

Infinite Data

- Often assumed.



Finite Data



Infinite Data

- Often assumed.

Finite Data

- An additional constraint.

Infinite Data

- Often assumed.
 - Compute is the bottleneck.
-

Finite Data

- An additional constraint.
-

Infinite Data

- Often assumed.
 - Compute is the bottleneck.
-

Finite Data

- An additional constraint.
 - Data is the bottleneck.
-

Infinite Data

- Often assumed.
 - Compute is the bottleneck.
 - Use each example once.
-

Finite Data

- An additional constraint.
 - Data is the bottleneck.
-

Infinite Data

- Often assumed.
 - Compute is the bottleneck.
 - Use each example once.
-

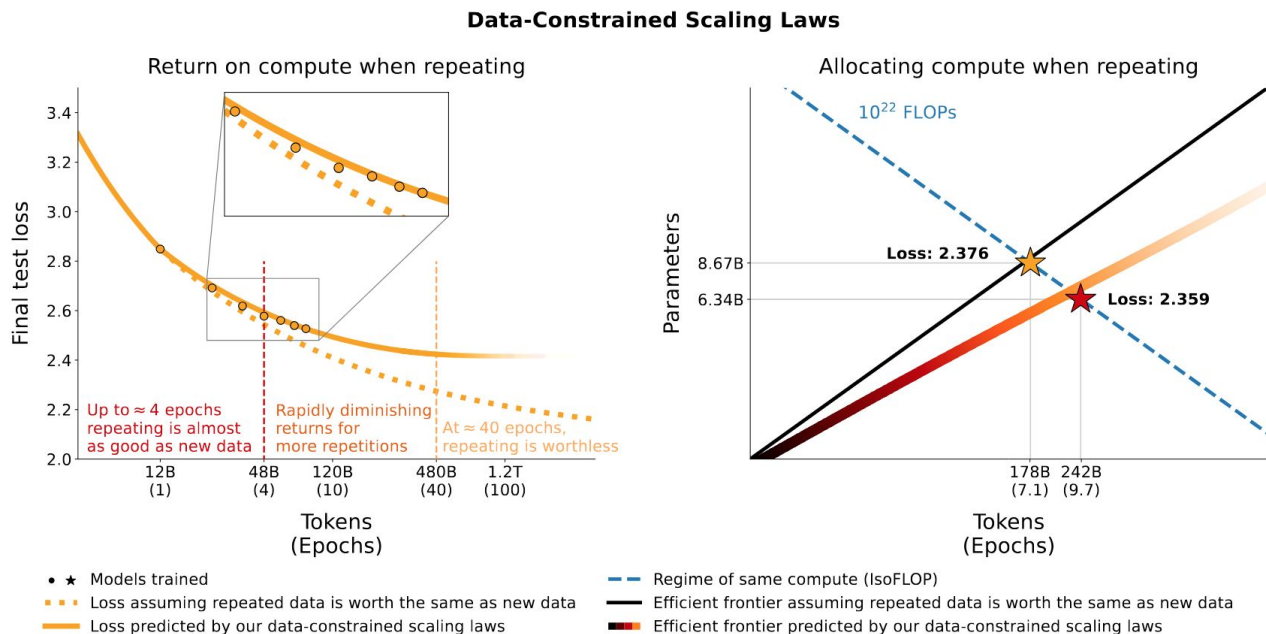
Finite Data

- An additional constraint.
 - Data is the bottleneck.
 - Use examples many times.
-

How does
scaling work
with **repeated** *data*?

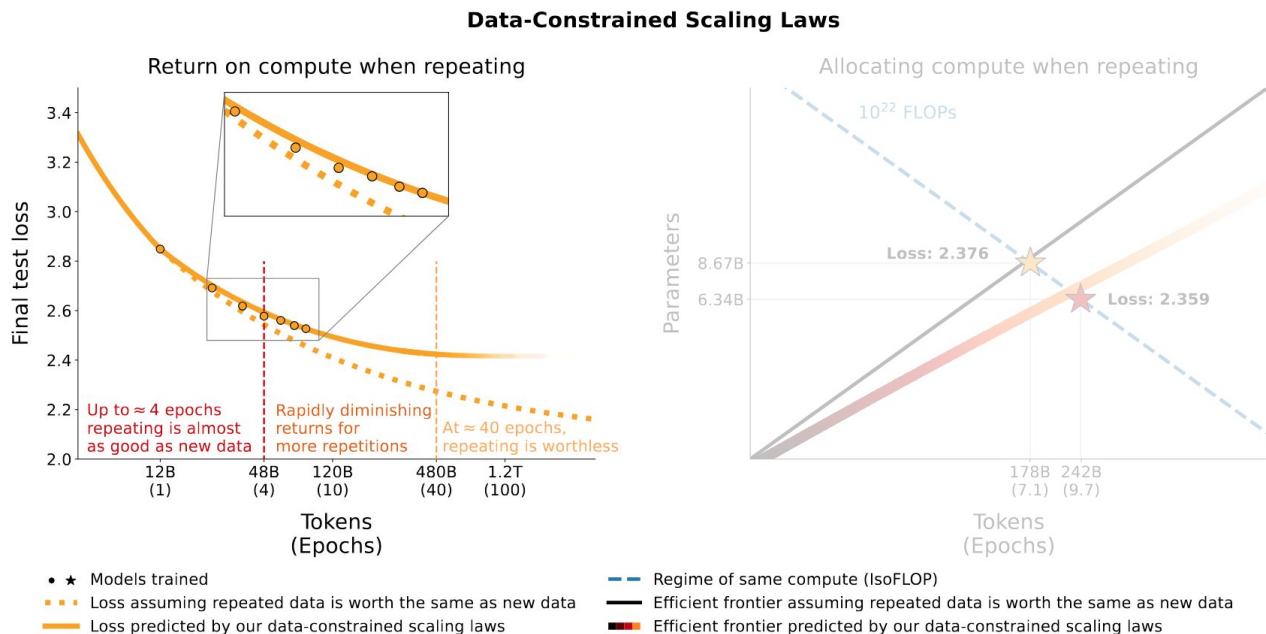
Using Scaling Laws

Scaling Data-Constrained Language Models (Muennighoff et al., 2023)



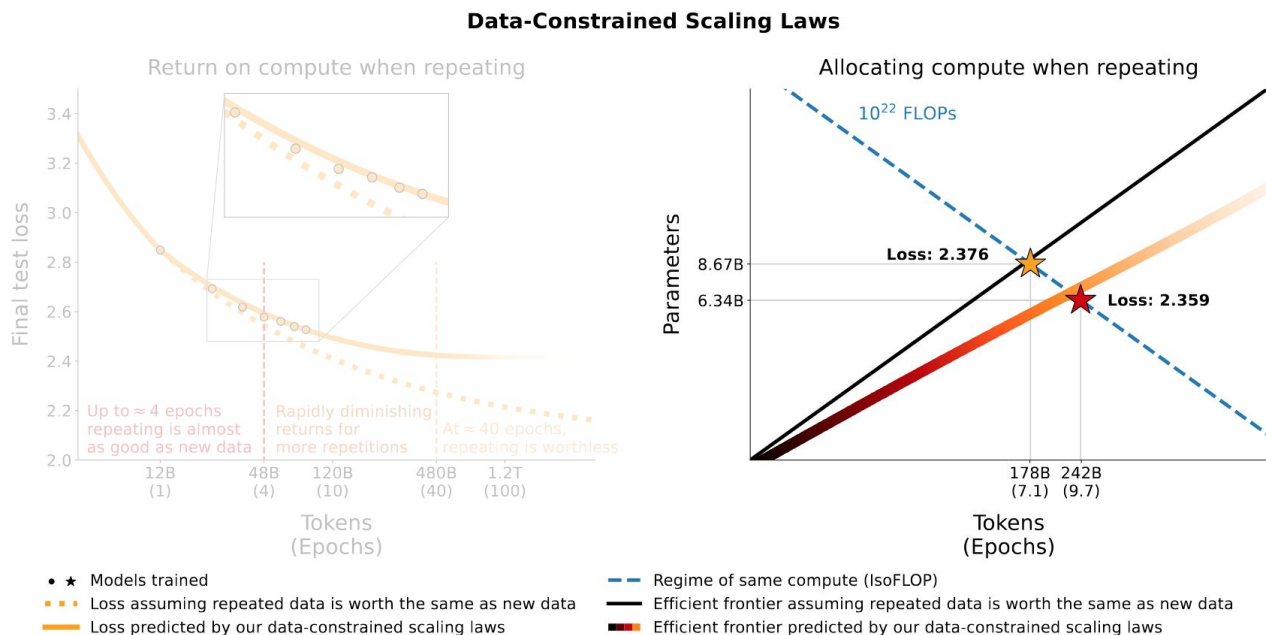
Using Scaling Laws

Scaling Data-Constrained Language Models (Muennighoff et al., 2023)



Using Scaling Laws

Scaling Data-Constrained Language Models (Muennighoff et al., 2023)



Using Scaling Laws

$$(1 - 1/R_D^*)$$

Scaling Data-Constrained Language Models (Muennighoff et al., 2023)

Using Scaling Laws

$$D' = U_D + U_D R_D^* (1 - e^{\frac{-R_D}{R_D^*}})$$

Scaling Data-Constrained Language Models (Muennighoff et al., 2023)

Using Scaling Laws

$$N' = U_N + U_N R_N^* (1 - e^{\frac{-R_N}{R_N^*}})$$

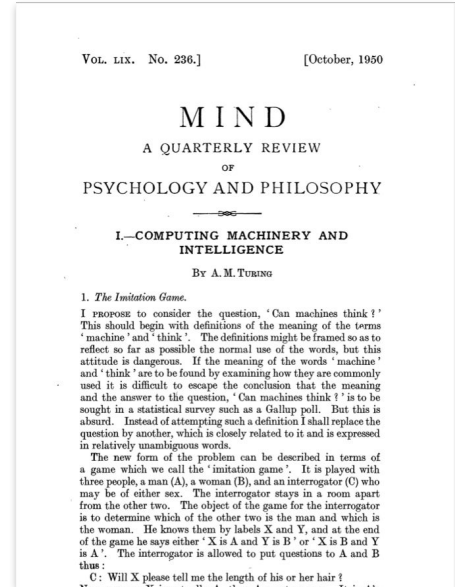
Scaling Data-Constrained Language Models (Muennighoff et al., 2023)

In the End...

Where does that **leave** us?

What are
the main
takeaways?

“Is there a corresponding phenomenon for minds, and is there one for machines?”

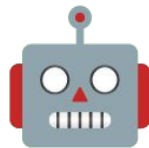


Computing Machinery and Intelligence, Alan M. Turing

How a method *scales* is **central**.



Intelligence = Data + Compute



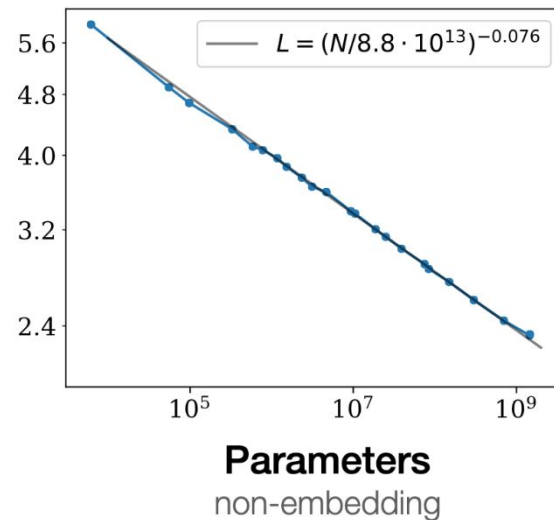
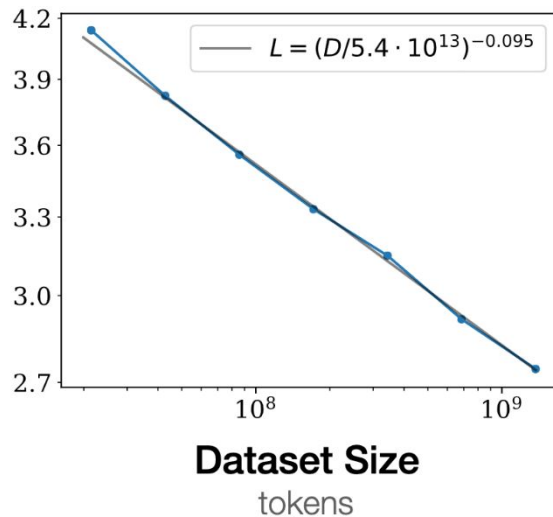
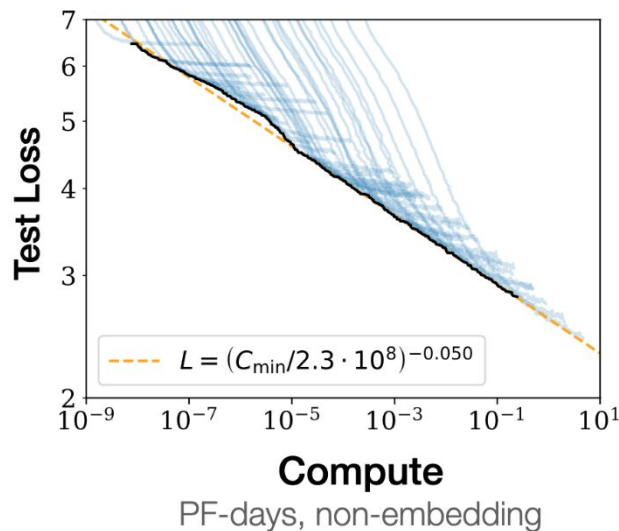
In the End...

compute $\propto$ parameters $\times$ data

*Measure compute by counting **batches** and **parameters**.*

In the End...

Scaling Laws for Neural Language Models (Kaplan et al., 2020)



It's not about the **best model**,
but which model *scales* **most efficiently**.

How will /
use **scaling laws**?

In the End...

How will I use scaling laws?

In the End...

How will I use scaling laws?

- Designing models and setting hyperparameters.

In the End...

How will I use scaling laws?

- Designing models and setting hyperparameters.
- Evaluating new ideas.

In the End...

How will I use scaling laws?

- Designing models and setting hyperparameters.
- Evaluating new ideas.
- Anticipating the cost vs. benefit.

Is it the **end**
of *scaling*?

In the End...

The *end* of scaling?

In the End...

~~The *end* of scaling?~~

The *end* of **economic** scaling?

Scaling laws give
a **simple theory**
of foundation models.

Thanks. 

beep boop beep!

References

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